

Microsoft ADC Cybersecurity Skilling Program

Week 9 Assignment 20

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Student ID: ADC-CSS02-25051

Securing Azure SQL Database

Introduction

In this lab, I will be working on **AZ-500 Lab 07: Key Vault – Implementing Secure Data by Setting Up Always Encrypted**. The objective of this lab is to enhance data security by leveraging **Azure Key Vault** in conjunction with **Azure SQL Database** to protect sensitive information using encryption technologies. I will begin by deploying the foundational infrastructure, including a virtual machine and SQL database, through an Azure Resource Manager (ARM) template. After that, I will configure an Azure Key Vault with the necessary keys and secrets to support encryption tasks. Subsequently, I will set up the SQL database and a sample data-driven application that will demonstrate real-world encryption practices. Finally, I will integrate the Azure Key Vault with the SQL database to demonstrate how **Always Encrypted** can be used to secure data at rest and ensure that sensitive data remains confidential, even from database administrators.

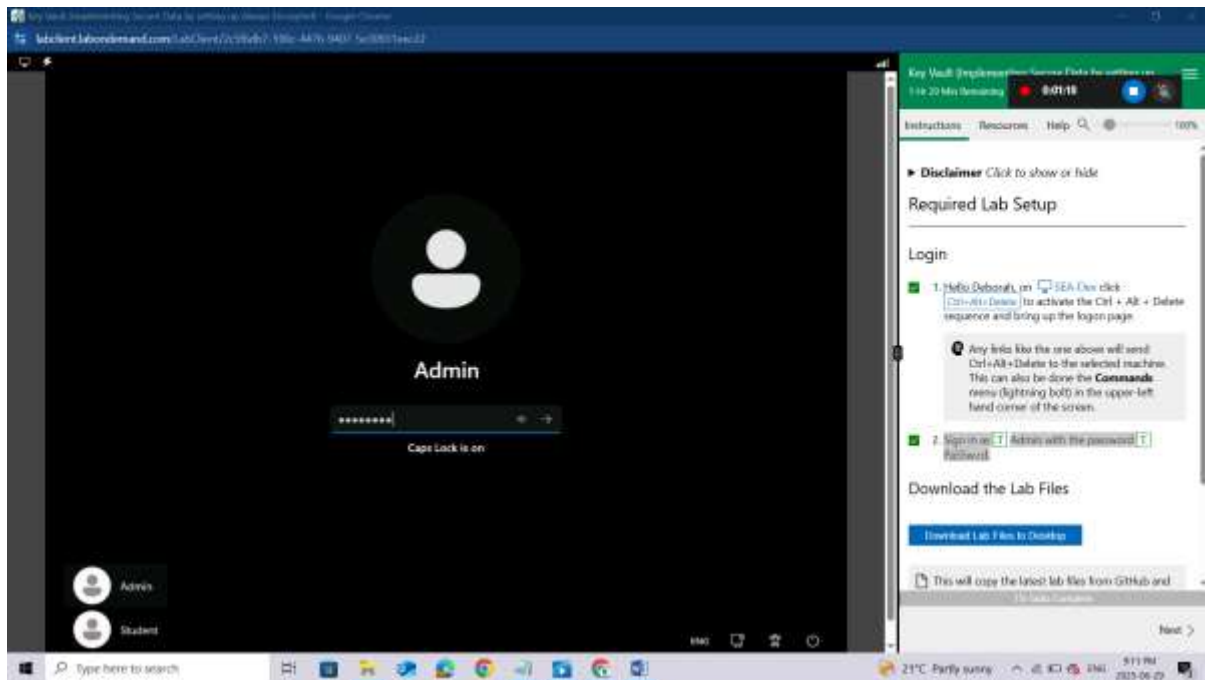
Tasks Completed

Required Lab Setup

Hello Deborah, on [SEA-Dev](#) click [Ctrl+Alt+Delete](#) to activate the Ctrl + Alt + Delete sequence and bring up the logon page.

*Any links like the one above will send Ctrl+Alt+Delete to the selected machine. This can also be done the **Commands** menu (lightning bolt) in the upper-left hand corner of the screen.*

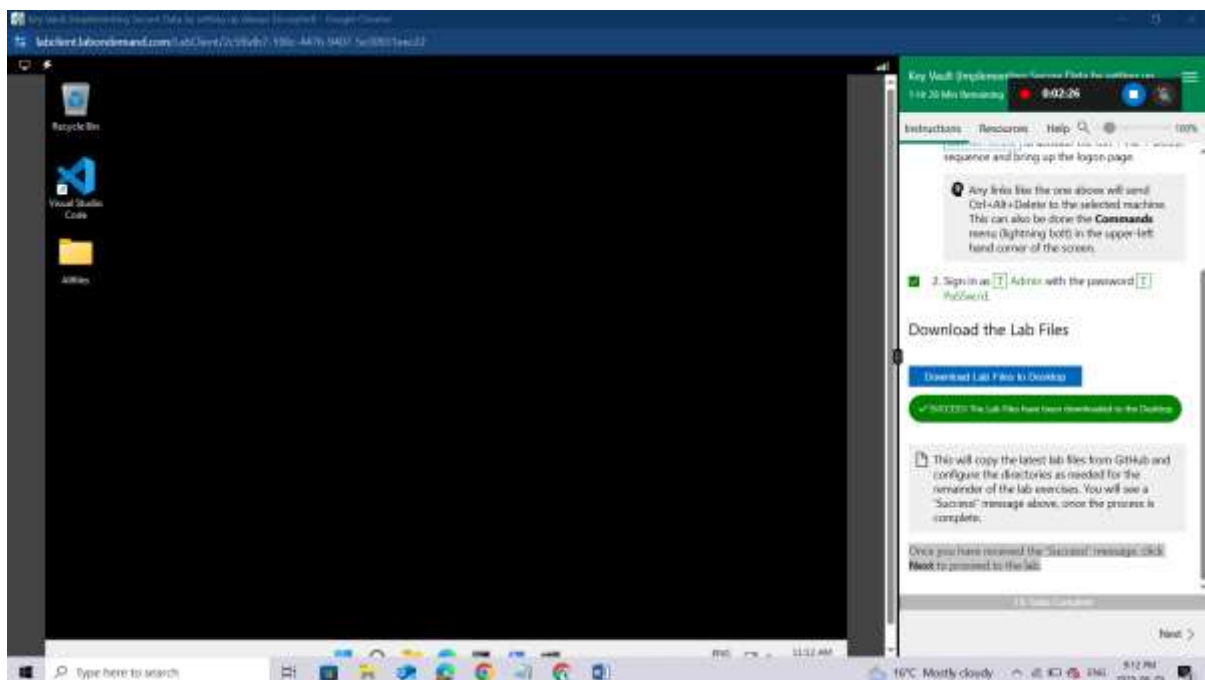
Sign in as **Admin** with the password **Pa55w.rd**.



Download the Lab Files

This will copy the latest lab files from GitHub and configure the directories as needed for the remainder of the lab exercises. You will see a 'Success!' message above, once the process is complete.

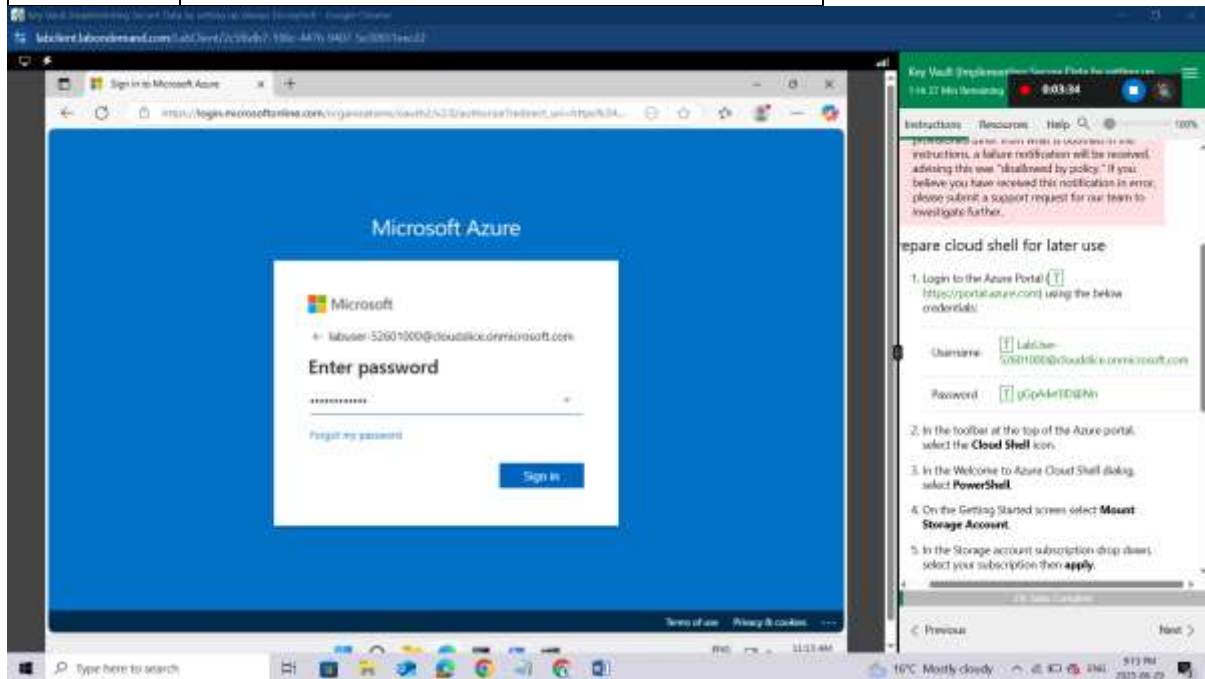
Once you have received the 'Success!' message, click **Next** to proceed to the lab.



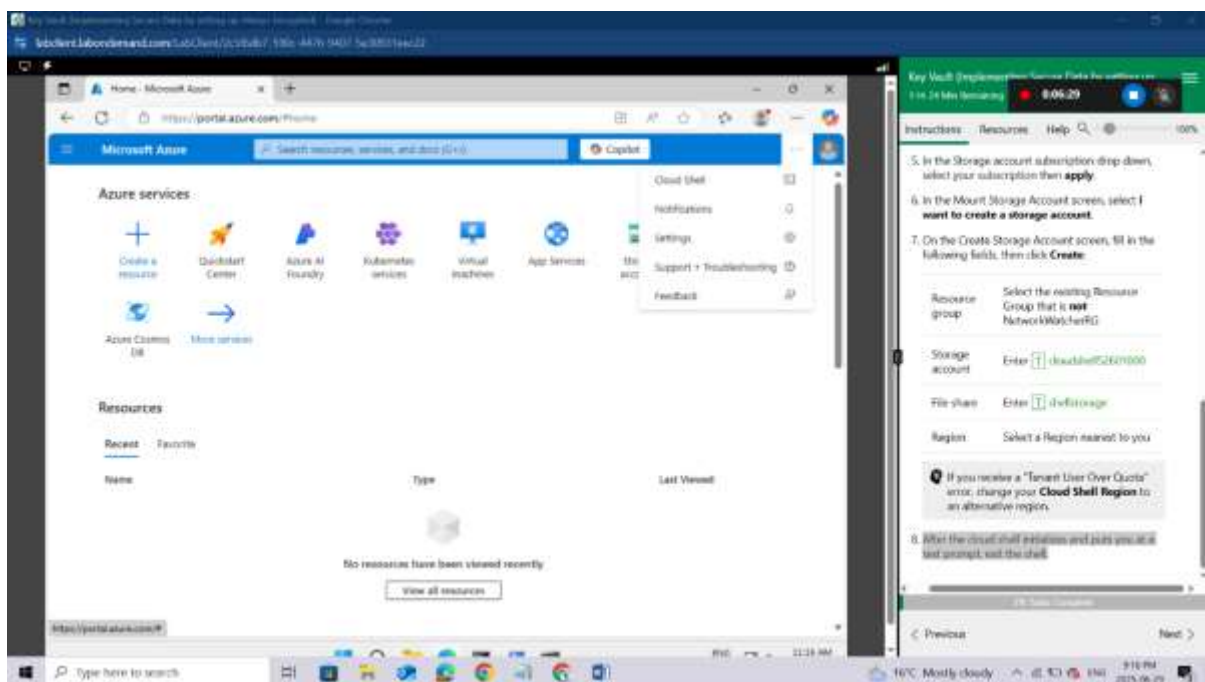
Prepare cloud shell for later use

Login to the Azure Portal (<https://portal.azure.com>) using the below credentials:

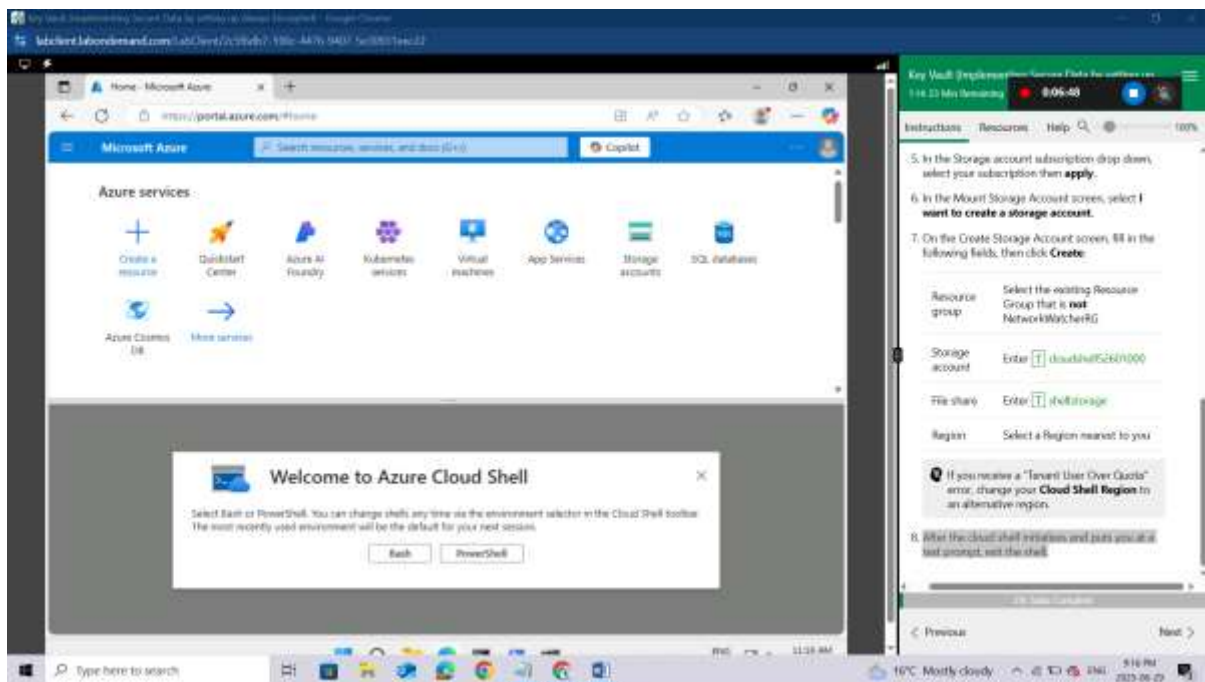
Username	LabUser-52601000@cloudslice.onmicrosoft.com
Password	gGpA4e!0D@Nn



In the toolbar at the top of the Azure portal, select the **Cloud Shell** icon.

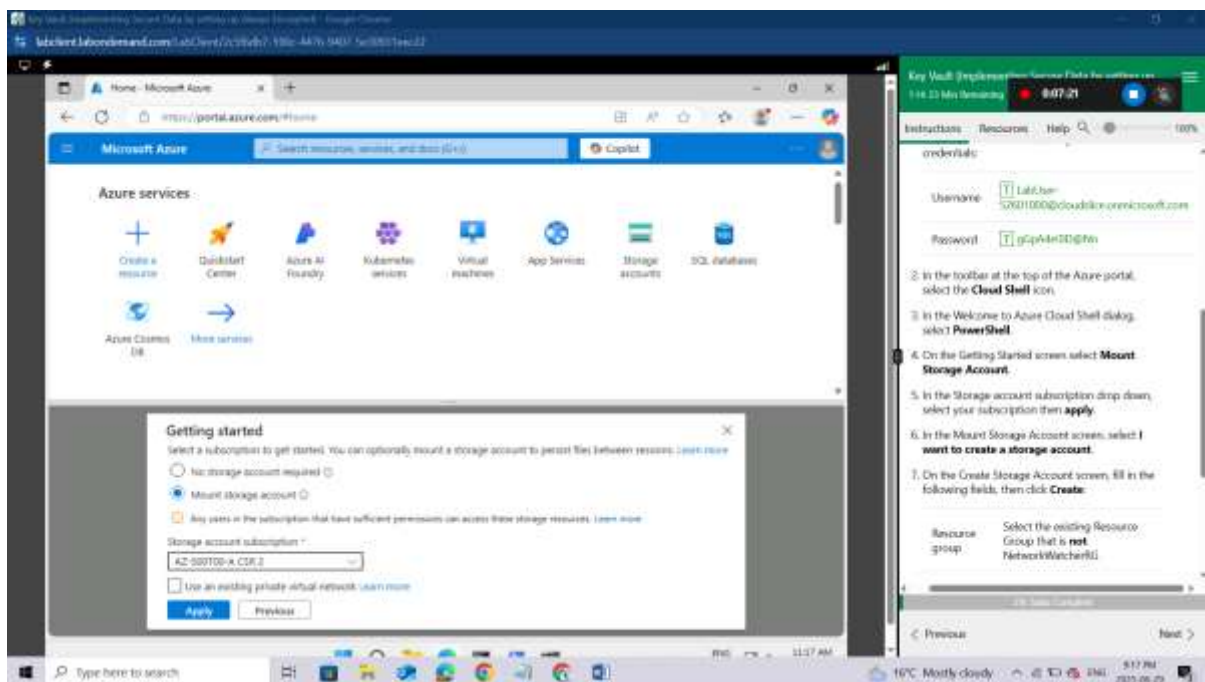


In the Welcome to Azure Cloud Shell dialog, select **PowerShell**.

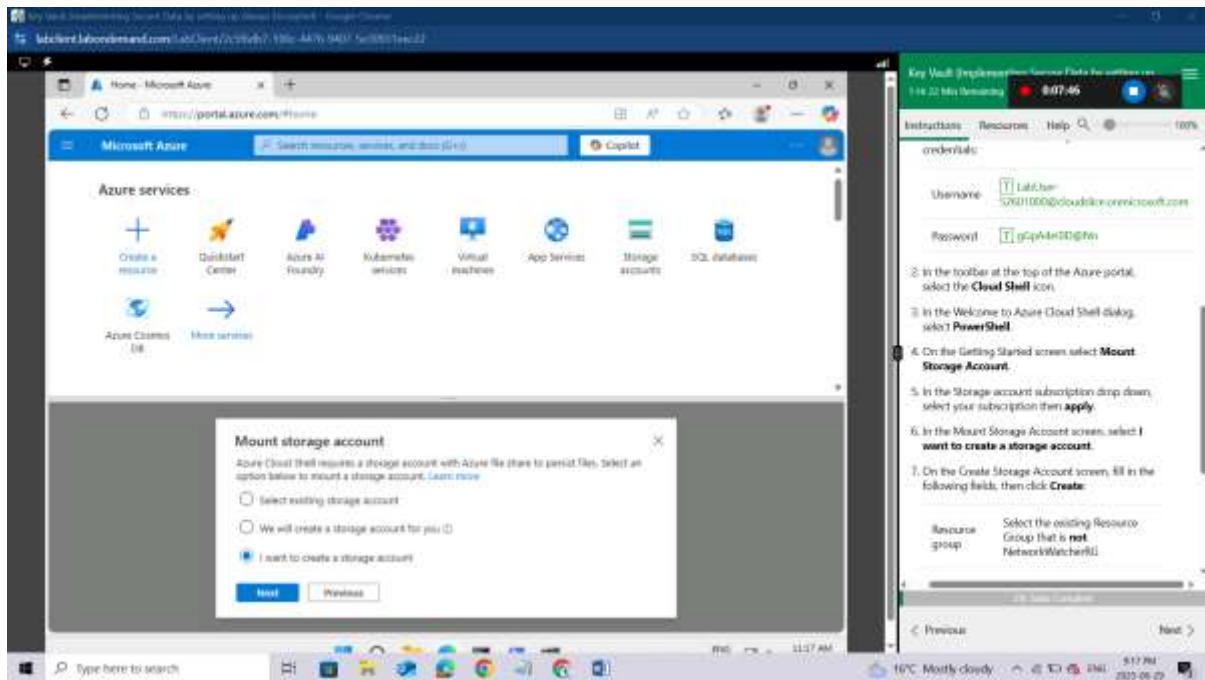


On the Getting Started screen select **Mount Storage Account**.

In the Storage account subscription drop down, select your subscription then **apply**.



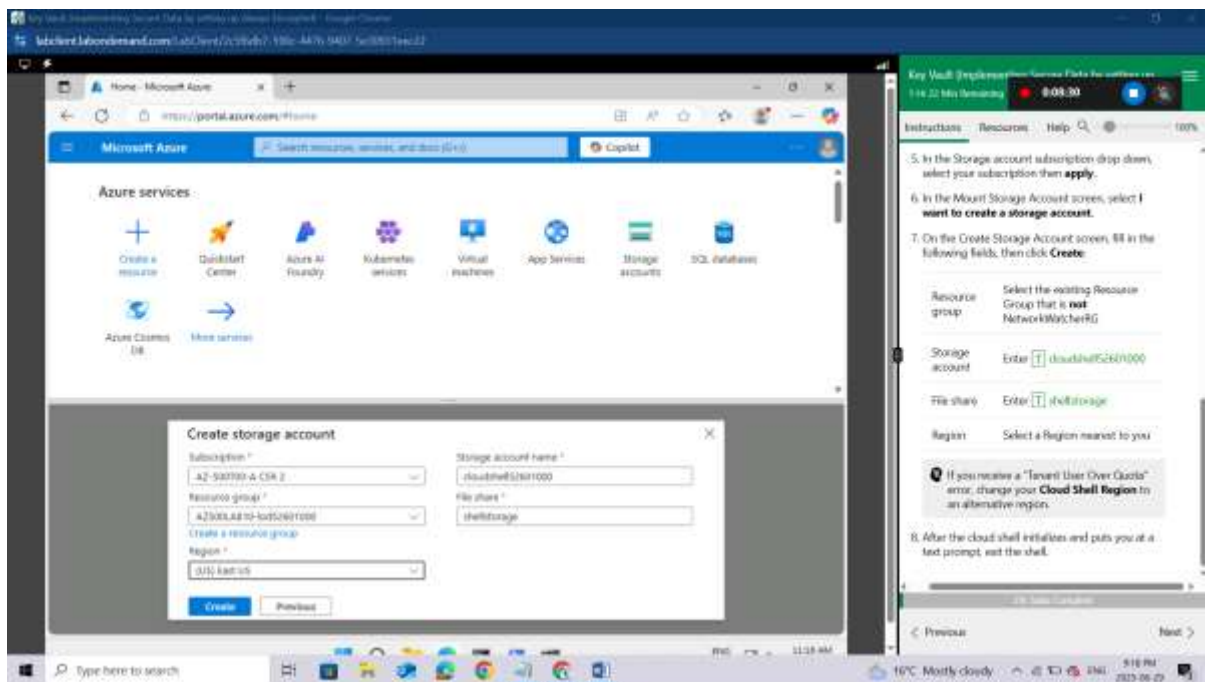
In the Mount Storage Account screen, select **I want to create a storage account**.



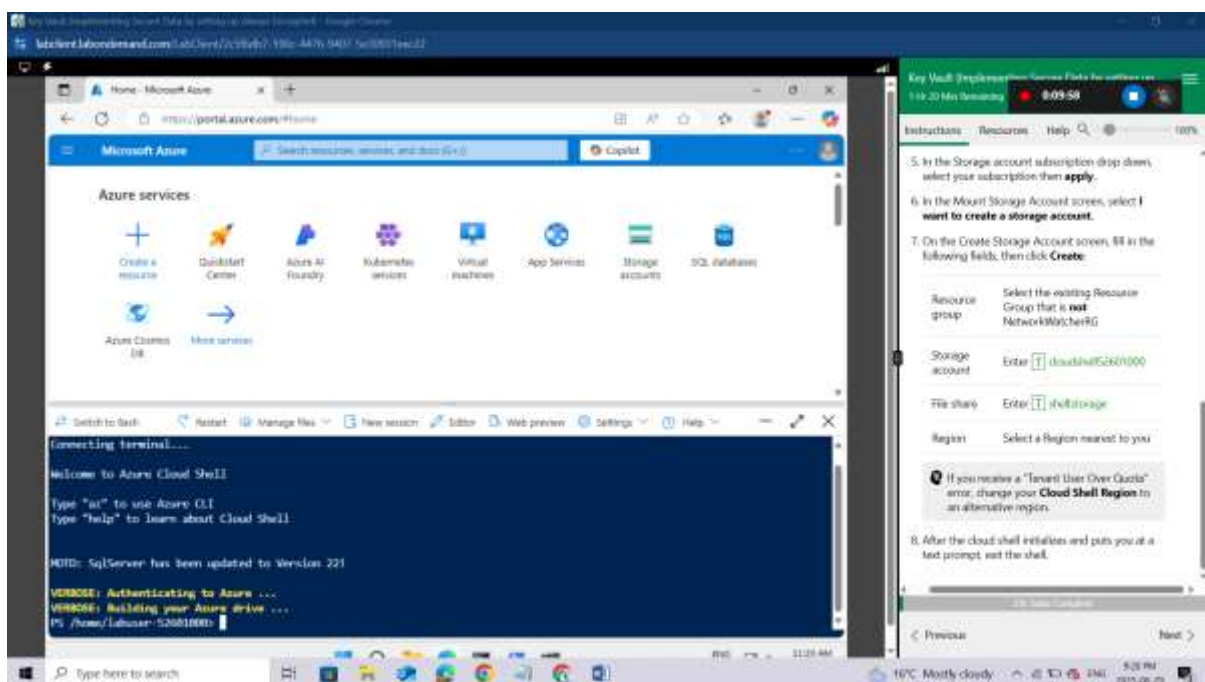
On the Create Storage Account screen, fill in the following fields, then click **Create**:

Resource group	Select the existing Resource Group that is not NetworkWatcherRG
Storage account	Enter cloudshell52601000
File share	Enter shellstorage
Region	Select a Region nearest to you

If you receive a "Tenant User Over Quota" error, change your **Cloud Shell Region** to an alternative region.



After the cloud shell initializes and puts you at a text prompt, exit the shell.



Lab 07: Key Vault (Implementing Secure Data by setting up Always Encrypted)

Student lab manual

Lab scenario

You have been asked to create a proof of concept application that makes use of the Azure SQL Database support for Always Encrypted functionality. All of the secrets and keys used in this scenario should be stored in Key Vault. The application should be registered in Microsoft Entra ID in order to enhance its security posture. To accomplish these objectives, the proof of concept should include:

- Creating an Azure Key Vault and storing keys and secrets in the vault.

- Create a SQL Database and encrypting content of columns in database tables by using Always Encrypted.

For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.

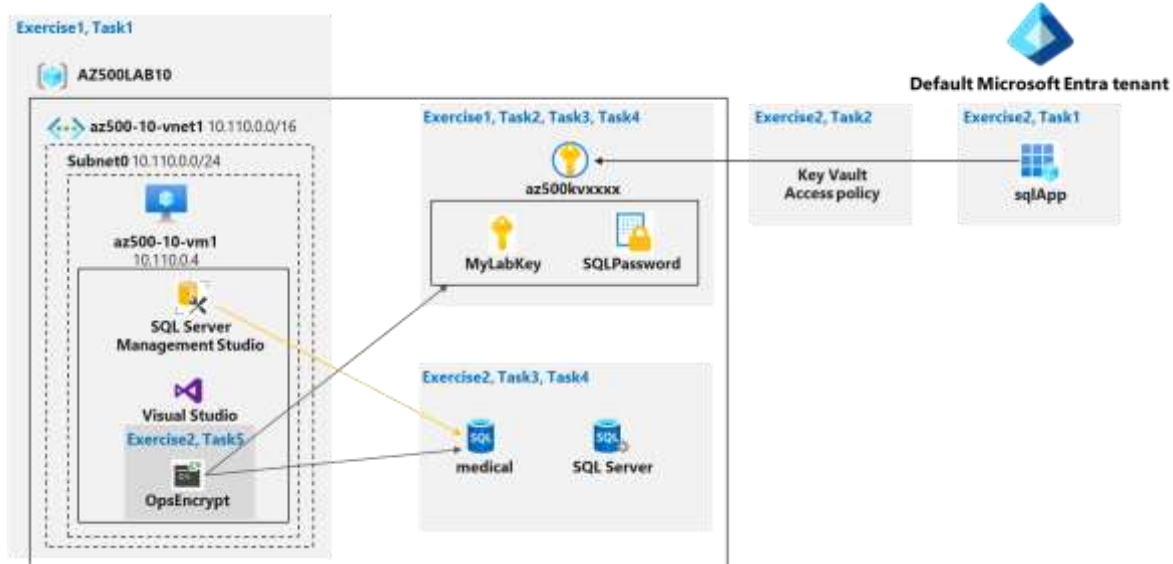
To keep the focus on the security aspects of Azure, related to building this proof of concept, you will start from an automated ARM template deployment, setting up a Virtual Machine with Visual Studio 2019 and SQL Server Management Studio 19.

Lab objectives

In this lab, you will complete the following exercises:

- Exercise 1: Deploy the base infrastructure from an ARM template
- Exercise 2: Configure the Key Vault resource with a key and a secret
- Exercise 3: Configure an Azure SQL database and a data-driven application
- Exercise 4: Demonstrate the use of Azure Key Vault in encrypting the Azure SQL database

Key Vault diagram



Instructions

Lab files:

- \\Allfiles\Labs\10\az-500-10_azuredeploy.json
- \\Allfiles\Labs\10\program.cs

Total Lab Time estimate: 60 minutes

Exercise 1: Deploy the base infrastructure from an ARM template

In this exercise, you will complete the following tasks:

- Task 1: Deploy an Azure VM and an Azure SQL database

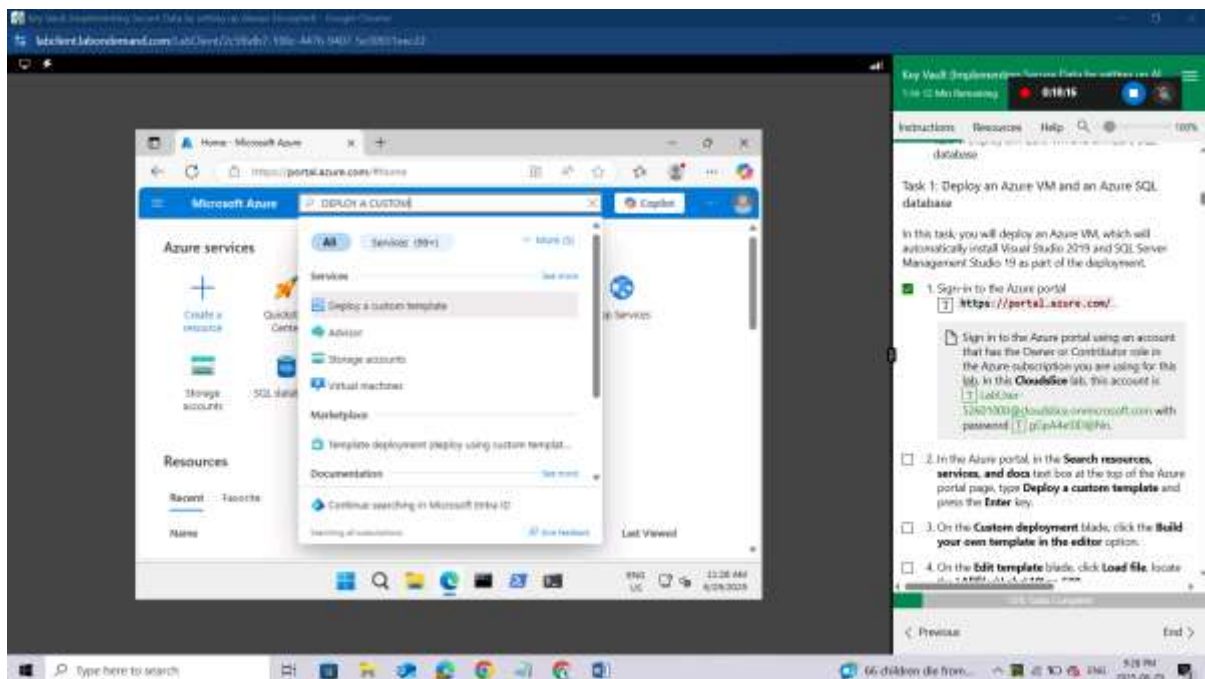
Task 1: Deploy an Azure VM and an Azure SQL database

In this task, you will deploy an Azure VM, which will automatically install Visual Studio 2019 and SQL Server Management Studio 19 as part of the deployment.

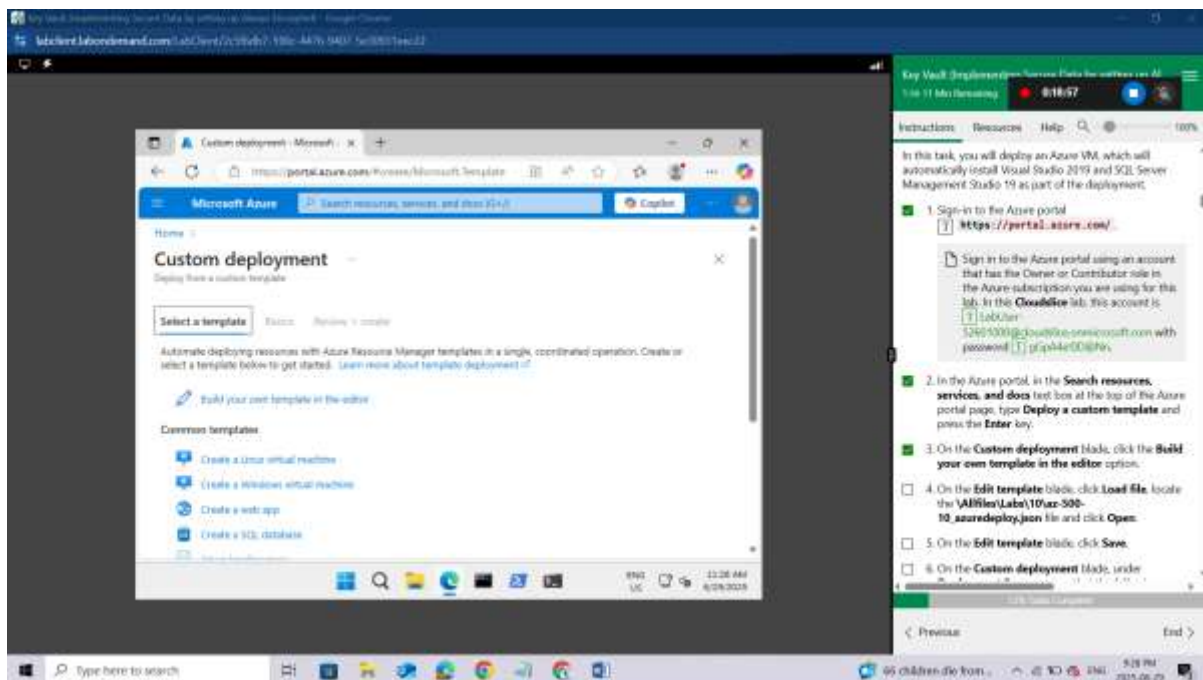
Sign-in to the Azure portal <https://portal.azure.com/>.

Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab. In this **Cloudslice** lab, this account is **LabUser-52601000@cloudslice.onmicrosoft.com** with password **gGpA4e!0D@Nn**.

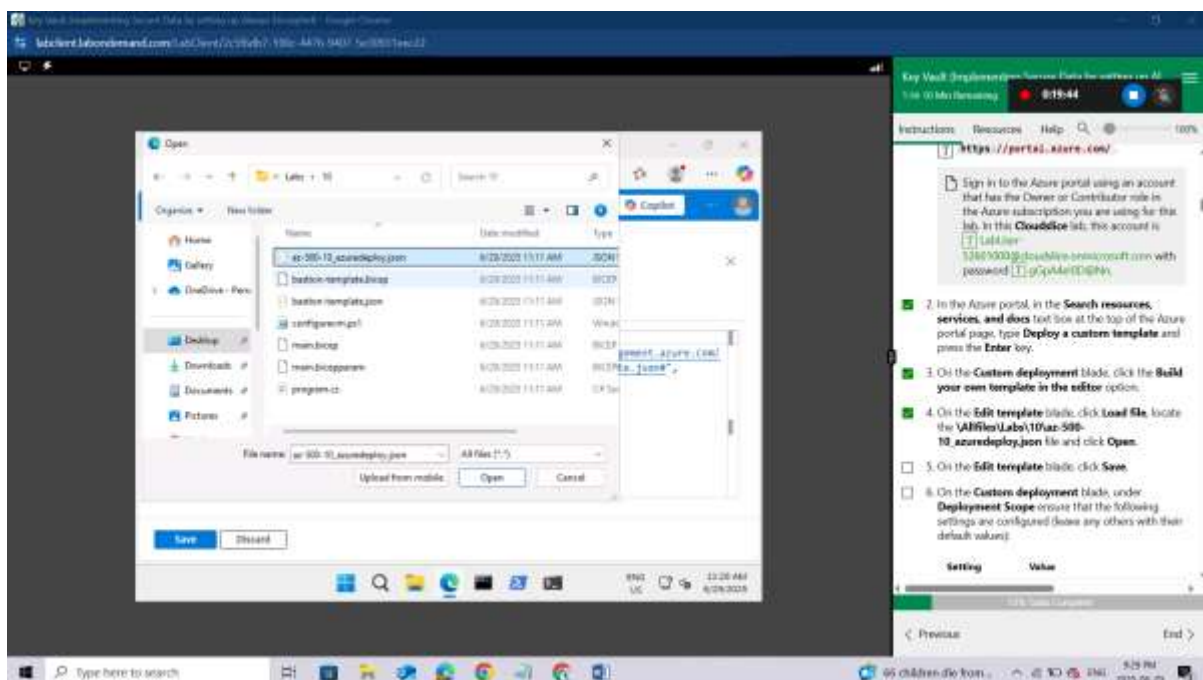
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Deploy a custom template** and press the **Enter** key.



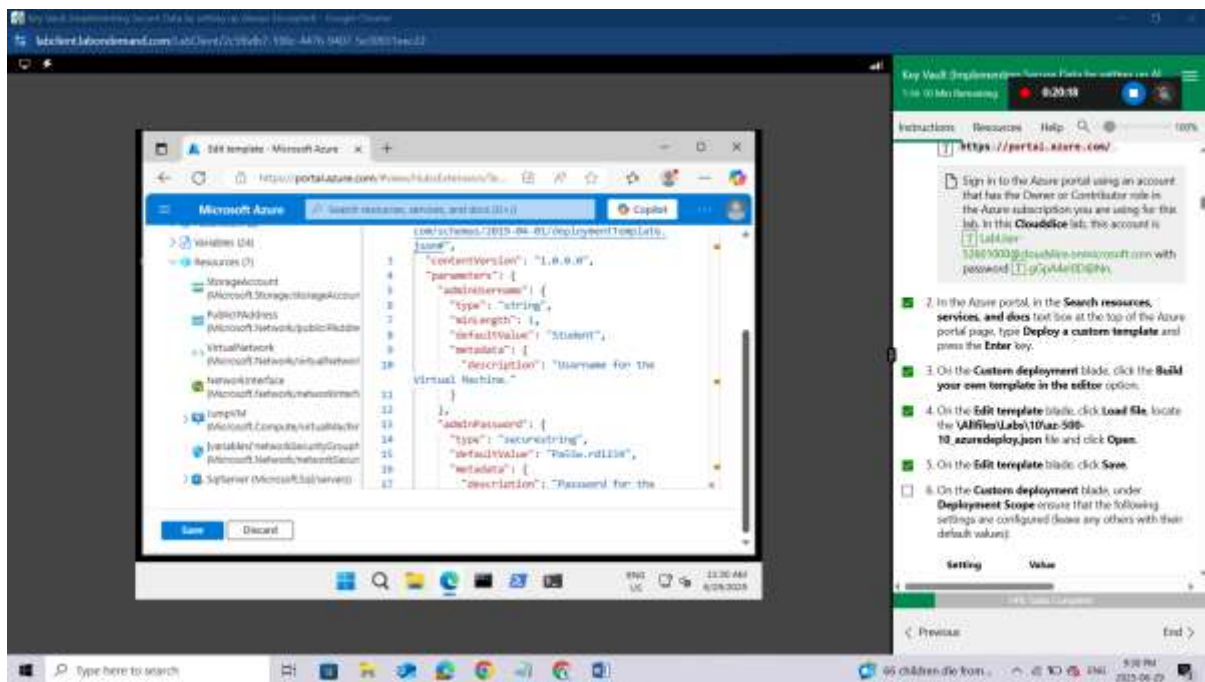
On the **Custom deployment** blade, click the **Build your own template in the editor** option.



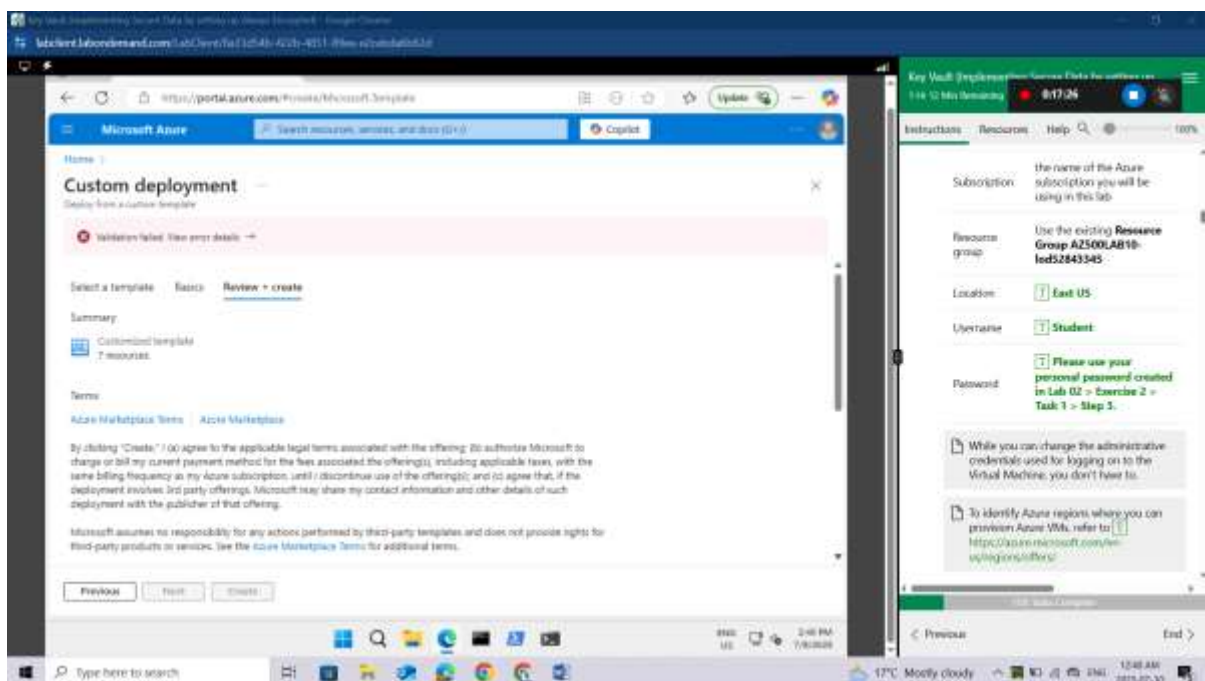
On the **Edit template** blade, click **Load file**, locate the **\\Allfiles\\Labs\\10\\az-500-10_azuredeploy.json** file and click **Open**.



On the **Edit template** blade, click **Save**.



If you get validation error, edit the json file you uploaded



To correct the error you can use this script:

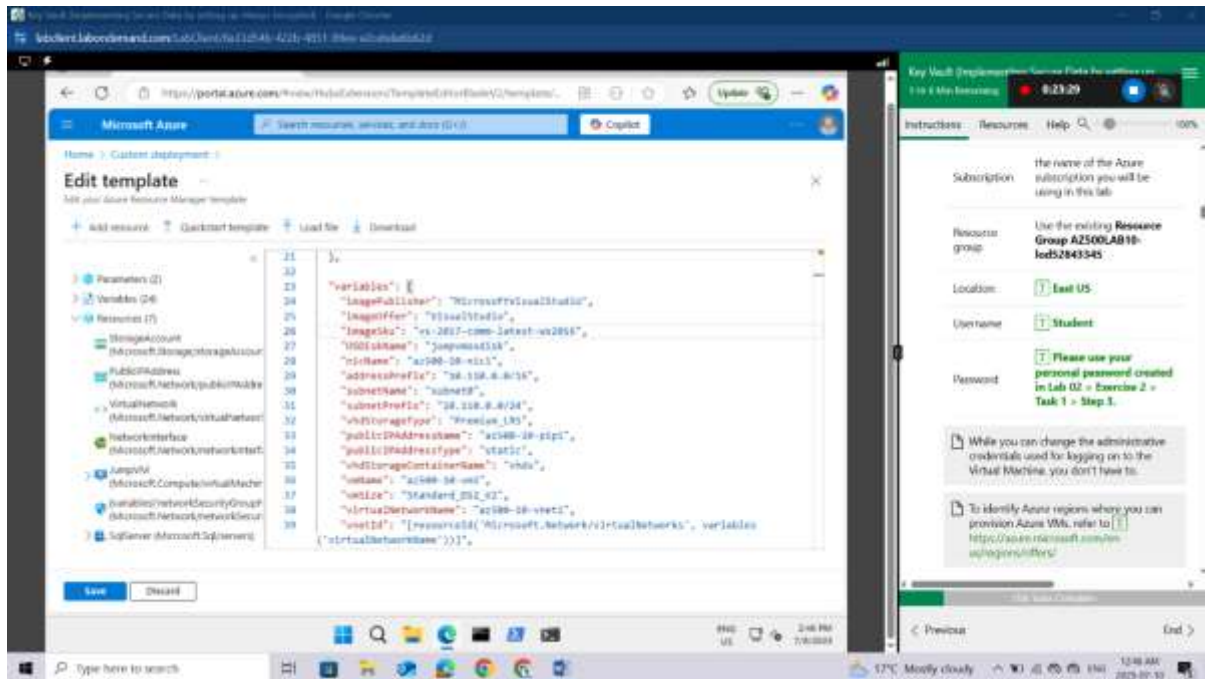
"imagePublisher": "MicrosoftVisualStudio",

"imageOffer": "VisualStudio",

"imageSku": "VS-2017-Comm-Latest-WS2016"

Correct this part using the above or the version that will work for you:

```
"variables": {
  "imagePublisher": "MicrosoftVisualStudio",
  "imageOffer": "VisualStudio2019latest",
  "imageSku": "vs-2019-comm-latest-ws2019",
```



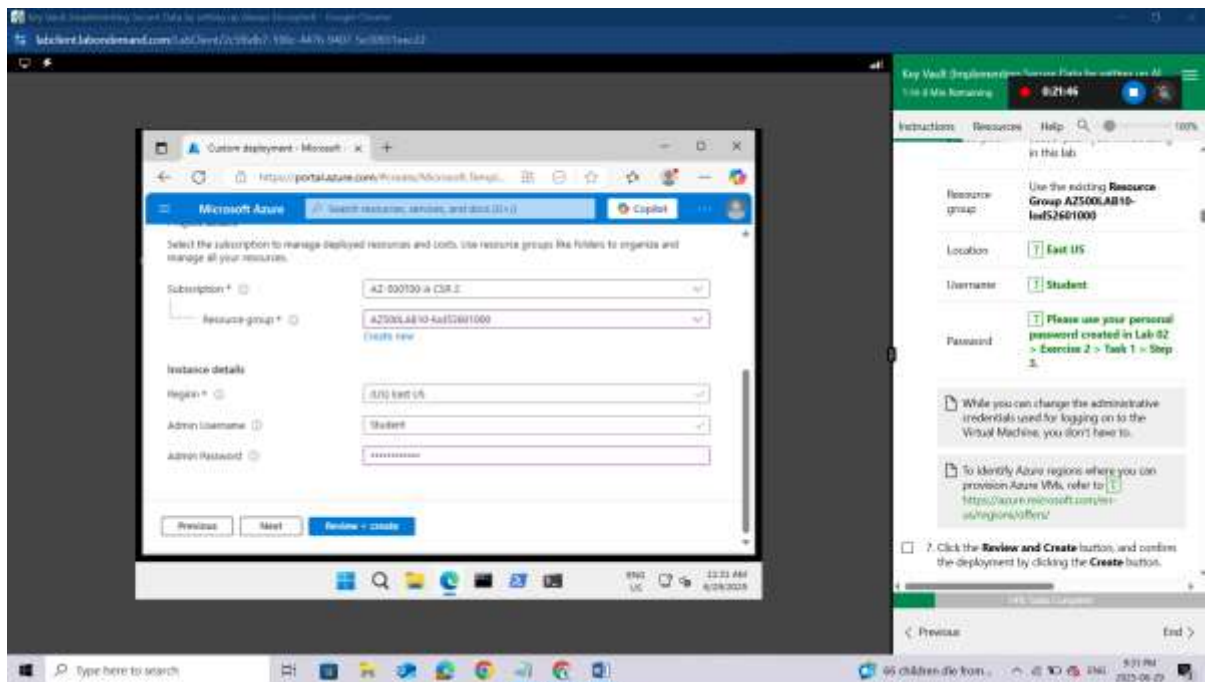
On the **Custom deployment** blade, under **Deployment Scope** ensure that the following settings are configured (leave any others with their default values):

While you can change the administrative credentials used for logging on to the Virtual Machine, you don't have to.

To identify Azure regions where you can provision Azure VMs, refer to <https://azure.microsoft.com/en-us/regions/offers/>

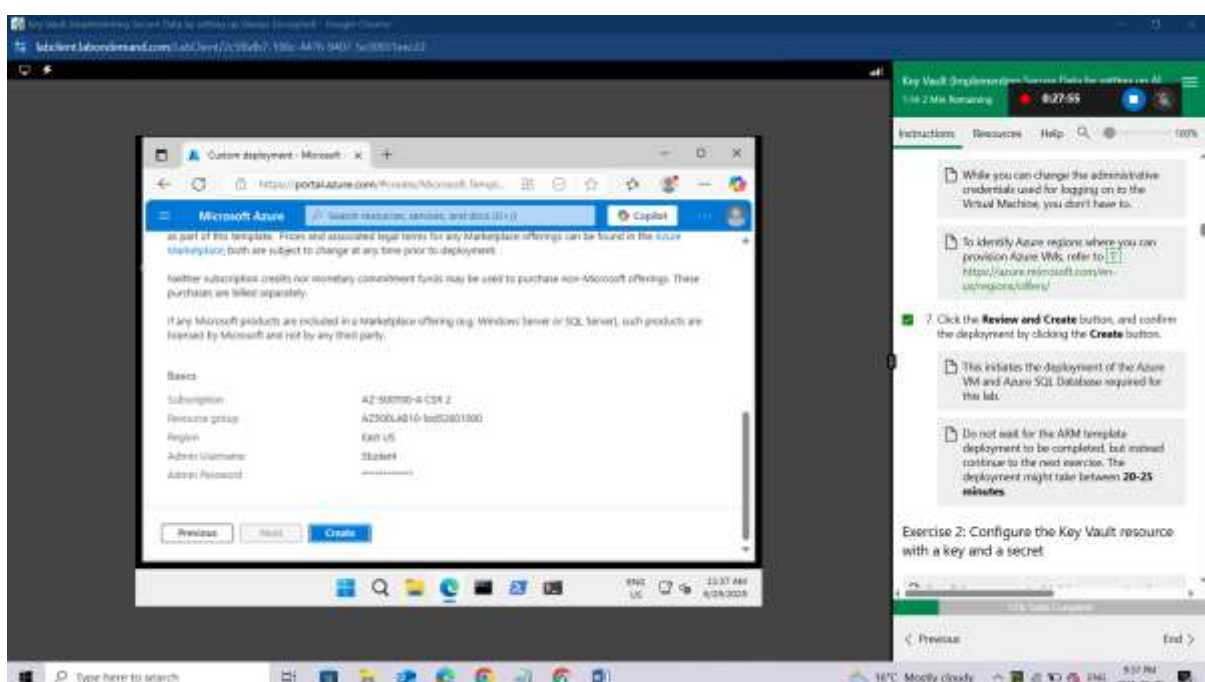
Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	Use the existing Resource Group AZ500LAB10-lod52601000
Location	East US
Username	Student

Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.



Click the **Review and Create** button, and confirm the deployment by clicking the **Create** button.

*This initiates the deployment of the Azure VM and Azure SQL Database required for this lab. Do not wait for the ARM template deployment to be completed, but instead continue to the next exercise. The deployment might take between **20-25 minutes**.*



Exercise 2: Configure the Key Vault resource with a key and a secret

For all the resources in this lab, we are using the **East (US)** region. Verify with your instructor this is region to use for you class.

In this exercise, you will complete the following tasks:

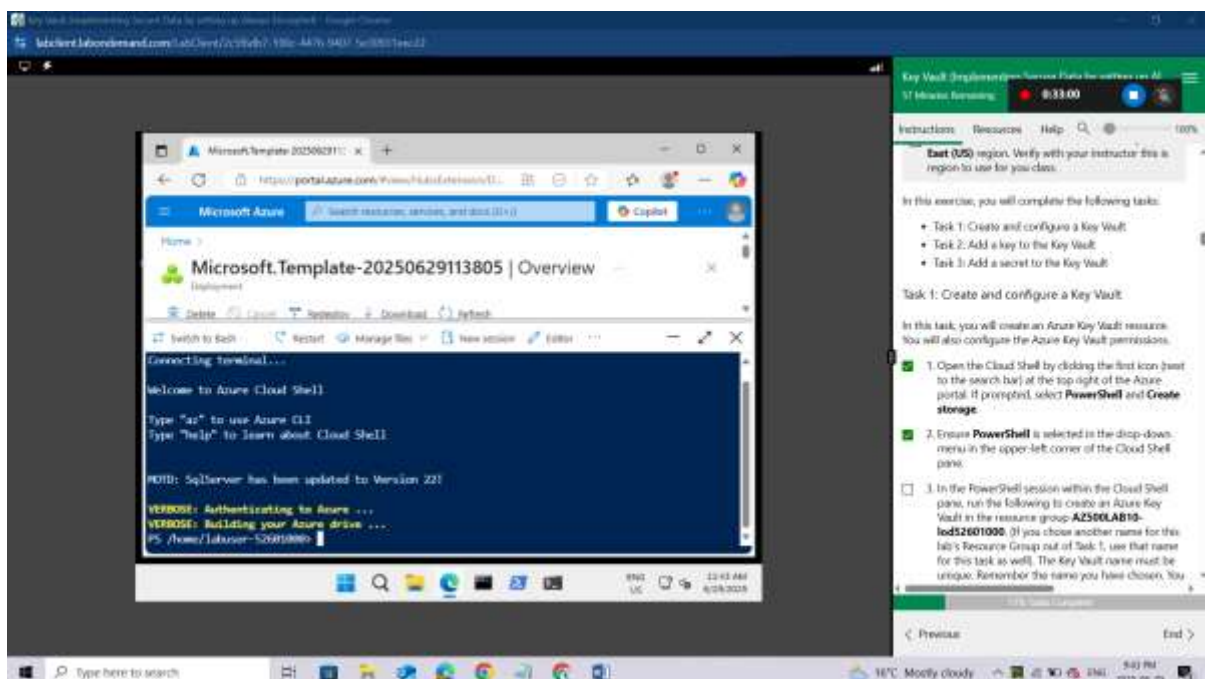
- Task 1: Create and configure a Key Vault
- Task 2: Add a key to the Key Vault
- Task 3: Add a secret to the Key Vault

Task 1: Create and configure a Key Vault

In this task, you will create an Azure Key Vault resource. You will also configure the Azure Key Vault permissions.

Open the Cloud Shell by clicking the first icon (next to the search bar) at the top right of the Azure portal. If prompted, select **PowerShell** and **Create storage**.

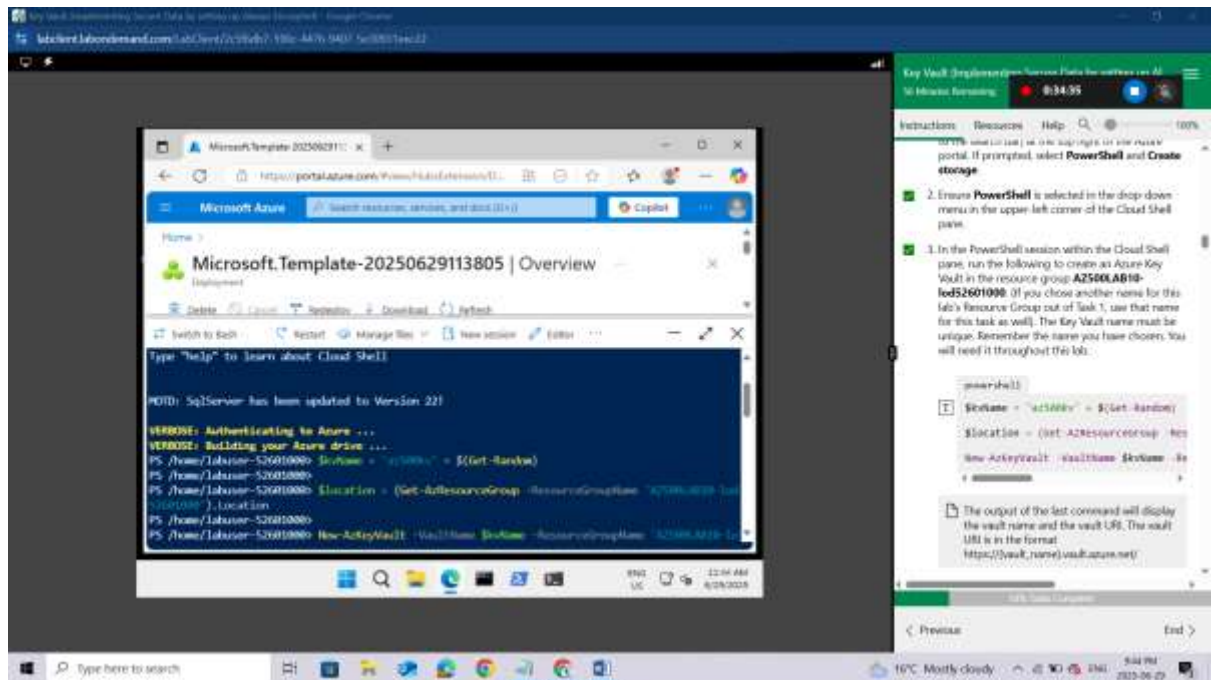
Ensure **PowerShell** is selected in the drop-down menu in the upper-left corner of the Cloud Shell pane.



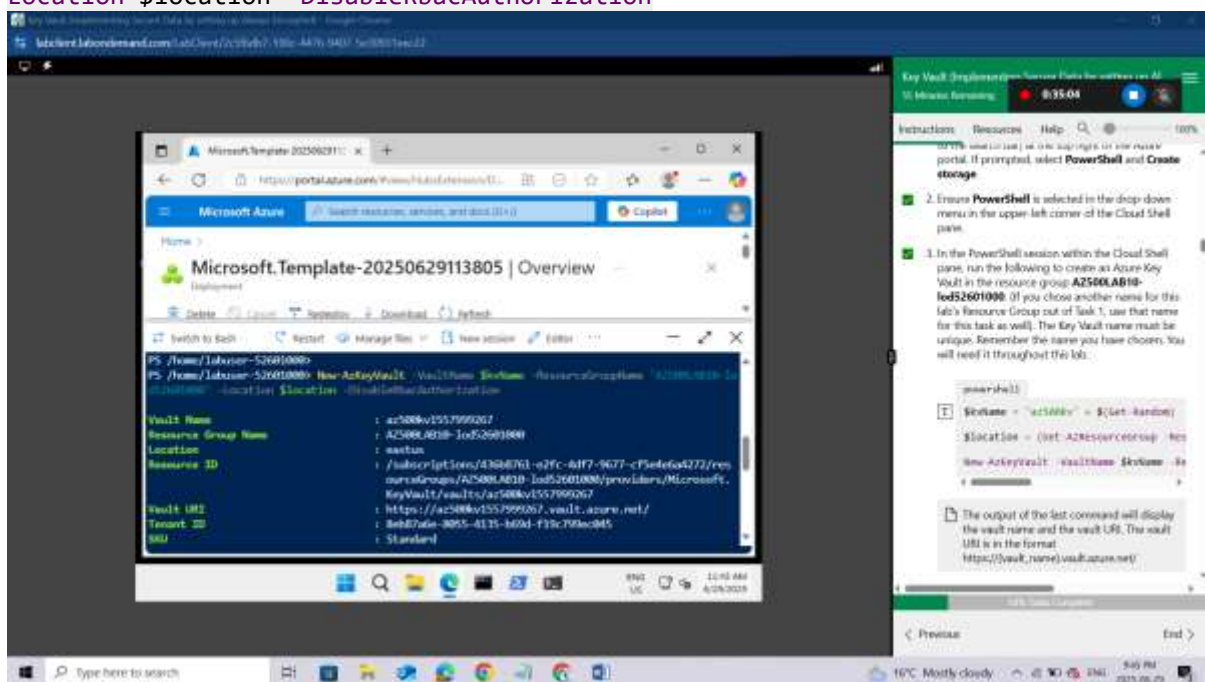
In the PowerShell session within the Cloud Shell pane, run the following to create an Azure Key Vault in the resource group **AZ500LAB10-lod52601000**. (If you chose another name for this lab's Resource Group out of Task 1, use that name for this task as well). The Key Vault name must be unique. Remember the name you have chosen. You will need it throughout this lab.

```
powershell
$kvName = 'az500kv' + $(Get-Random)

$location = (Get-AzResourceGroup -ResourceGroupName 'AZ500LAB10-lod52601000').Location
```

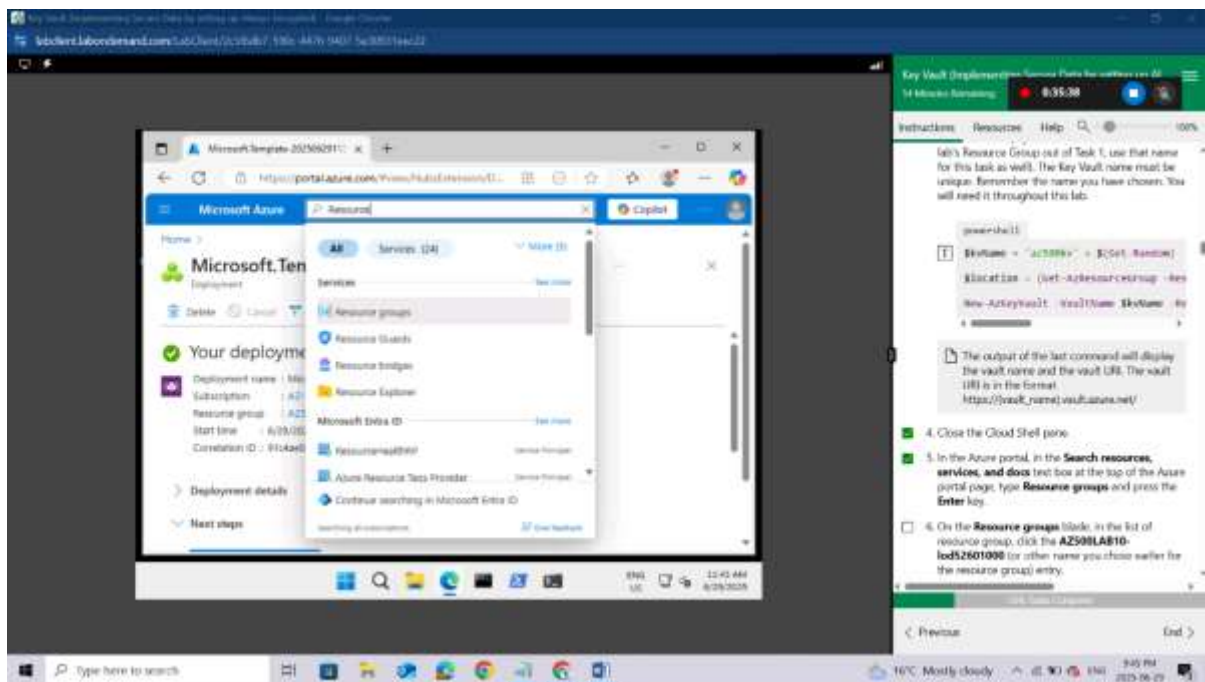
New-AzKeyVault -VaultName \$kvName -ResourceGroupName 'AZ500LAB10-lod52601000' -Location \$location -DisableRbacAuthorization



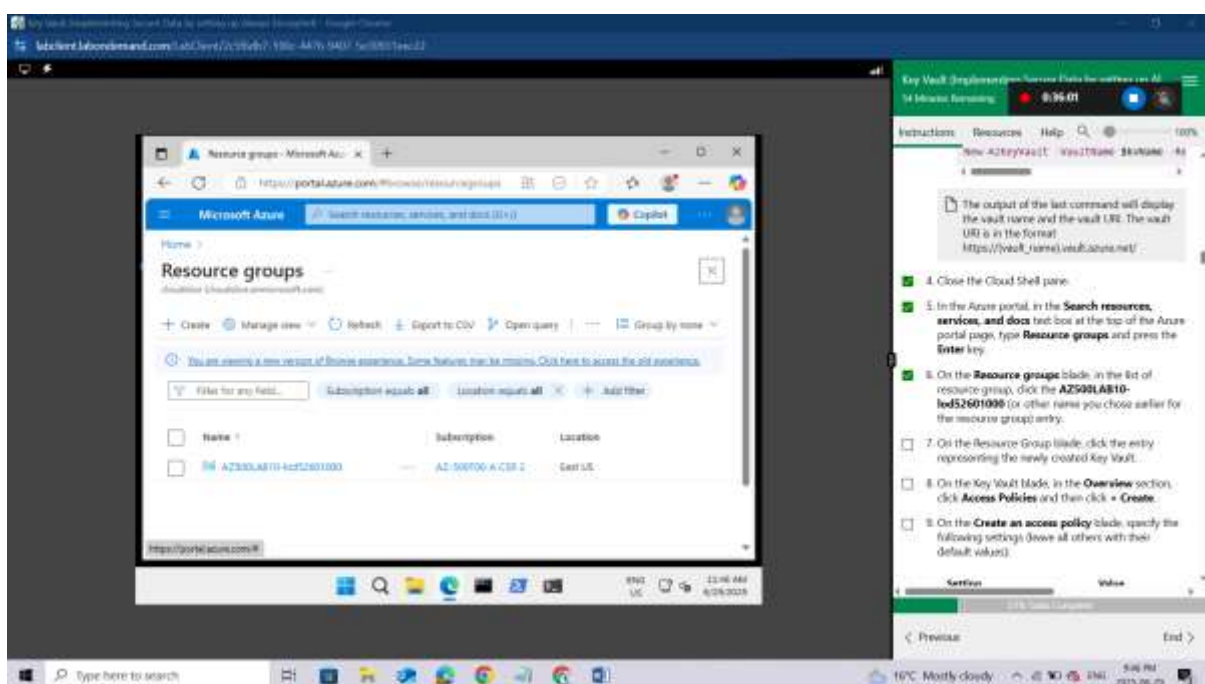
The output of the last command will display the vault name and the vault URI. The vault URI is in the format `https://(vault_name).vault.azure.net/`

Close the Cloud Shell pane.

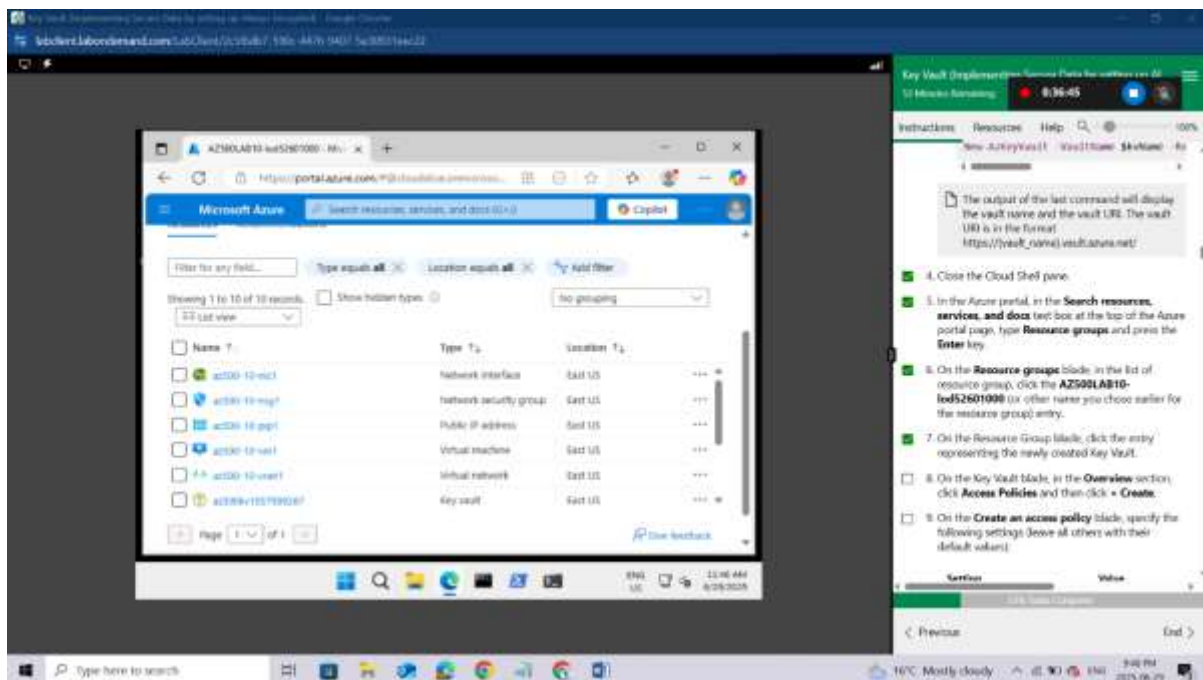
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the **Enter** key.



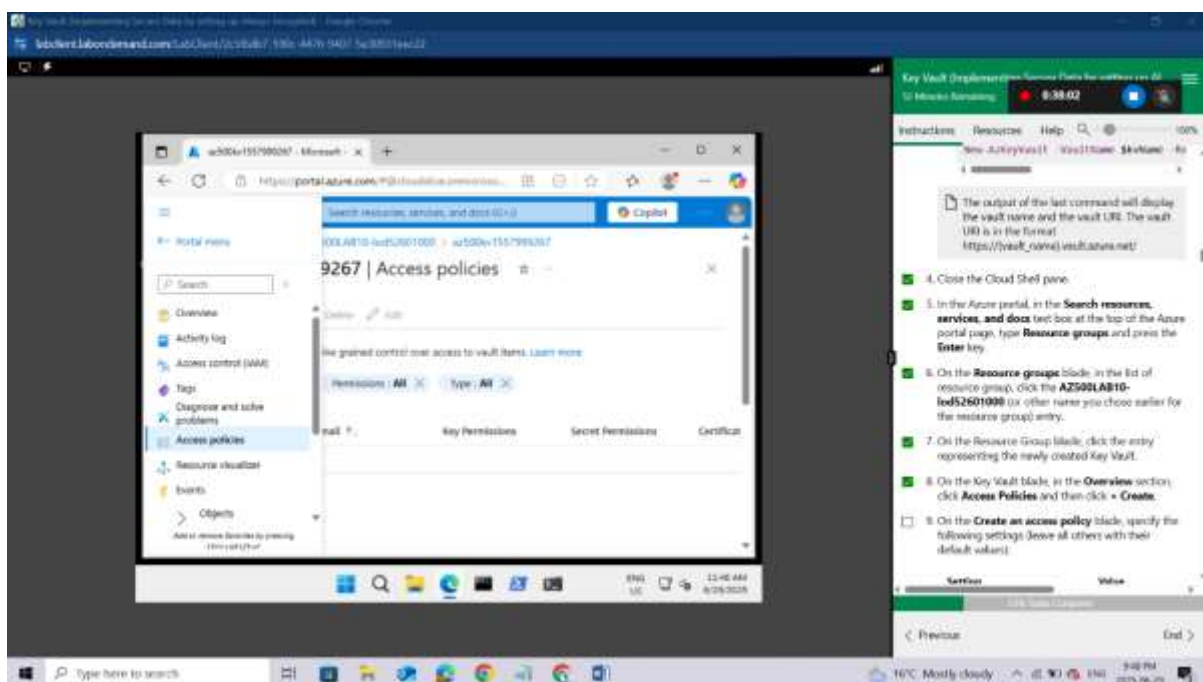
On the **Resource groups** blade, in the list of resource group, click the **AZ500LAB10-lod52601000** (or other name you chose earlier for the resource group) entry.



On the Resource Group blade, click the entry representing the newly created Key Vault.



On the Key Vault blade, in the **Overview** section, click **Access Policies** and then click + **Create**.

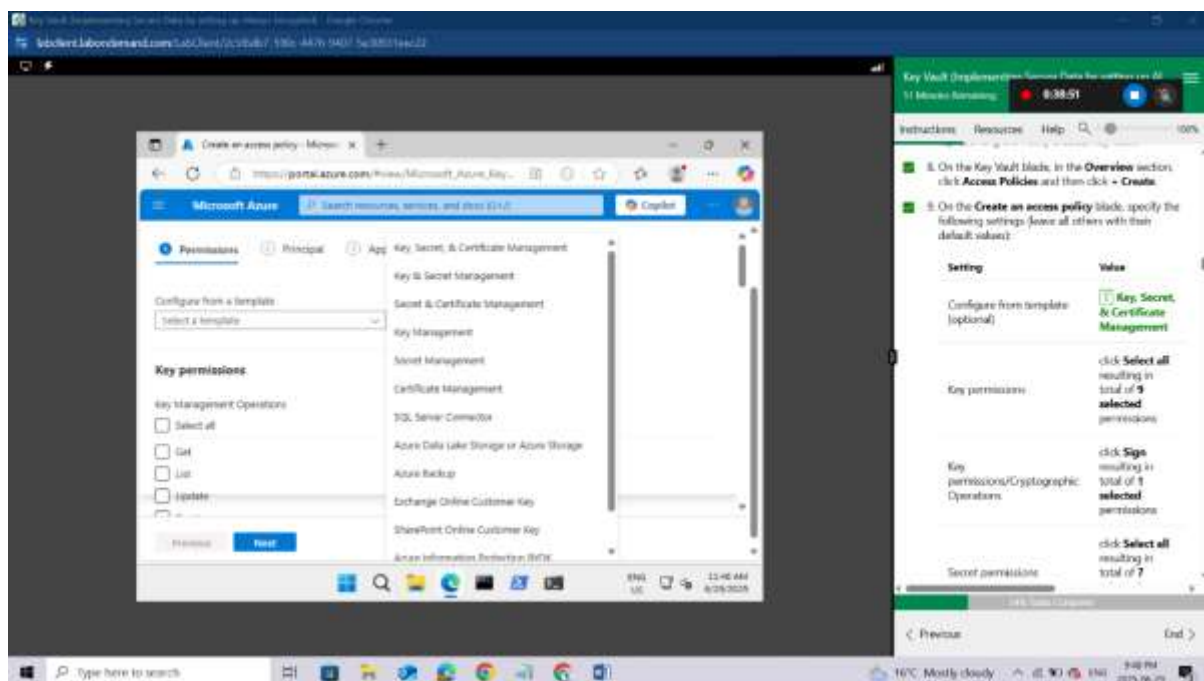


On the **Create an access policy** blade, specify the following settings (leave all others with their default values):

Setting	Value
Configure from template (optional)	Key, Secret, & Certificate Management

Setting	Value
Key permissions	click Select all resulting in total of 9 selected permissions
Key permissions/Cryptographic Operations	click Sign resulting in total of 1 selected permissions
Secret permissions	click Select all resulting in total of 7 selected permissions
Certification permissions	click Select all resulting in total of 15 selected permissions
Select principal	click None selected , on the Principal blade, select your user account, and click Next
Application (optional)	click Next
Review + create	click Create

The previous Review + create operation returns to the Access policies page that lists Application, Email, Key Permissions, Secret Permissions, and Certificate Permissions.



Key Vault (Implementation: Service Principal for authentication) - 50 Minutes Remaining: 8:40:36

Instructions: Resources: Help

1. On the Key Vault blade, in the Overview section, click **Access Policies** and then click **Create**.

2. On the **Create an access policy** blade, specify the following settings (leave all others with their default values):

Setting	Value
Configure from template (optional)	Key, Secret, & Certificate Management
Key permissions	click Select all resulting in total of 9 selected permissions
Key permissions/Cryptographic Operations	click Sign resulting in total of 1 selected permissions
Secret permissions	click Select all resulting in total of 7

< Previous End >

Microsoft Azure

Search resources, services, and docs (CTRL+K)

Copy

Key Management Operations

- ☒ Select all
- ☒ Get
- ☒ List
- ☒ Update
- ☒ Create
- ☒ Import
- ☒ Delete
- ☒ Recover
- ☒ Backup
- ☒ Restore

Previous Next

16°C Mostly cloudy 9:38 PM 2025-06-29

Key Vault (Implementation: Service Principal for authentication) - 50 Minutes Remaining: 8:40:36

Instructions: Resources: Help

1. On the Key Vault blade, in the Overview section, click **Access Policies** and then click **Create**.

2. On the **Create an access policy** blade, specify the following settings (leave all others with their default values):

Setting	Value
Configure from template (optional)	Key, Secret, & Certificate Management
Key permissions	click Select all resulting in total of 9 selected permissions
Key permissions/Cryptographic Operations	click Sign resulting in total of 1 selected permissions
Secret permissions	click Select all resulting in total of 7

< Previous End >

Microsoft Azure

Search resources, services, and docs (CTRL+K)

Copy

Cryptographic Operations

- ☐ Select all
- ☐ Decrypt
- ☐ Encrypt
- ☐ Wrap Key
- ☐ Unwrap Key
- ☐ Sign

Privileged Key Operations

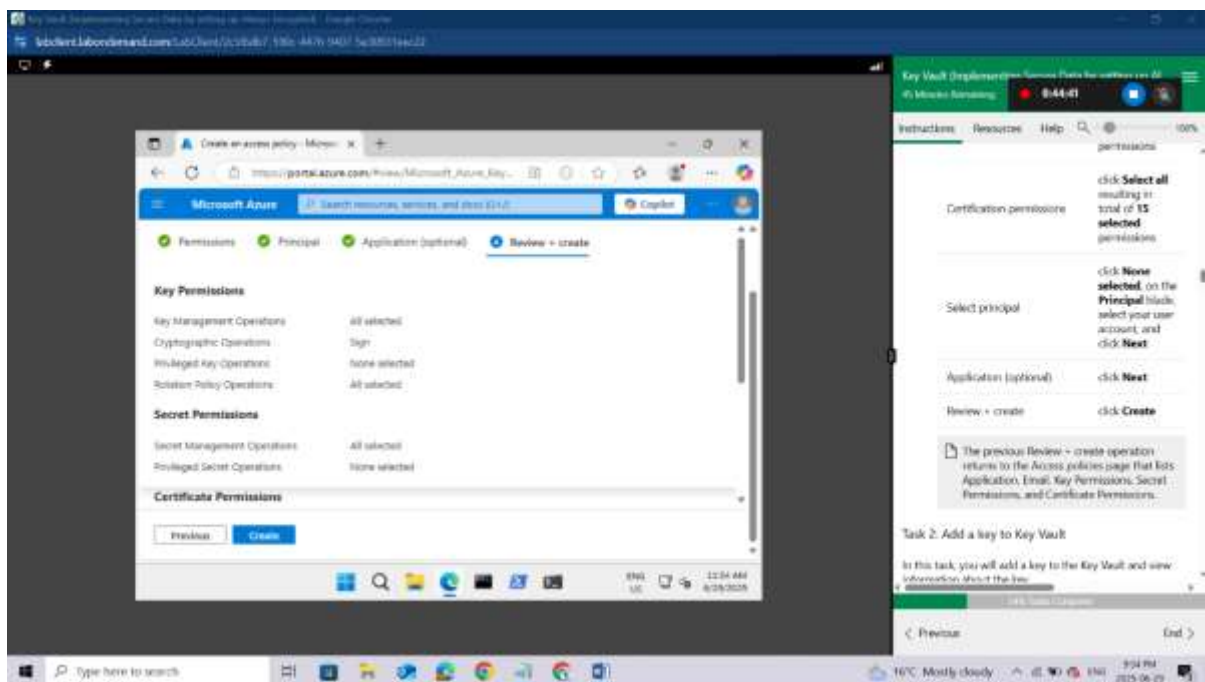
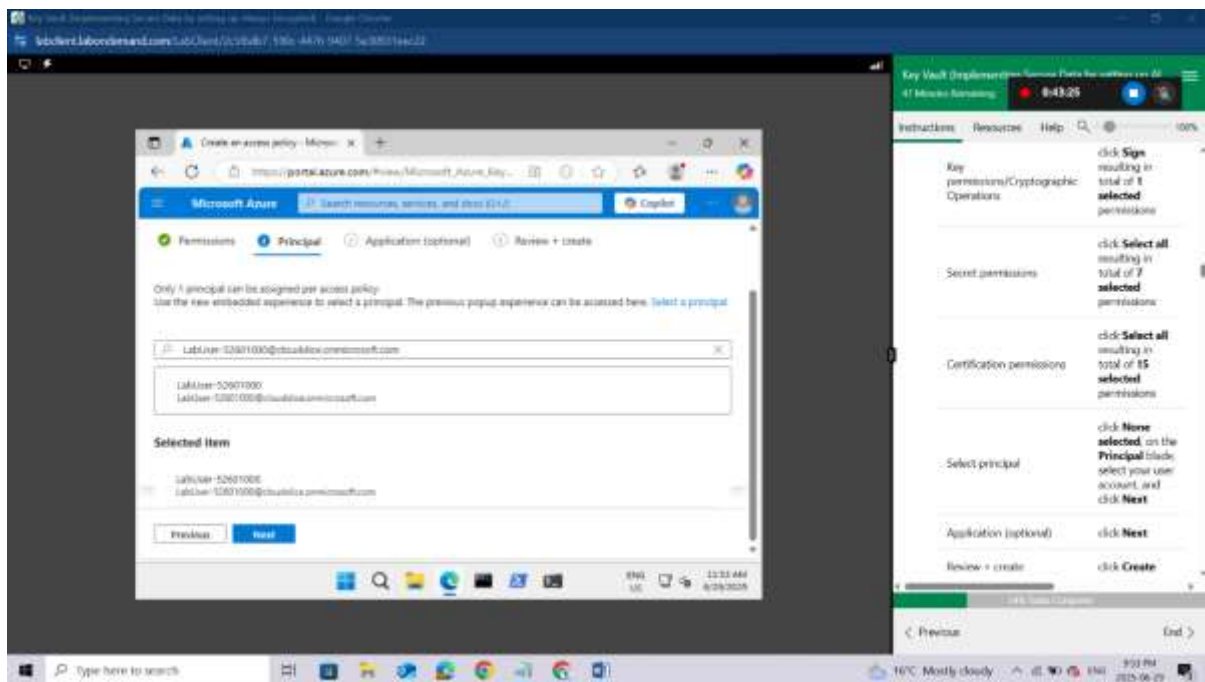
- ☐ Select all
- ☐ Purge
- ☐ Release

Network I: Internet access

Previous Next

16°C Mostly cloudy 9:38 PM 2025-06-29

The screenshot displays the Microsoft Azure portal interface. The main window shows the 'Create an Azure policy' wizard. The 'Certificate Management Operations' section is expanded, revealing a list of permissions with checkboxes. The 'Next' button is highlighted. To the right, a sidebar shows the 'Key Vault' configuration page with instructions and a 'Next' button.



Task 2: Add a key to Key Vault

In this task, you will add a key to the Key Vault and view information about the key.

In the Azure portal, open a PowerShell session in the Cloud Shell pane.

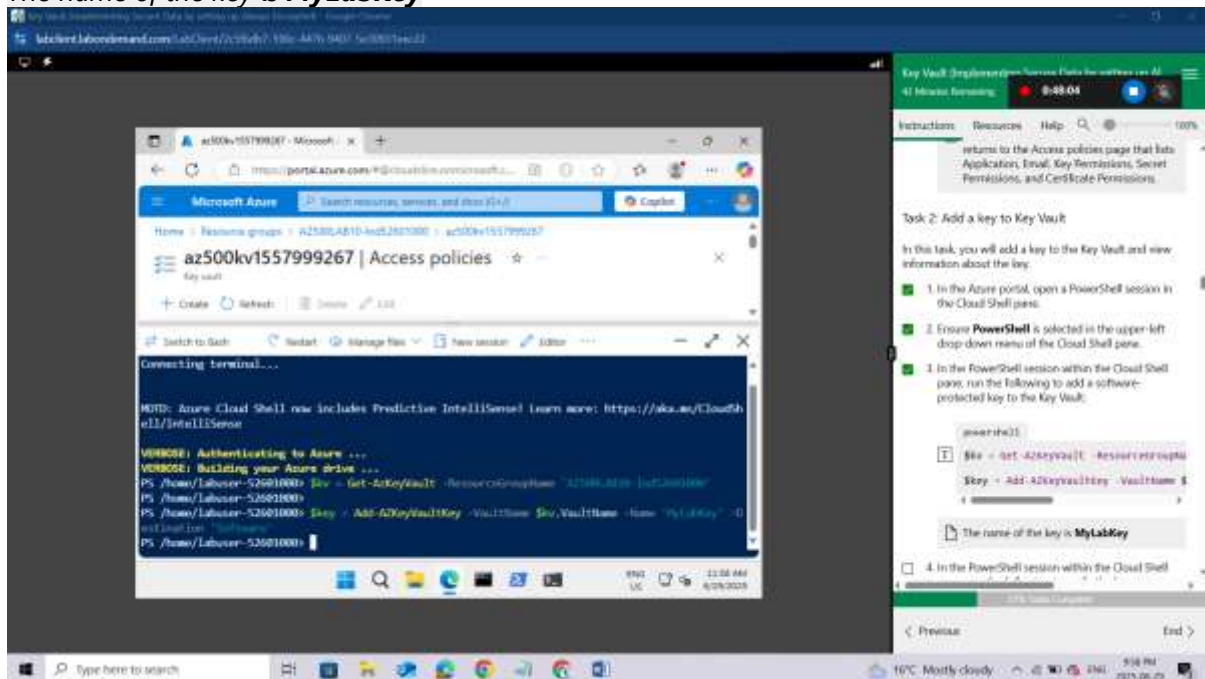
Ensure **PowerShell** is selected in the upper-left drop-down menu of the Cloud Shell pane.

In the PowerShell session within the Cloud Shell pane, run the following to add a software-protected key to the Key Vault:

```
powershell
$kv = Get-AzKeyVault -ResourceGroupName 'AZ500LAB10-lod52601000'
```

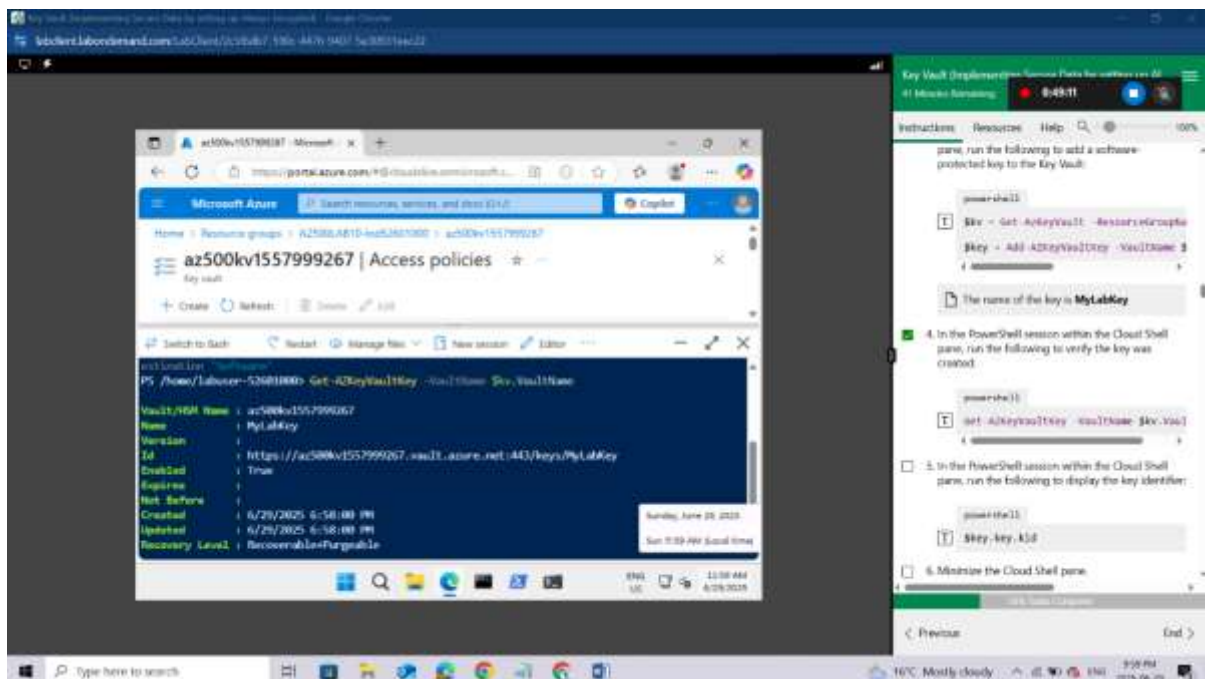
```
$key = Add-AZKeyVaultKey -VaultName $kv.VaultName -Name 'MyLabKey' -Destination 'Software'
```

The name of the key is **MyLabKey**



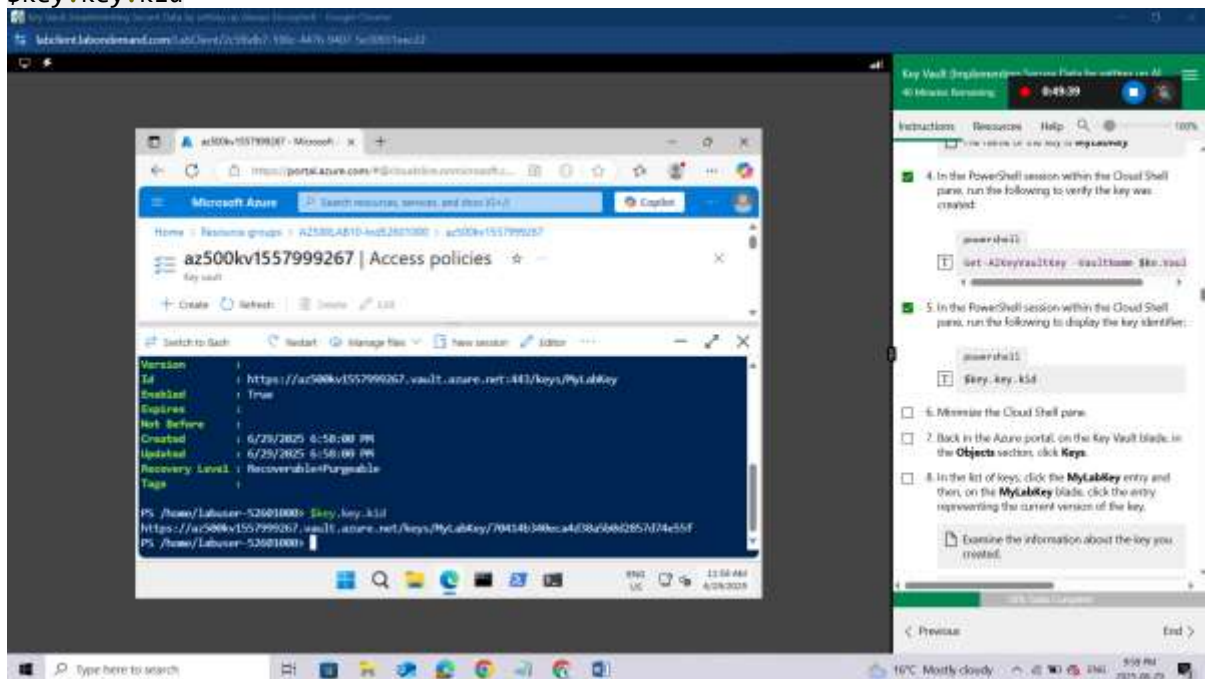
In the PowerShell session within the Cloud Shell pane, run the following to verify the key was created:

```
powershell
Get-AZKeyVaultKey -VaultName $kv.VaultName
```



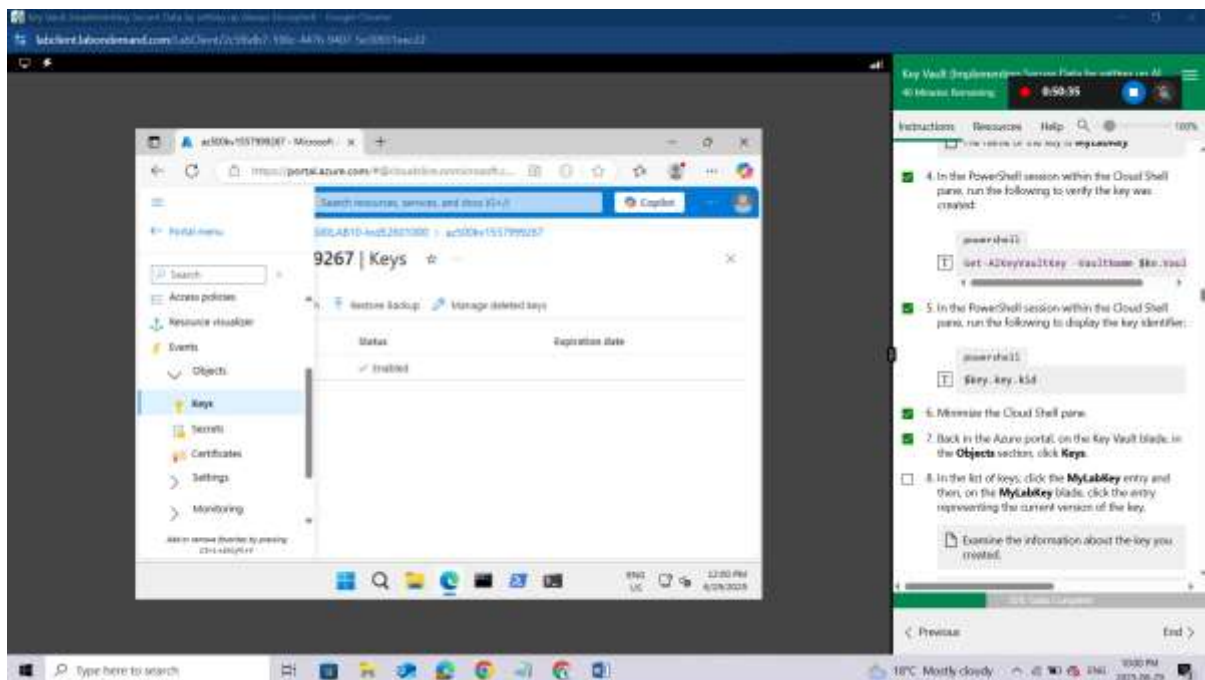
In the PowerShell session within the Cloud Shell pane, run the following to display the key identifier:

```
powershell
$key.key.kid
```

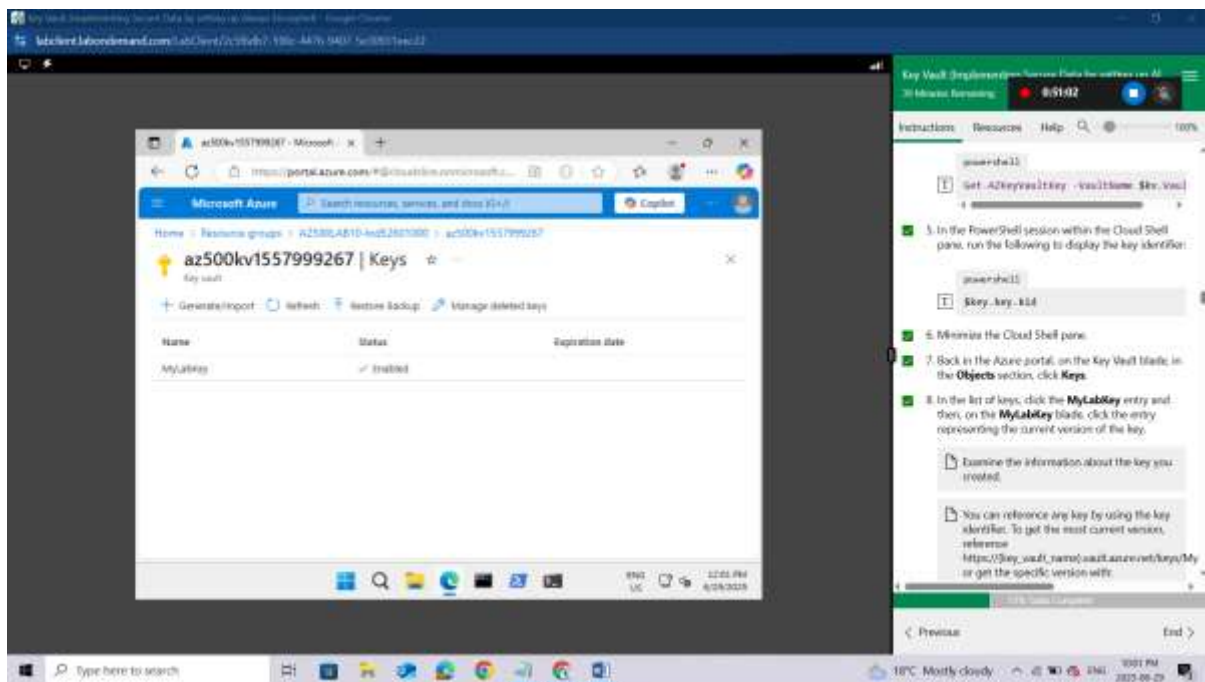


Minimize the Cloud Shell pane.

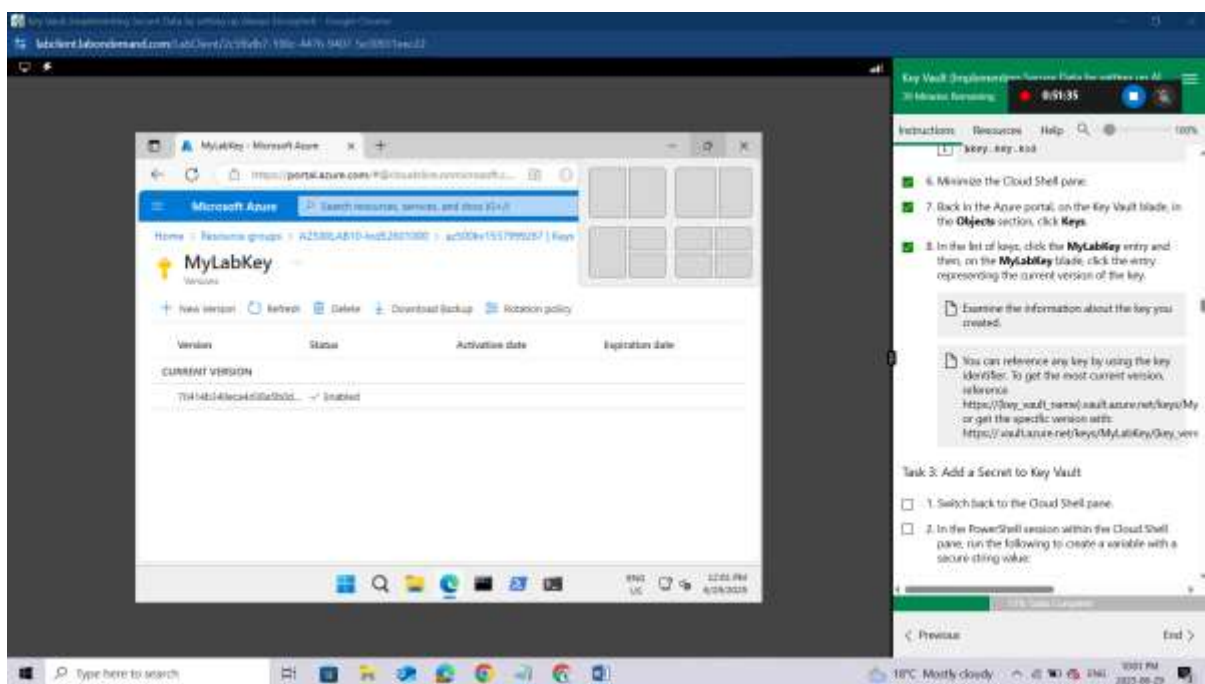
Back in the Azure portal, on the Key Vault blade, in the **Objects** section, click **Keys**.



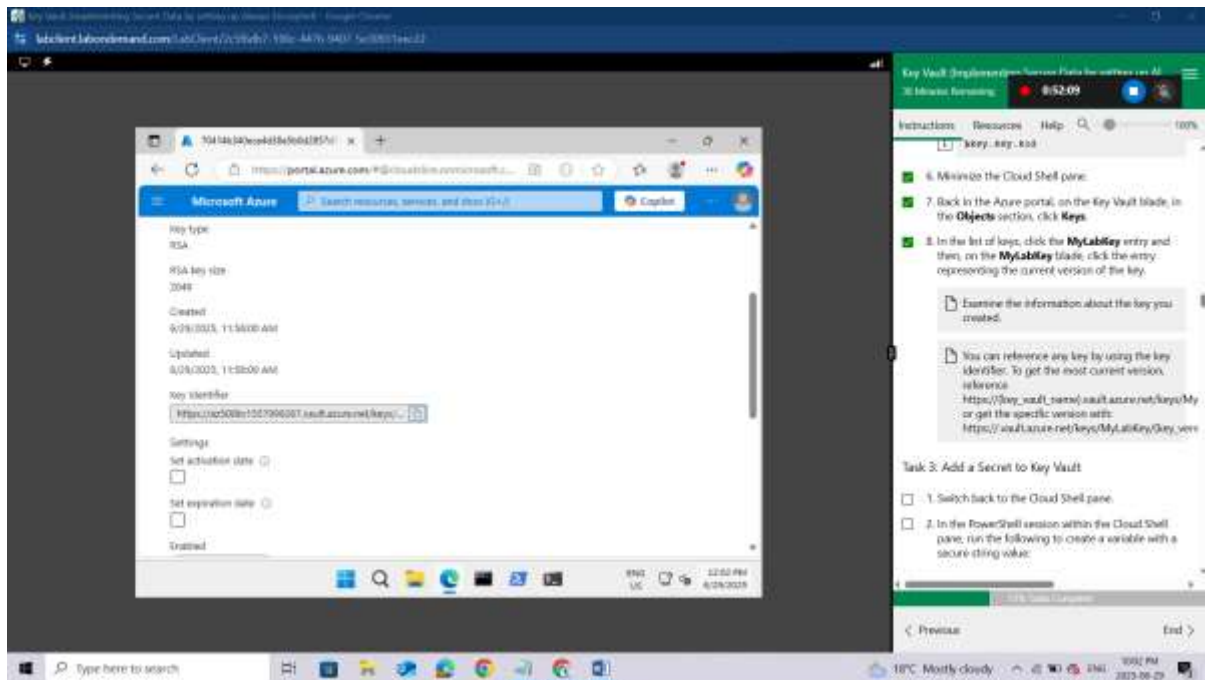
In the list of keys, click the **MyLabKey** entry and then,



on the **MyLabKey** blade, click the entry representing the current version of the key.



Examine the information about the key you created.
You can reference any key by using the key identifier. To get the most current version, reference [https://\(key_vault_name\).vault.azure.net/keys/MyLabKey](https://(key_vault_name).vault.azure.net/keys/MyLabKey) or get the specific version with: [https://.vault.azure.net/keys/MyLabKey/\(key_version\)](https://.vault.azure.net/keys/MyLabKey/(key_version))



Task 3: Add a Secret to Key Vault

Switch back to the Cloud Shell pane.

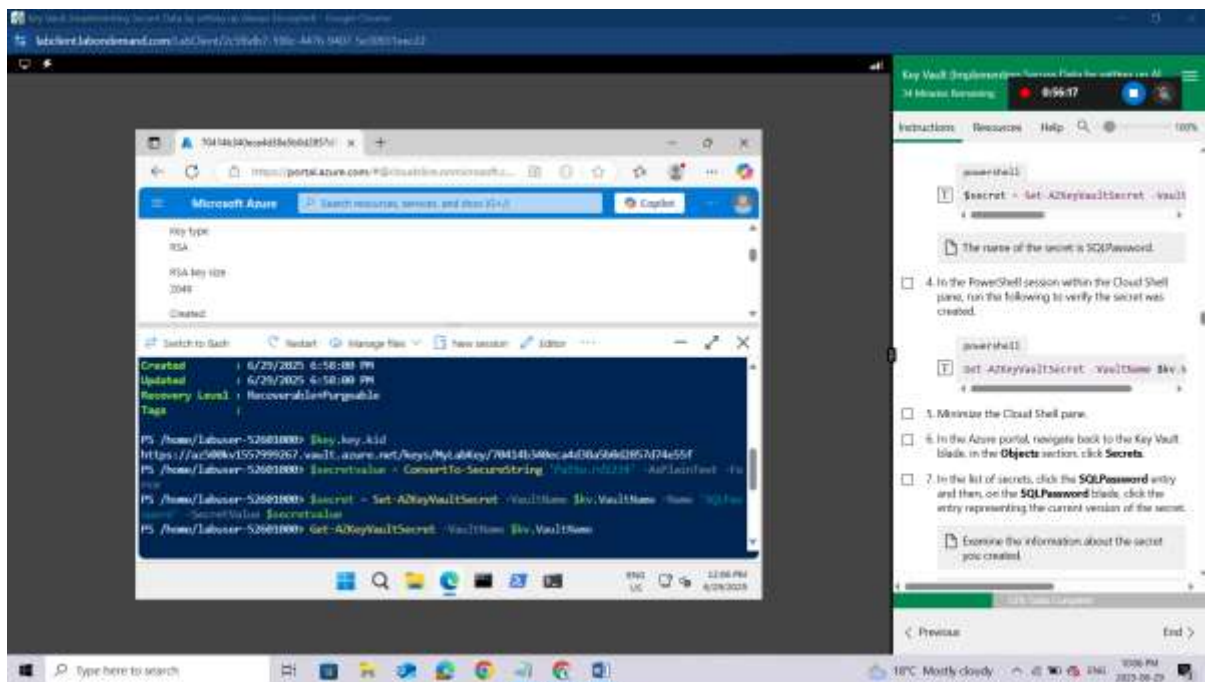
In the PowerShell session within the Cloud Shell pane, run the following to create a variable with a secure string value:

```
powershell
$secretvalue = ConvertTo-SecureString 'Pa55w.rd1234' -AsPlainText
```

In the PowerShell session within the Cloud Shell pane, run the following to add the secret to the vault:

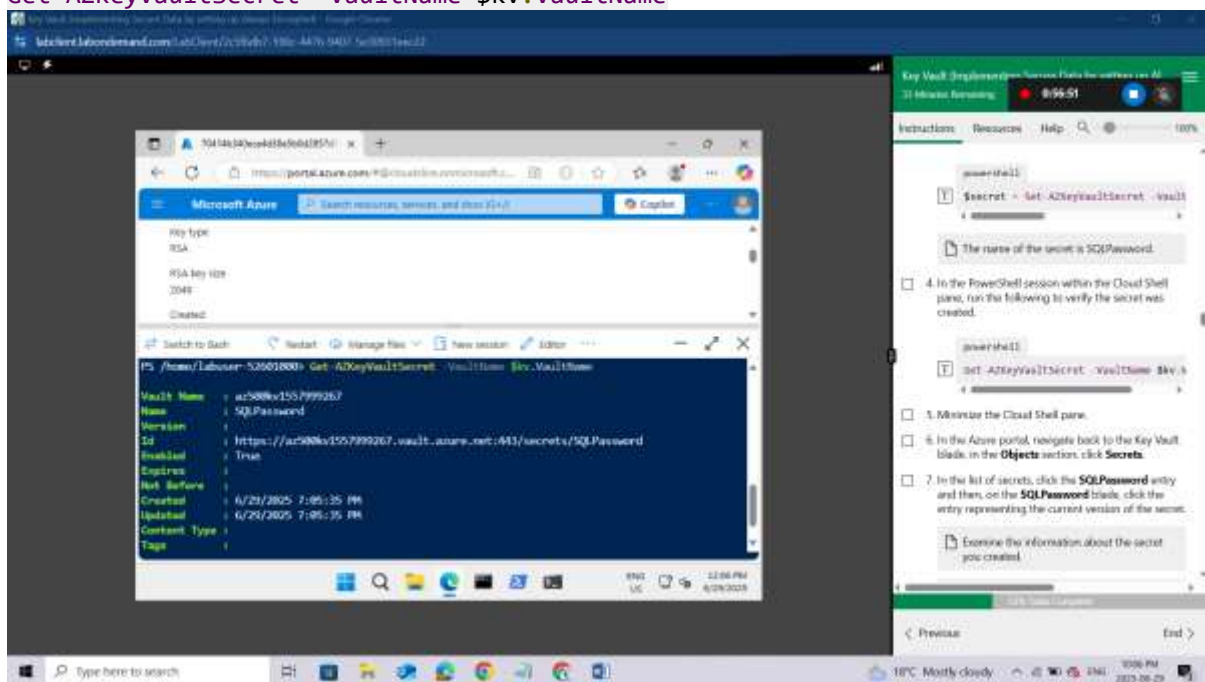
```
powershell
$secret = Set-AZKeyVaultSecret -VaultName $kv.VaultName -Name 'SQLPassword' -
SecretValue $secretvalue
```

The name of the secret is SQLPassword.



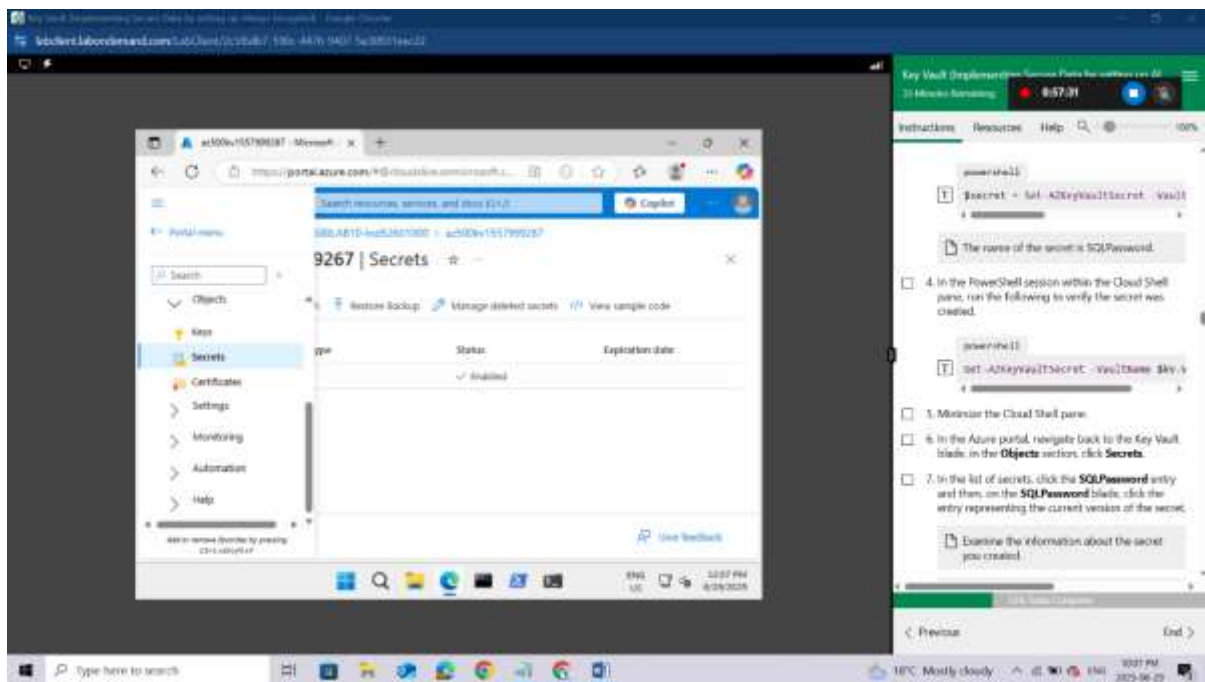
In the PowerShell session within the Cloud Shell pane, run the following to verify the secret was created.

```
powershell
Get-AZKeyVaultSecret -VaultName $kv.VaultName
```

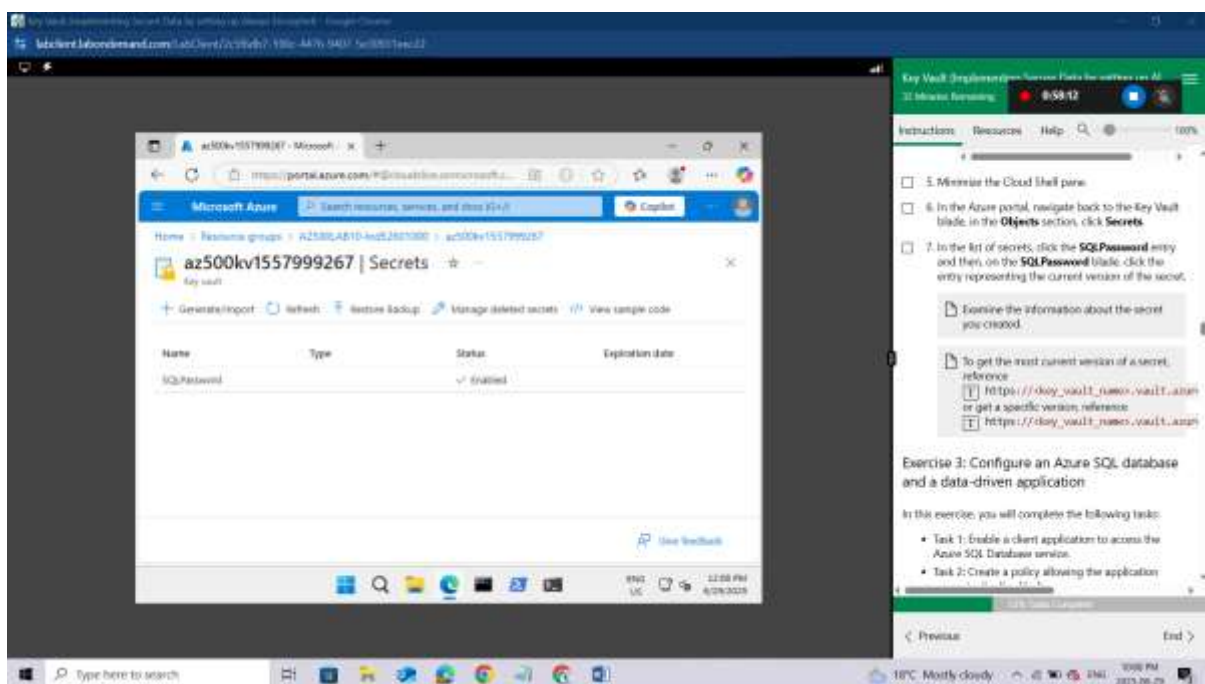


Minimize the Cloud Shell pane.

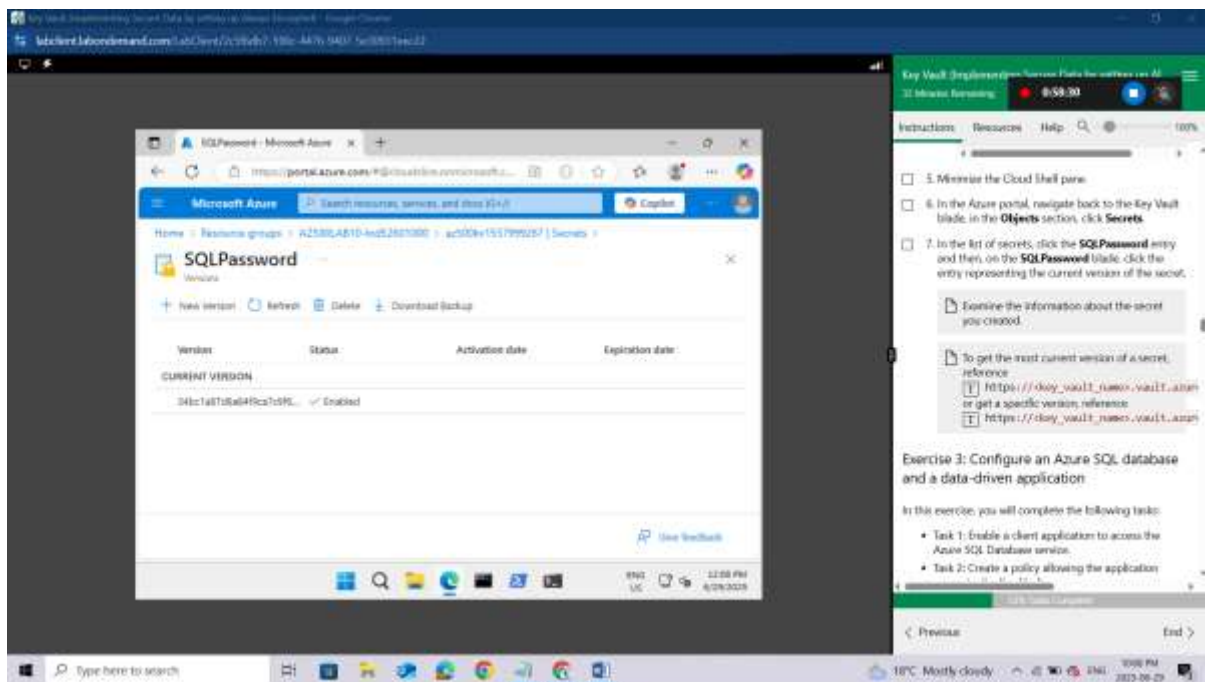
In the Azure portal, navigate back to the Key Vault blade, in the **Objects** section, click **Secrets**.



In the list of secrets, click the **SQLPassword** entry and then,



on the **SQLPassword** blade, click the entry representing the current version of the secret.

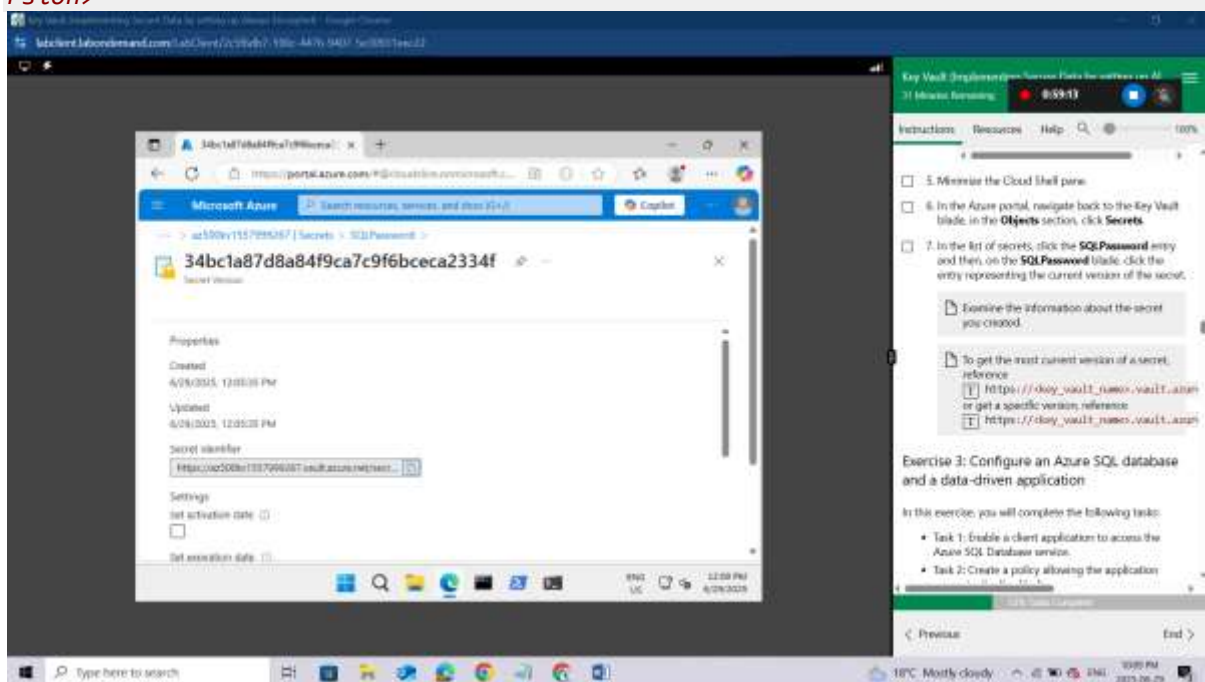


Examine the information about the secret you created.

To get the most current version of a secret,

reference https://<key_vault_name>.vault.azure.net/secrets/<secret_name> or get a specific version,

reference https://<key_vault_name>.vault.azure.net/secrets/<secret_name>/<secret_version>



Exercise 3: Configure an Azure SQL database and a data-driven application

In this exercise, you will complete the following tasks:

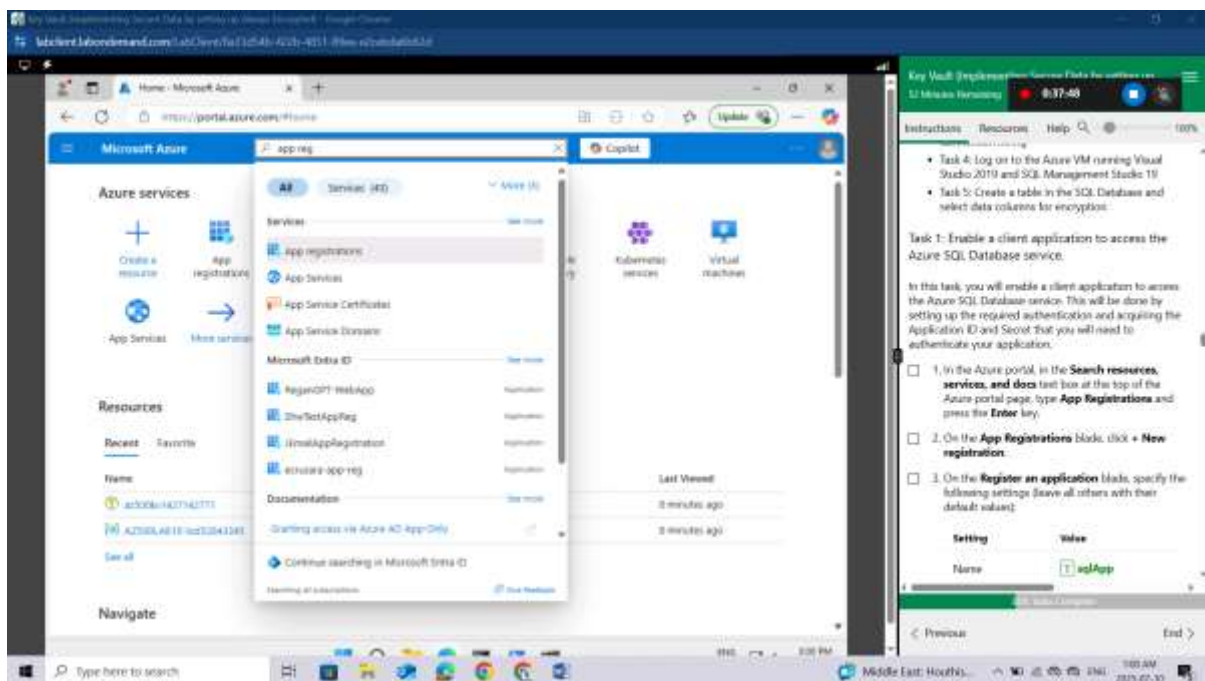
- Task 1: Enable a client application to access the Azure SQL Database service.
- Task 2: Create a policy allowing the application access to the Key Vault.

- Task 3: Retrieve SQL Azure database ADO.NET Connection String
- Task 4: Log on to the Azure VM running Visual Studio 2019 and SQL Management Studio 19
- Task 5: Create a table in the SQL Database and select data columns for encryption

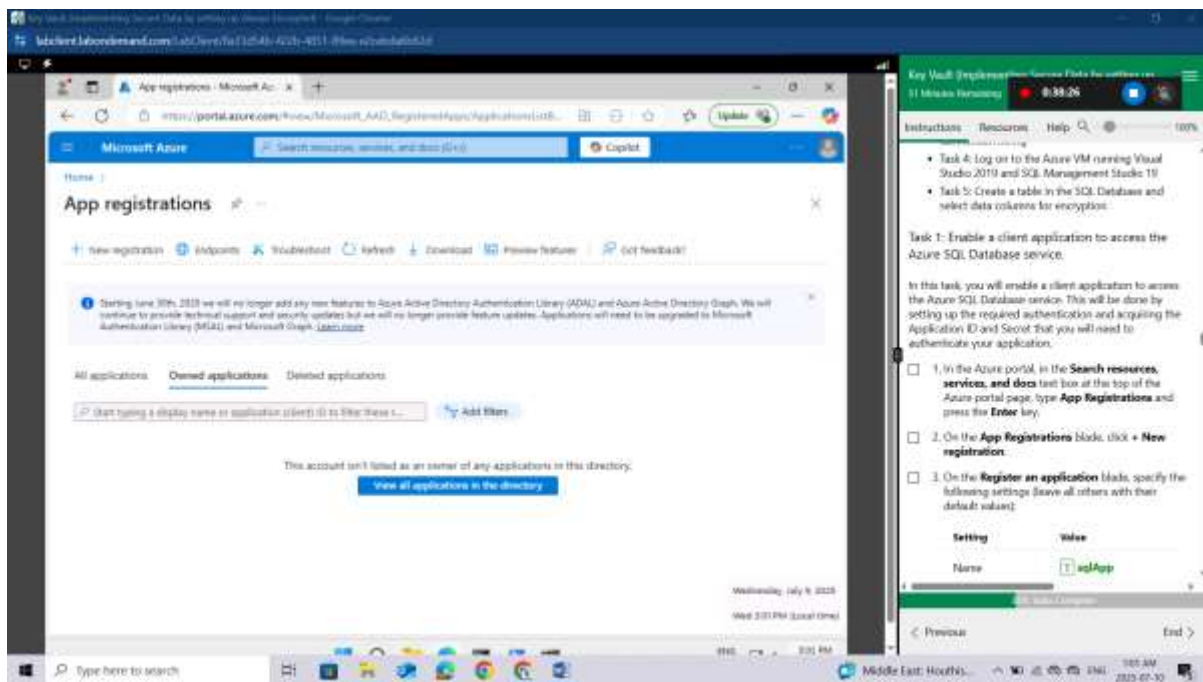
Task 1: Enable a client application to access the Azure SQL Database service.

In this task, you will enable a client application to access the Azure SQL Database service. This will be done by setting up the required authentication and acquiring the Application ID and Secret that you will need to authenticate your application.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **App Registrations** and press the **Enter** key. If you do not locate it go to Microsoft Entra ID, then on left navigation go to manage, then locate App Registration

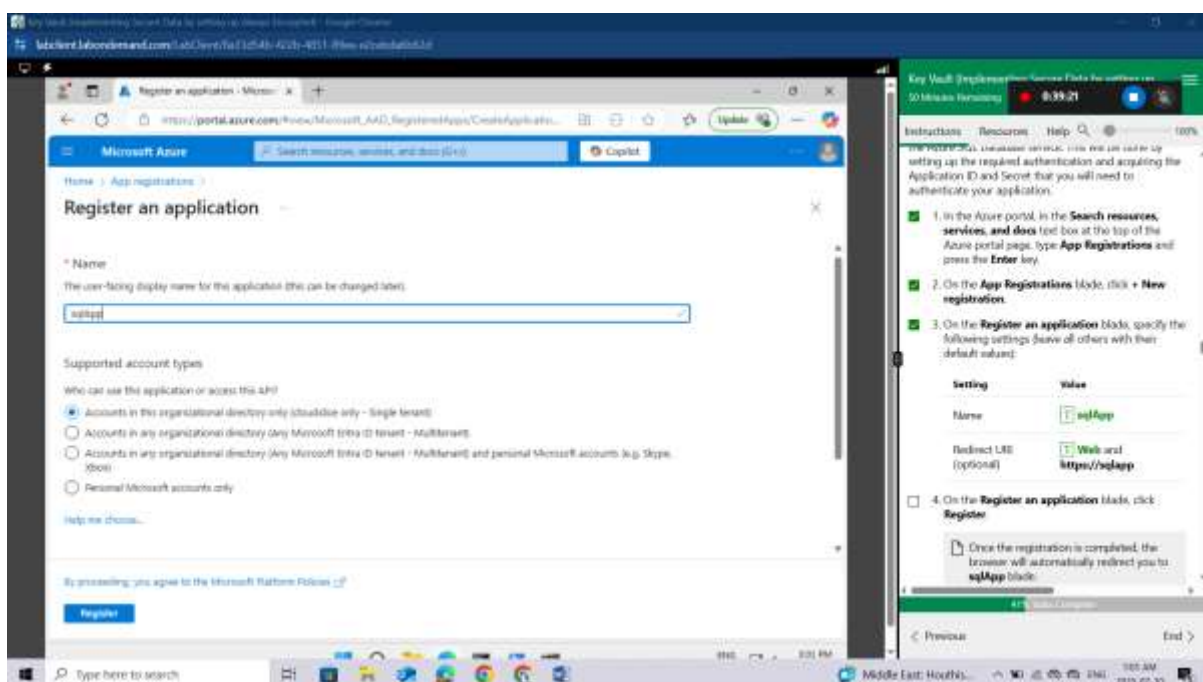


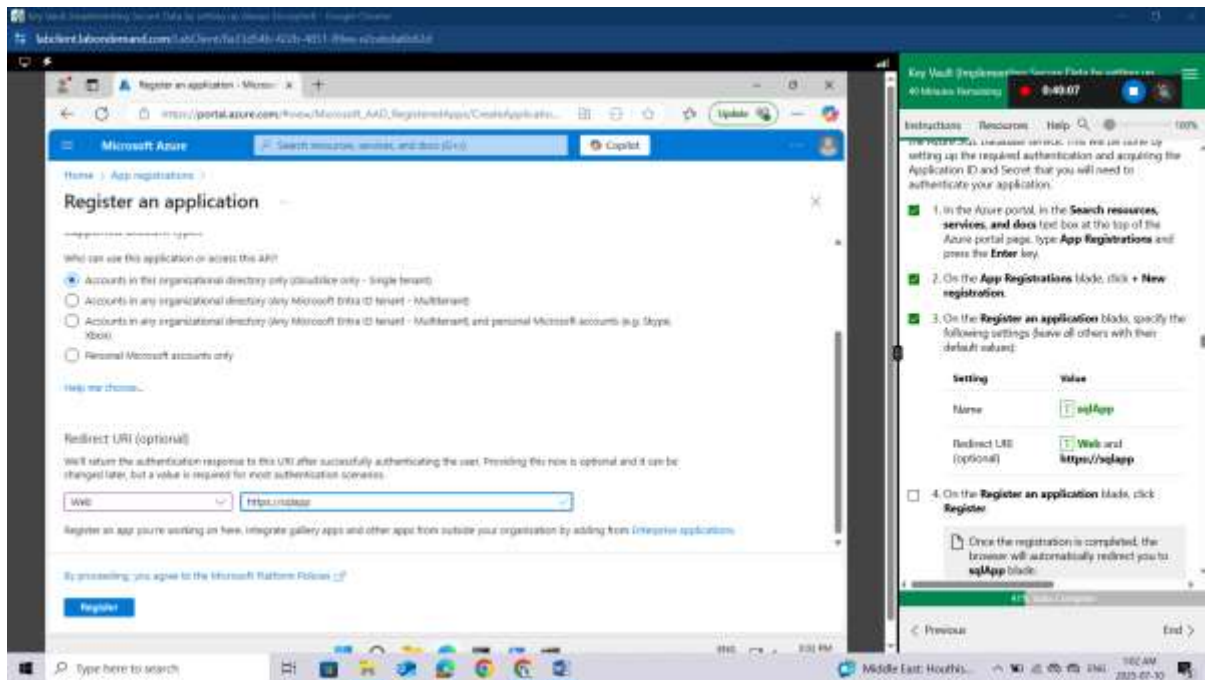
On the **App Registrations** blade, click + **New registration**.



On the **Register an application** blade, specify the following settings (leave all others with their default values):

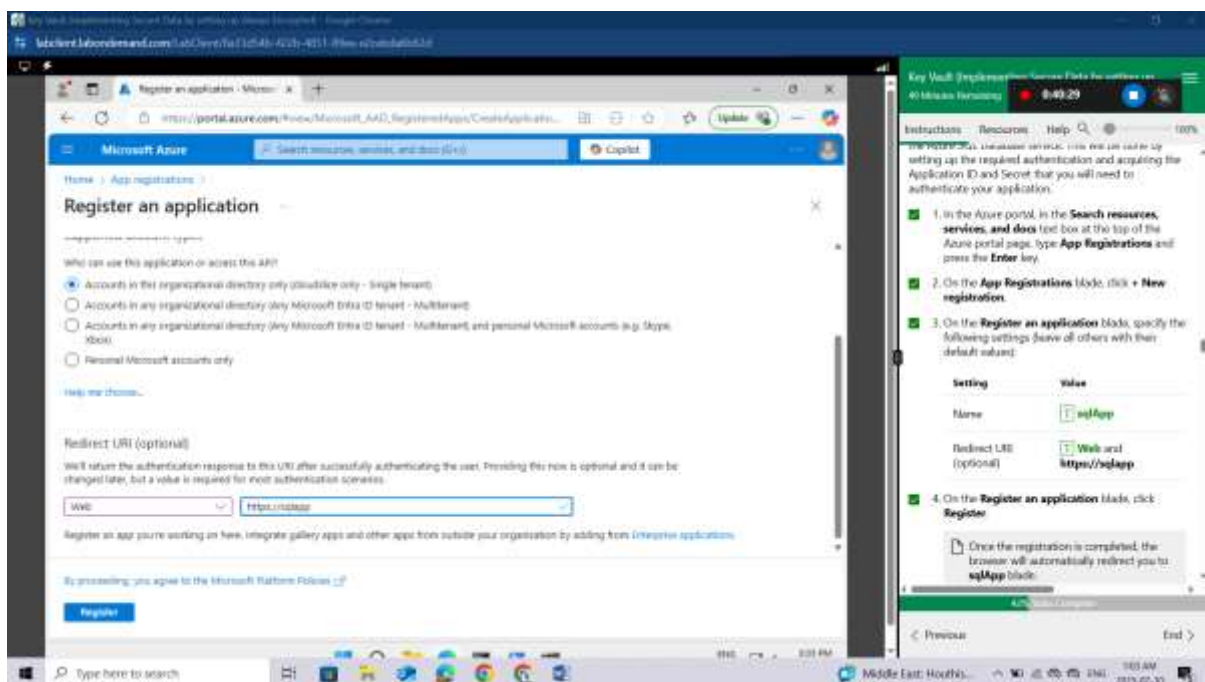
Setting	Value
Name	sqlApp
Redirect URI (optional)	Web and https://sqlapp





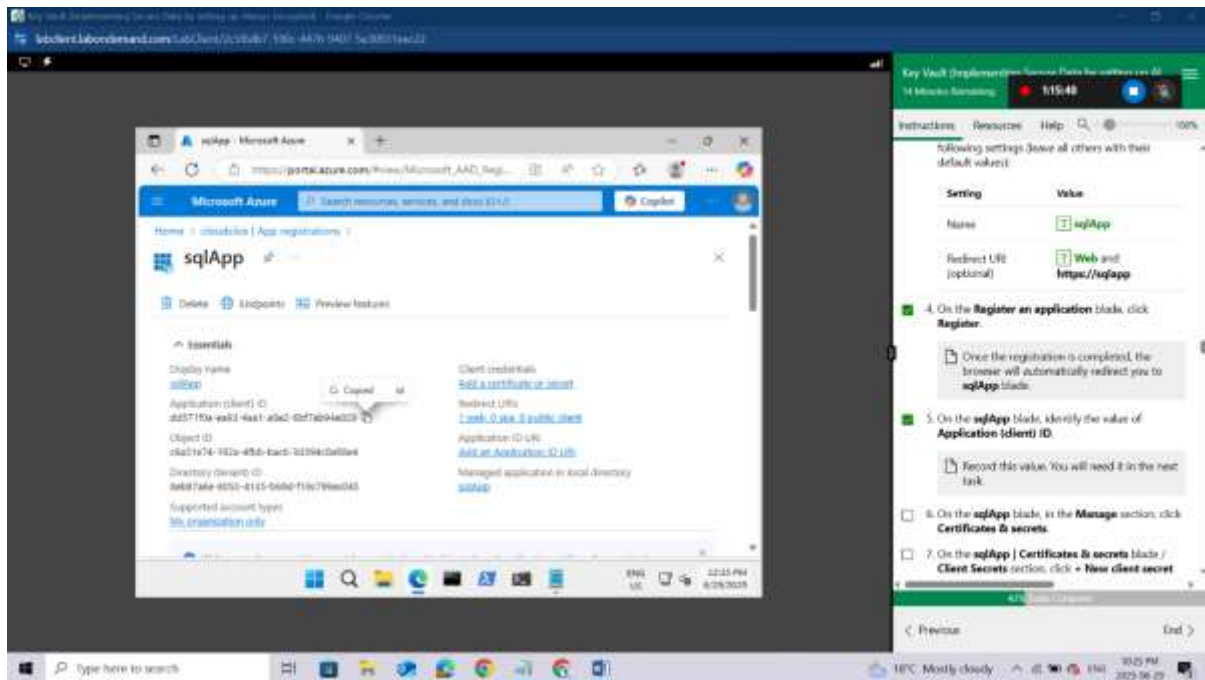
On the **Register an application** blade, click **Register**.

Once the registration is completed, the browser will automatically redirect you to **sqlApp** blade.

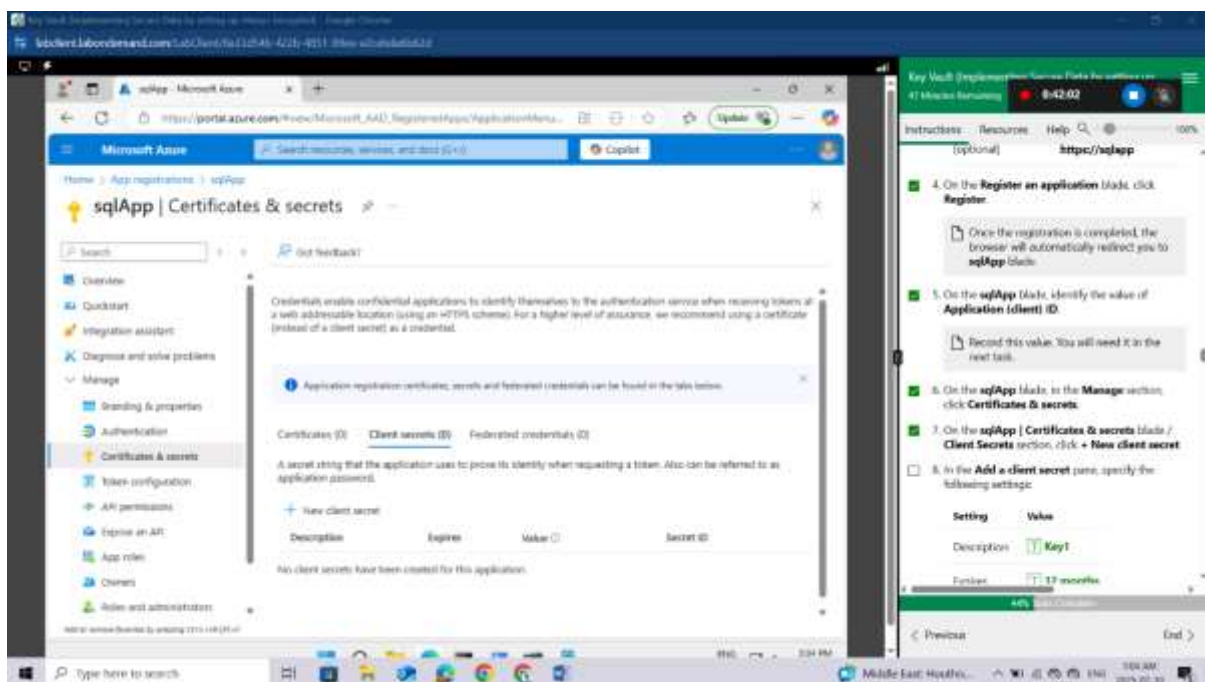


On the **sqlApp** blade, identify the value of **Application (client) ID**.

Record this value. You will need it in the next task.



On the **sqlApp** blade, in the **Manage** section, click **Certificates & secrets**.

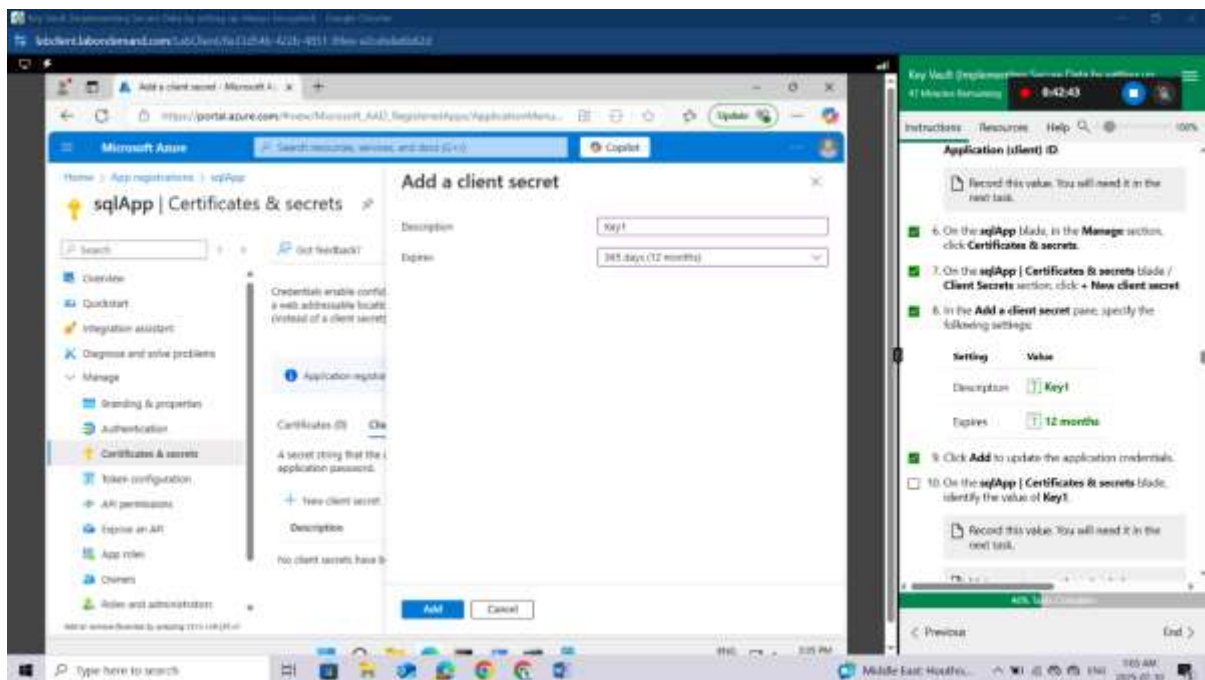


On the **sqlApp | Certificates & secrets** blade / **Client Secrets** section, click **+ New client secret**

In the **Add a client secret** pane, specify the following settings:

Setting	Value
Description	Key1
Expires	12 months

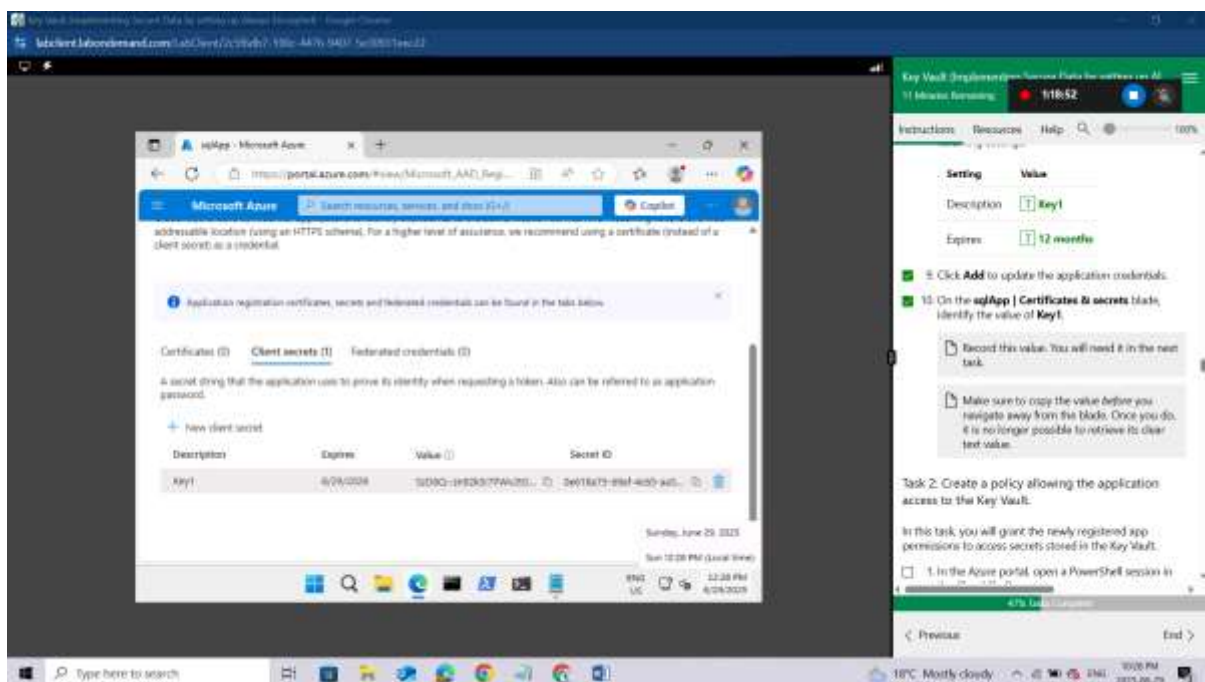
Click **Add** to update the application credentials.



On the **sqlApp | Certificates & secrets** blade, identify the value of **Key1**.

Record this value. You will need it in the next task.

Make sure to copy the value before you navigate away from the blade. Once you do, it is no longer possible to retrieve its clear text value.



Task 2: Create a policy allowing the application access to the Key Vault.

In this task, you will grant the newly registered app permissions to access secrets stored in the Key Vault.

In the Azure portal, open a PowerShell session in the Cloud Shell pane.

Ensure **PowerShell** is selected in the upper-left drop-down menu of the Cloud Shell pane.

In the PowerShell session within the Cloud Shell pane, run the following to create a variable storing the **Application (client) ID** you recorded in the previous task (replace the (Azure_AD_Application_ID) placeholder with the value of the **Application (client) ID**):

```
powershell
$applicationId = '<Azure_AD_Application_ID>'
```

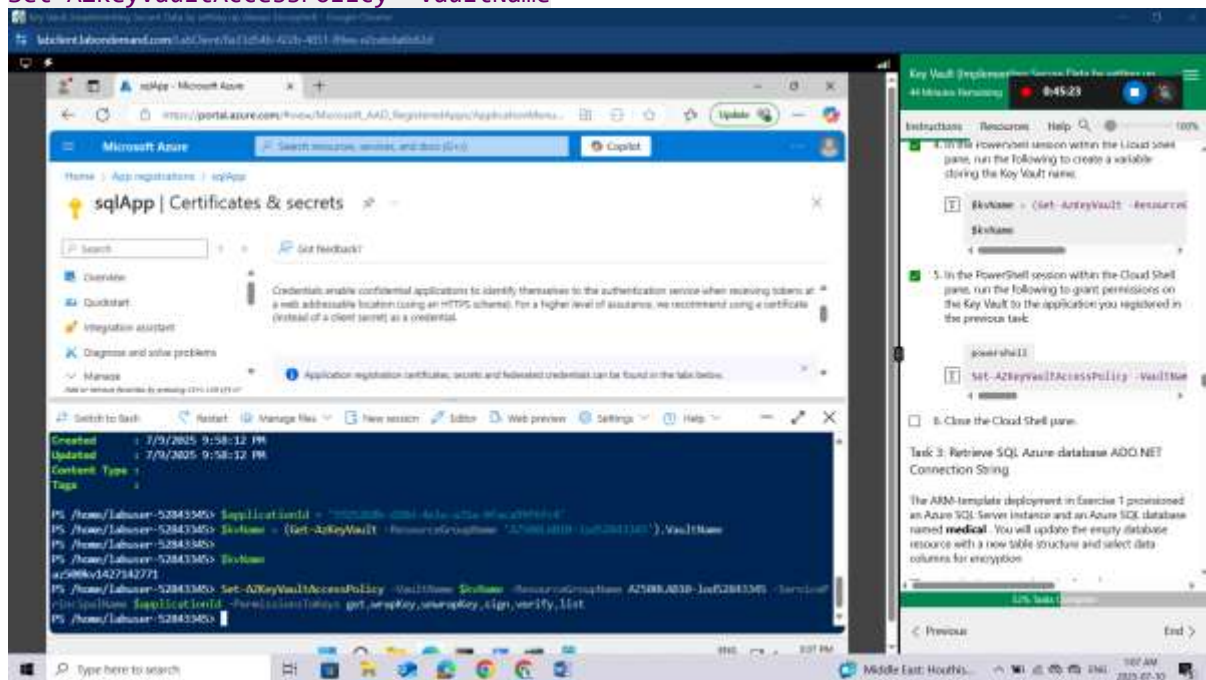
In the PowerShell session within the Cloud Shell pane, run the following to create a variable storing the Key Vault name.

```
$kvName = (Get-AzKeyVault -ResourceGroupName 'AZ500LAB10-1od52601000').VaultName
```

```
$kvName
```

In the PowerShell session within the Cloud Shell pane, run the following to grant permissions on the Key Vault to the application you registered in the previous task:

```
powershell
Set-AzKeyVaultAccessPolicy -VaultName
```

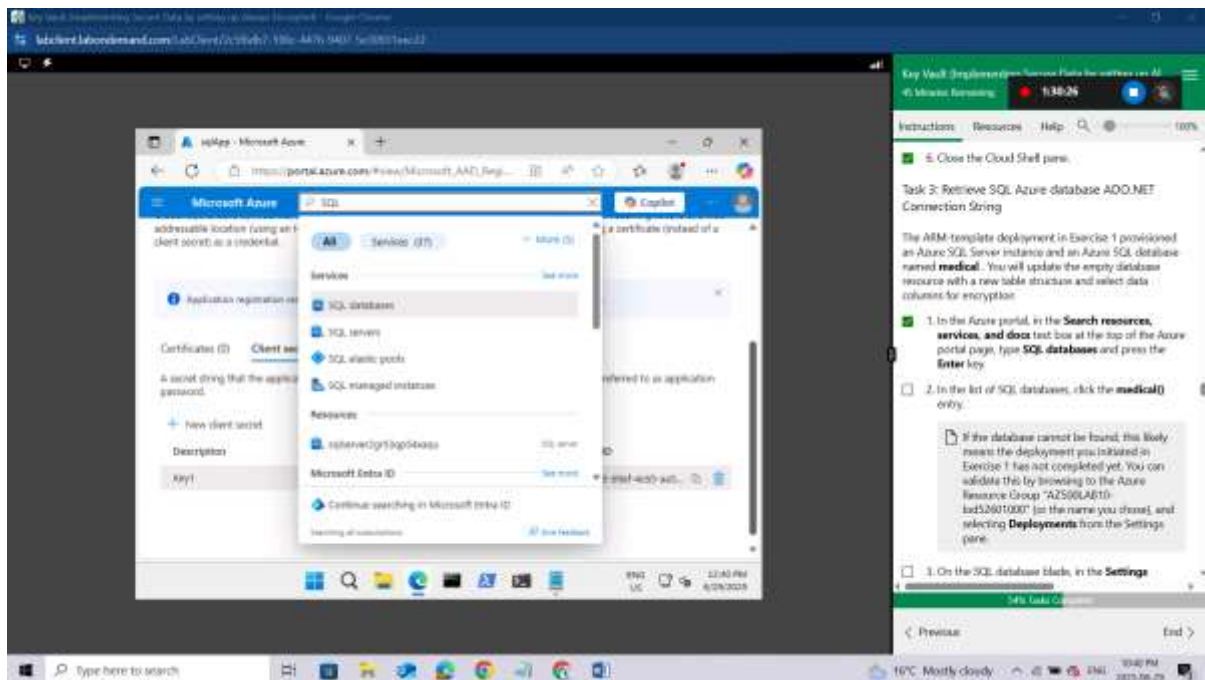


Close the Cloud Shell pane.

Task 3: Retrieve SQL Azure database ADO.NET Connection String

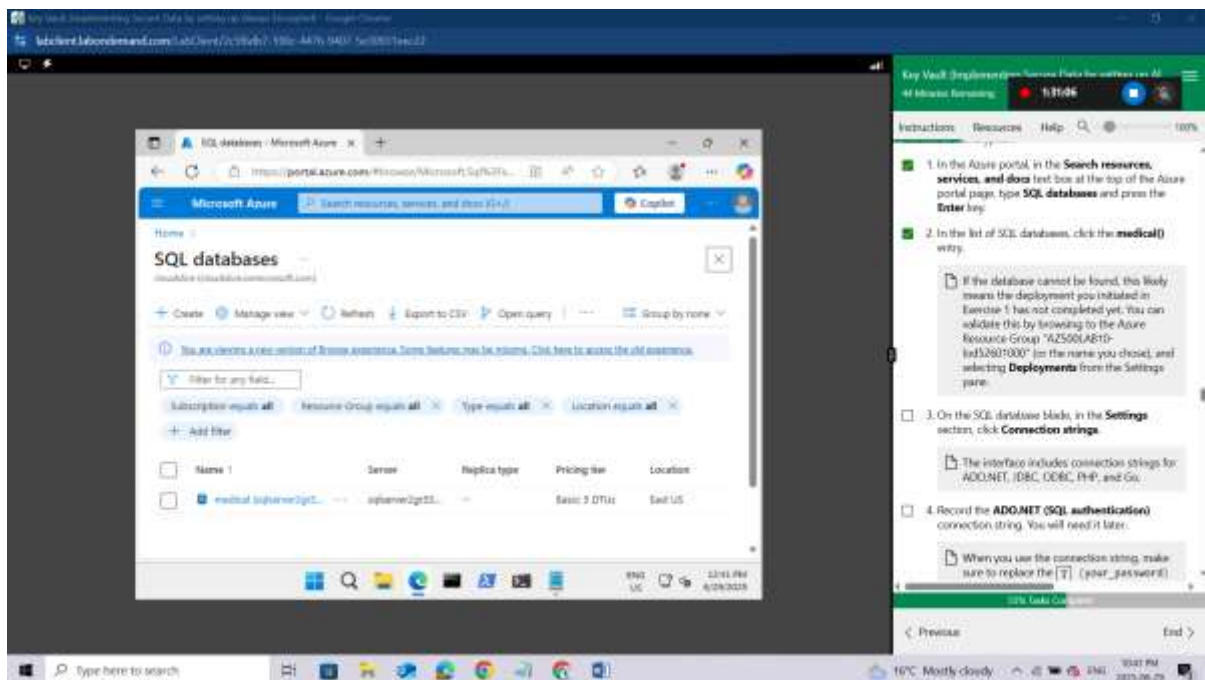
The ARM-template deployment in Exercise 1 provisioned an Azure SQL Server instance and an Azure SQL database named **medical**. You will update the empty database resource with a new table structure and select data columns for encryption

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **SQL databases** and press the **Enter** key.



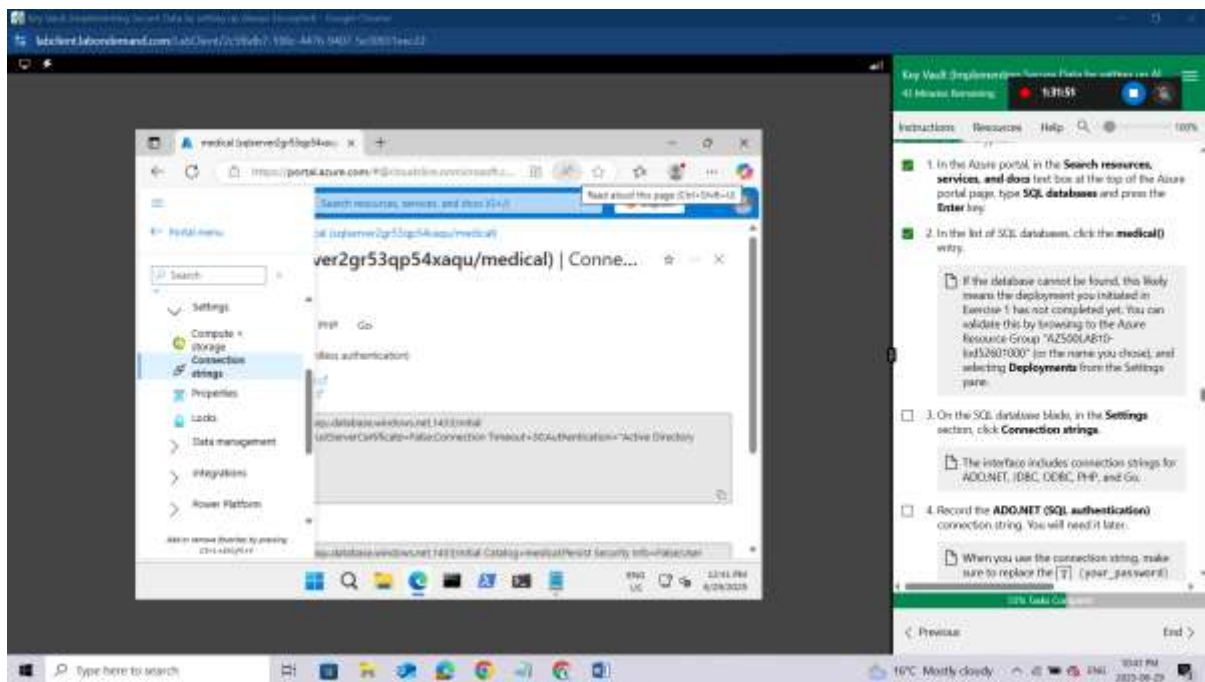
In the list of SQL databases, click the **medical()** entry.

*If the database cannot be found, this likely means the deployment you initiated in Exercise 1 has not completed yet. You can validate this by browsing to the Azure Resource Group "AZ500LAB10-lod52601000" (or the name you chose), and selecting **Deployments** from the Settings pane.*



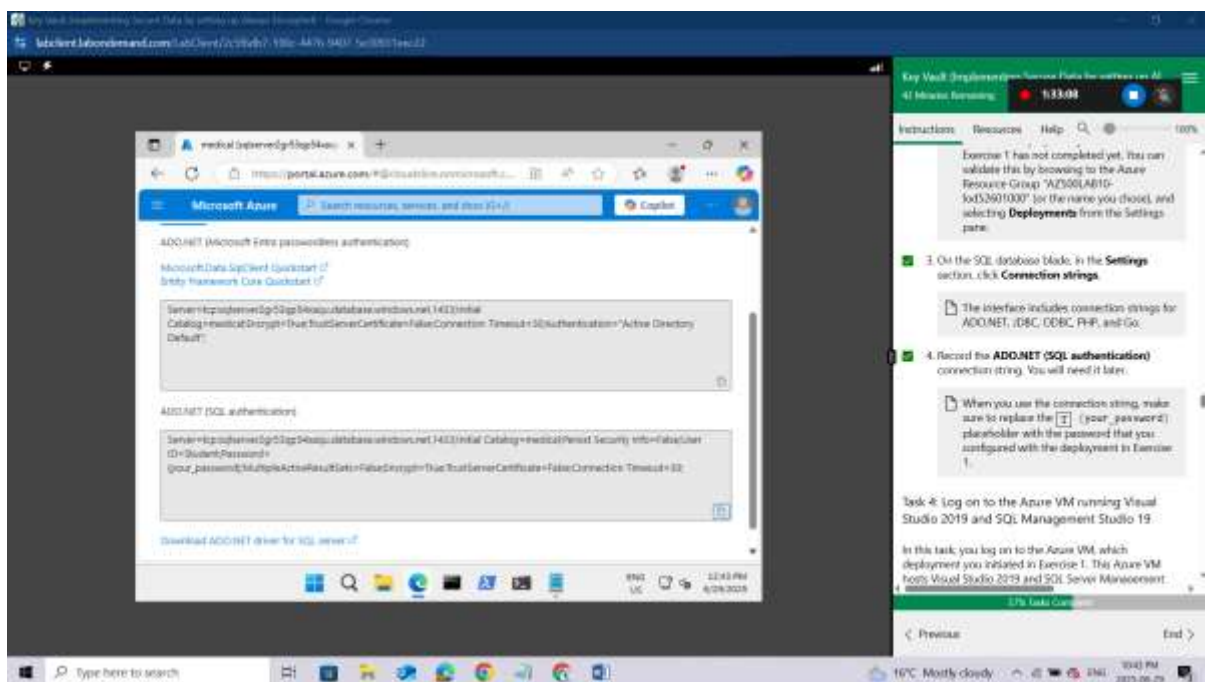
On the SQL database blade, in the **Settings** section, click **Connection strings**.

The interface includes connection strings for ADO.NET, JDBC, ODBC, PHP, and Go.



Record the **ADO.NET (SQL authentication)** connection string. You will need it later.

When you use the connection string, make sure to replace the {your_password} placeholder with the password that you configured with the deployment in Exercise 1.

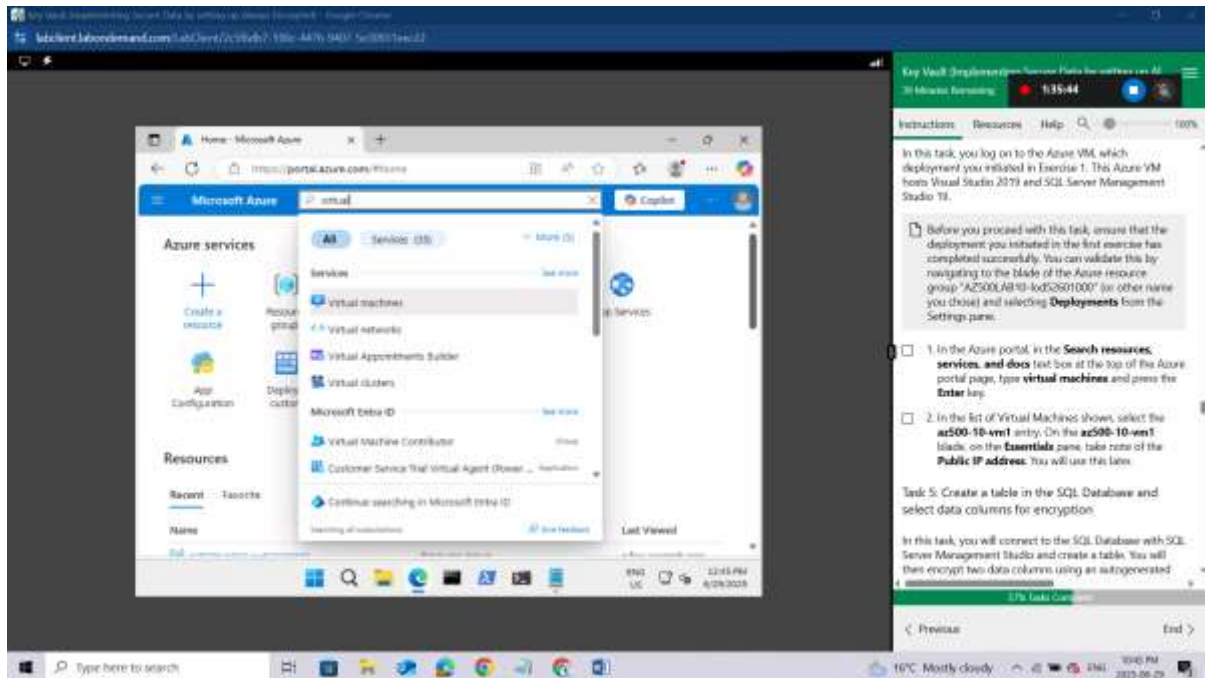


Task 4: Log on to the Azure VM running Visual Studio 2019 and SQL Management Studio 19

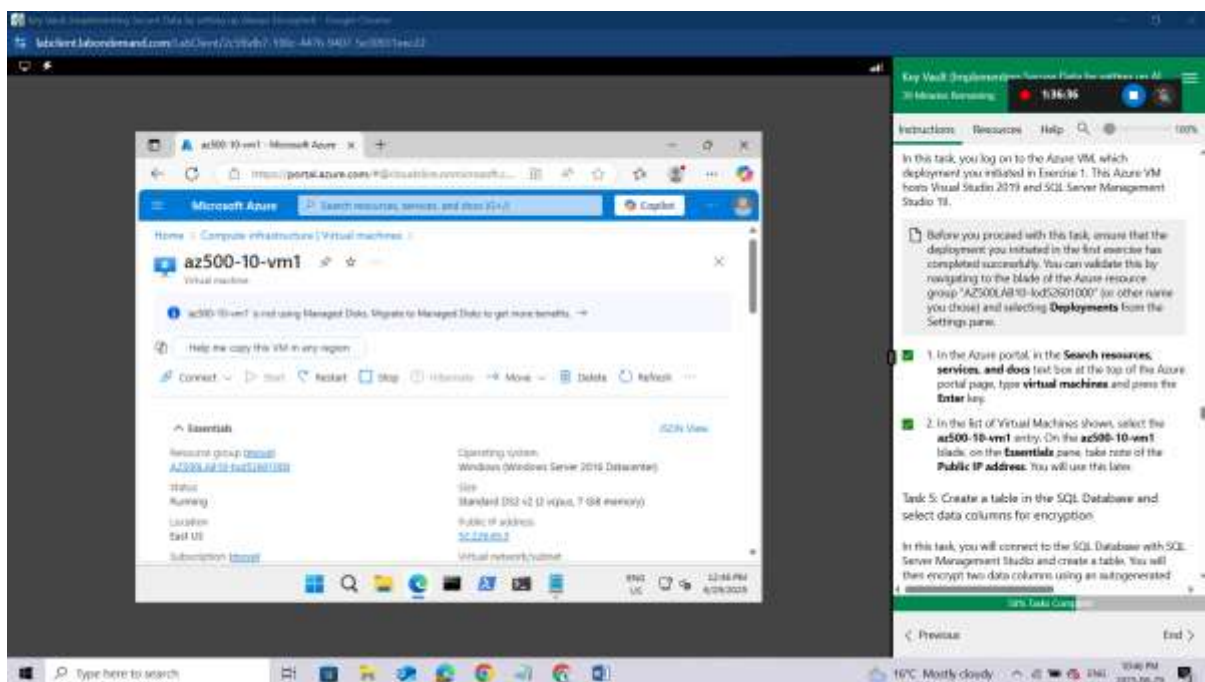
In this task, you log on to the Azure VM, which deployment you initiated in Exercise 1. This Azure VM hosts Visual Studio 2019 and SQL Server Management Studio 19.

Before you proceed with this task, ensure that the deployment you initiated in the first exercise has completed successfully. You can validate this by navigating to the blade of the Azure resource group "AZ500LAB10-lod52601000" (or other name you chose) and selecting **Deployments** from the Settings pane.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **virtual machines** and press the **Enter** key.



In the list of Virtual Machines shown, select the **az500-10-vm1** entry. On the **az500-10-vm1** blade, on the **Essentials** pane, take note of the **Public IP address**. You will use this later.

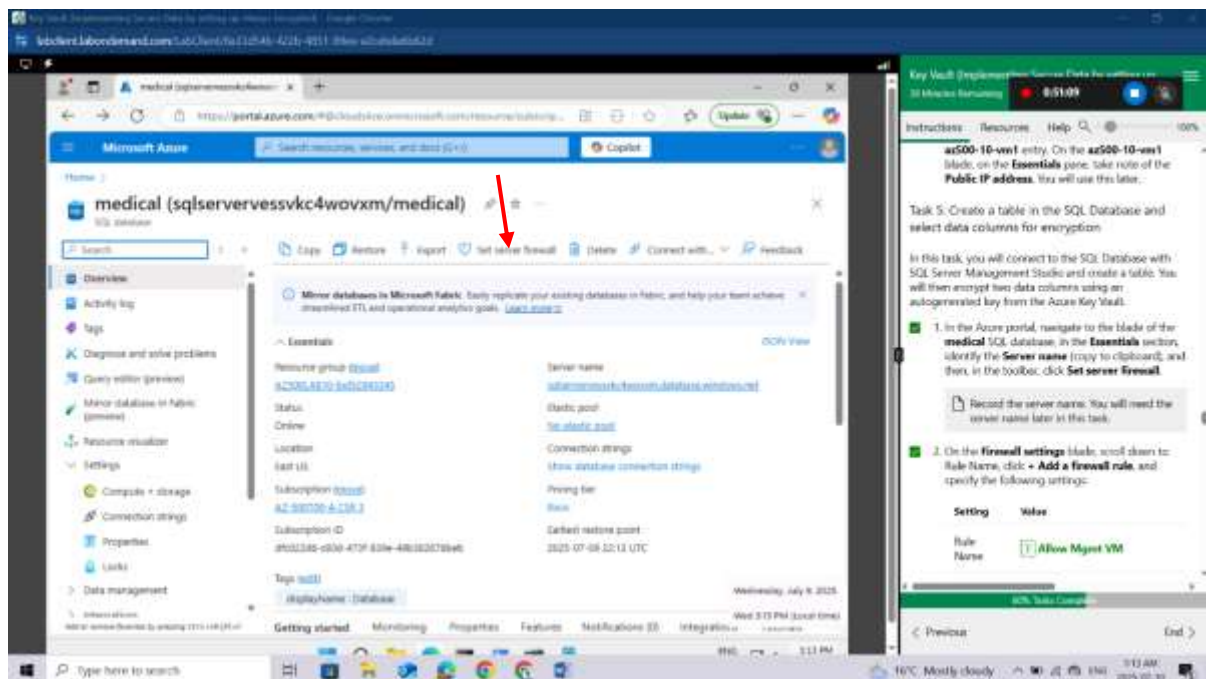


Task 5: Create a table in the SQL Database and select data columns for encryption

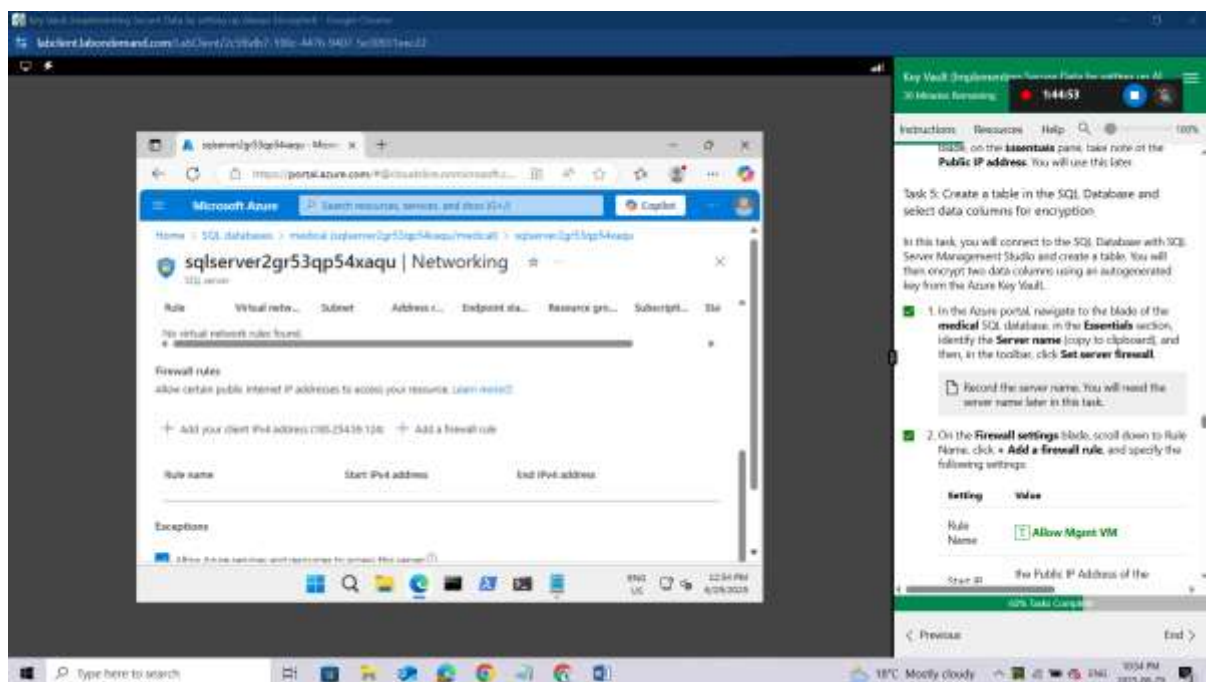
In this task, you will connect to the SQL Database with SQL Server Management Studio and create a table. You will then encrypt two data columns using an autogenerated key from the Azure Key Vault.

In the Azure portal, navigate to the blade of the **medical** SQL database, in the **Essentials** section, identify the **Server name** (copy to clipboard), and then, in the toolbar, click **Set server firewall**.

Record the server name. You will need the server name later in this task.

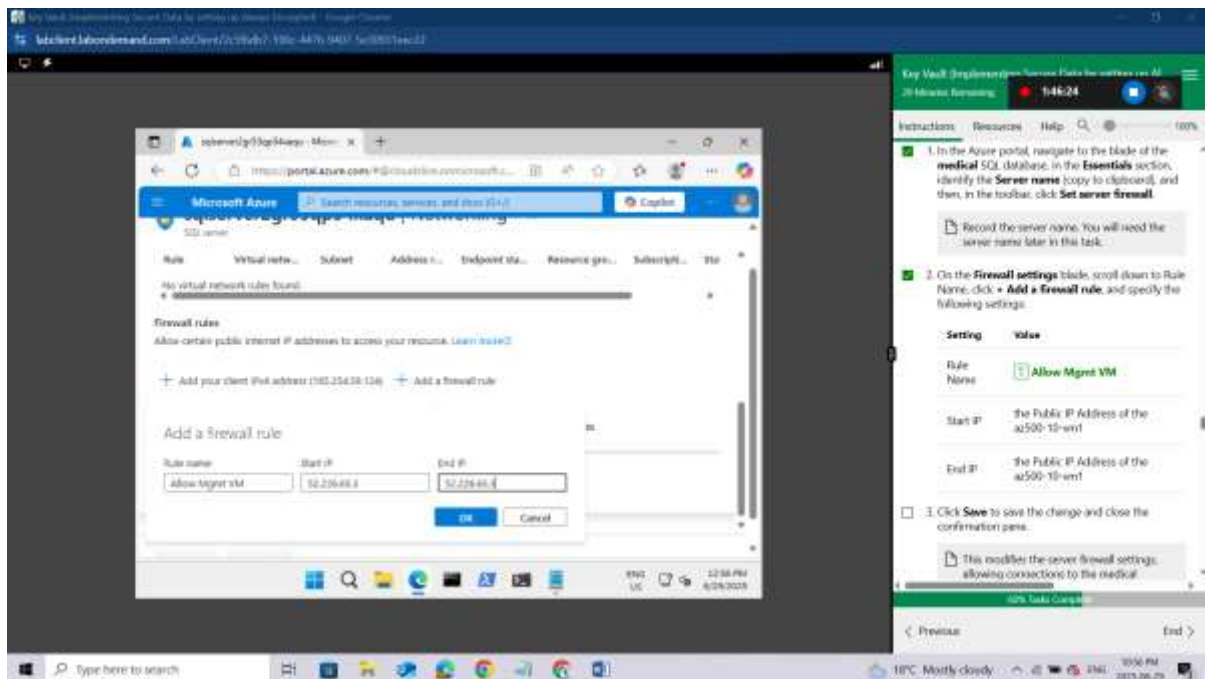


On the **Firewall settings** blade, scroll down to Rule Name, click + **Add a firewall rule**,



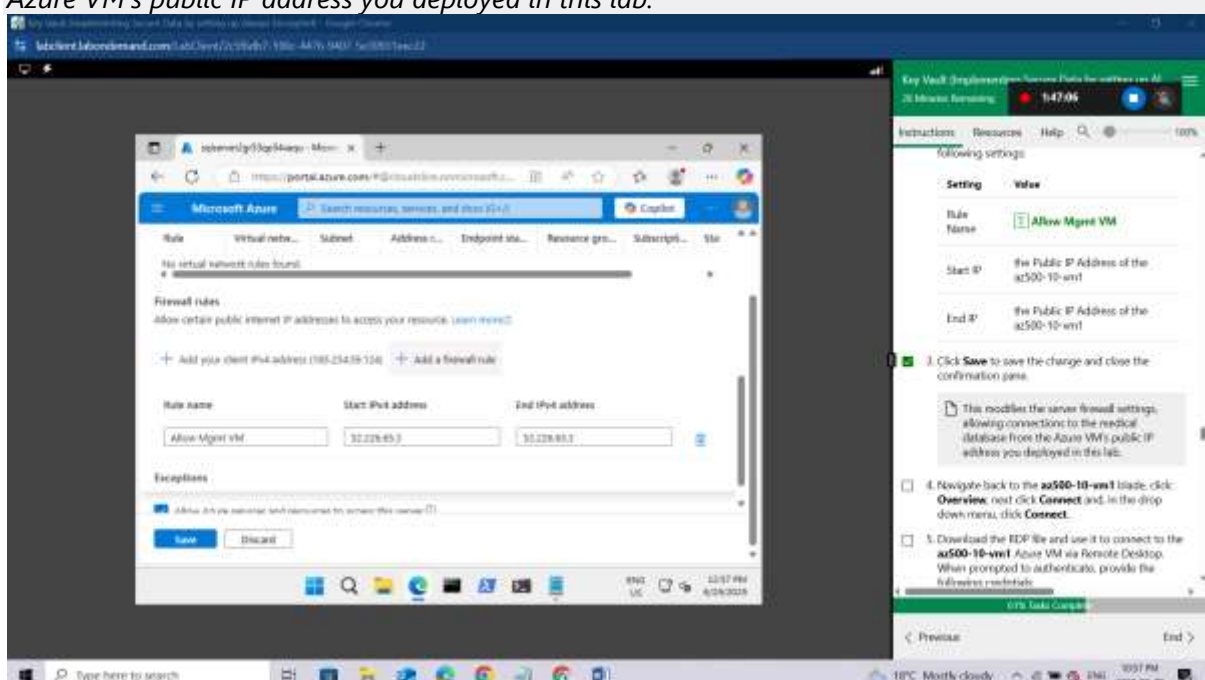
and specify the following settings:

Setting	Value
Rule Name	Allow Mgmt VM
Start IP	the Public IP Address of the az500-10-vm1
End IP	the Public IP Address of the az500-10-vm1

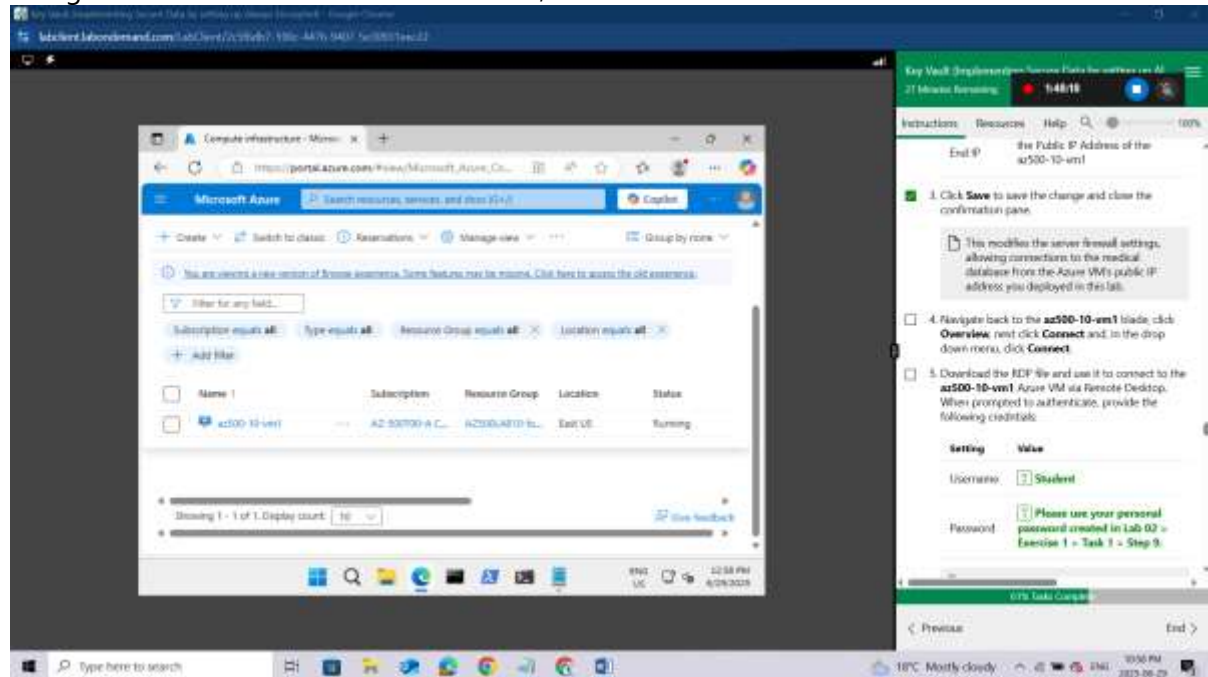


Click **Save** to save the change and close the confirmation pane.

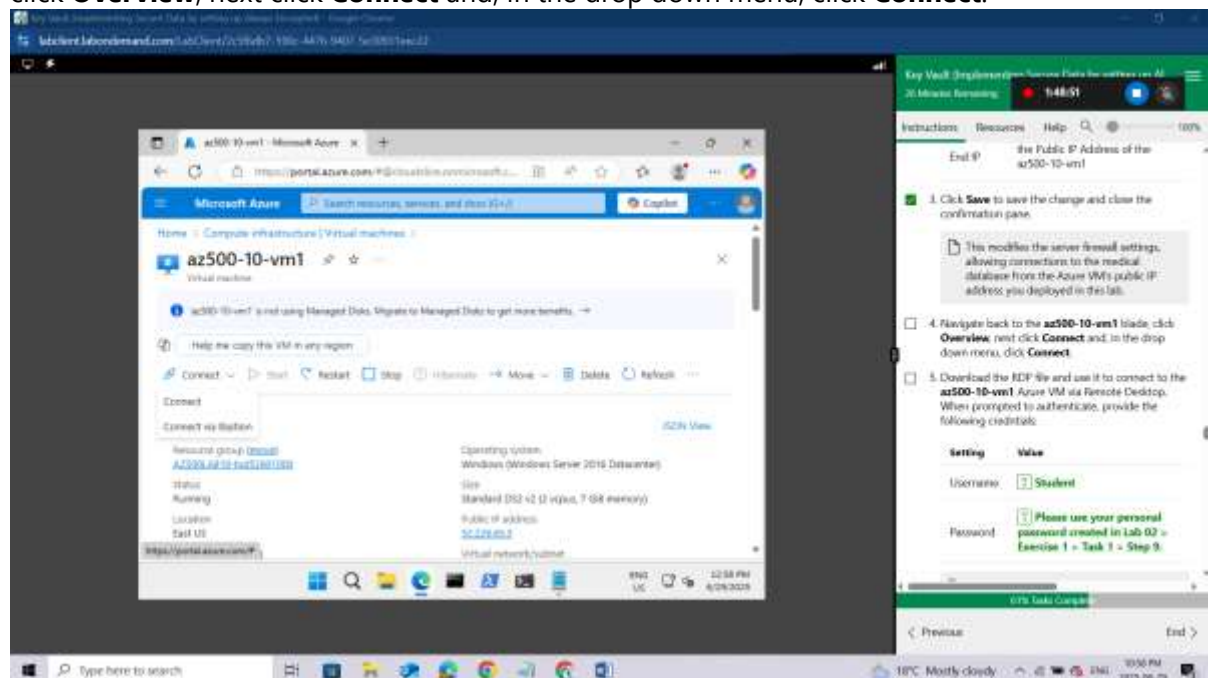
This modifies the server firewall settings, allowing connections to the medical database from the Azure VM's public IP address you deployed in this lab.



Navigate back to the **az500-10-vm1** blade,

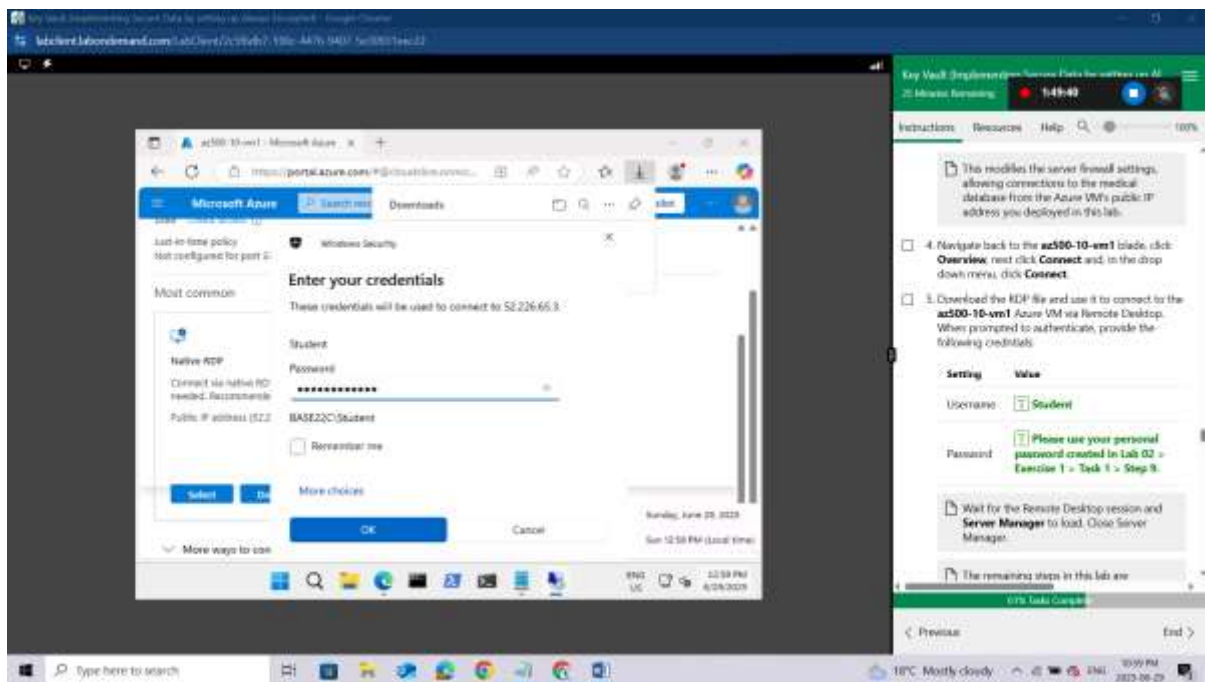


click **Overview**, next click **Connect** and, in the drop down menu, click **Connect**.



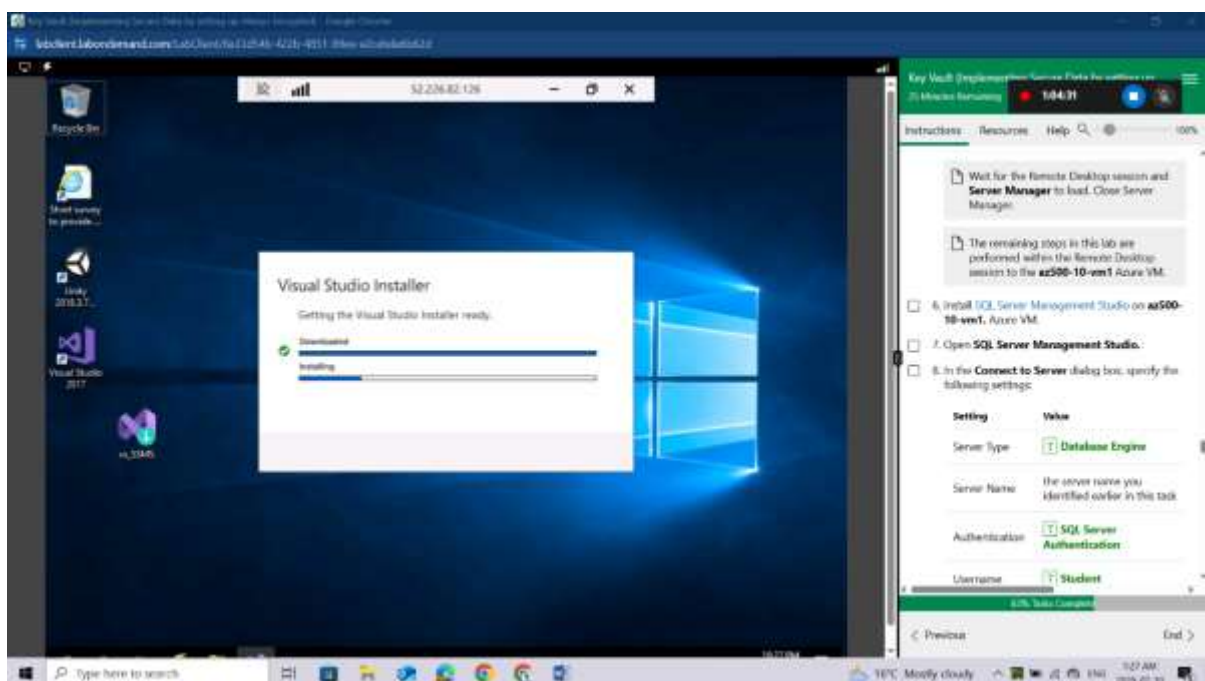
Download the RDP file and use it to connect to the **az500-10-vm1** Azure VM via Remote Desktop. When prompted to authenticate, provide the following credentials:

Setting	Value
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 1 > Task 1 > Step 9.

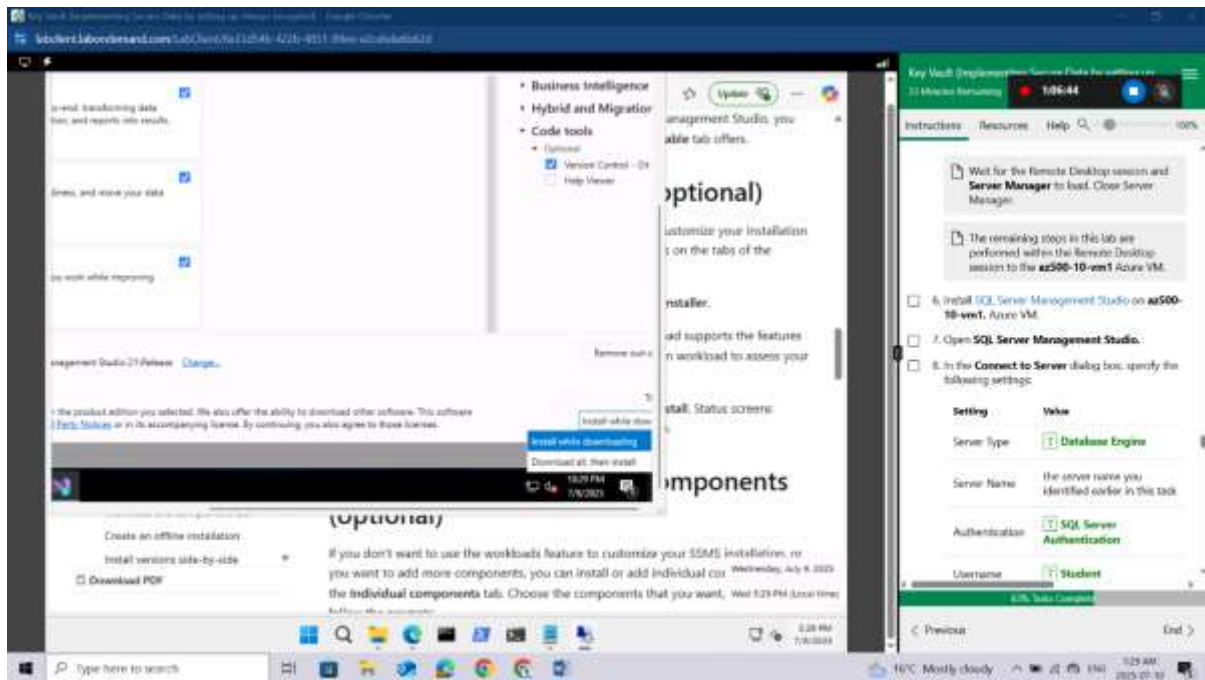


Wait for the Remote Desktop session and **Server Manager** to load. Close Server Manager. The remaining steps in this lab are performed within the Remote Desktop session to the **az500-10-vm1** Azure VM.

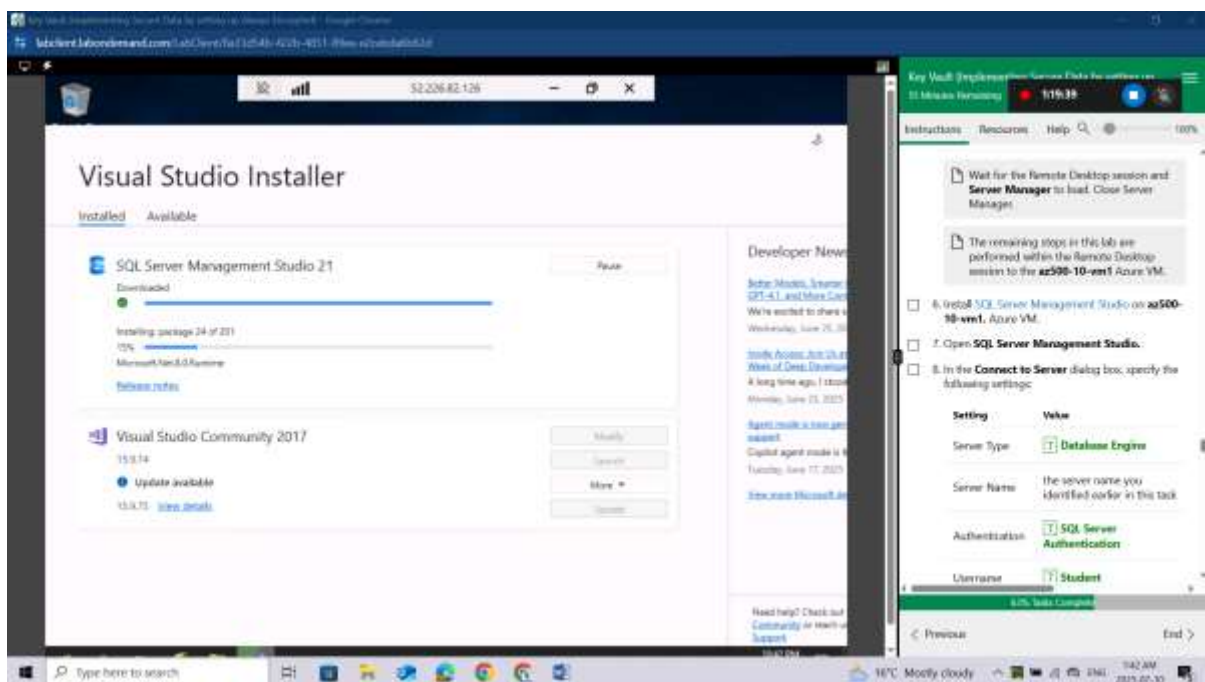
Install [SQL Server Management Studio](#) on **az500-10-vm1**. Azure VM.



Select your workloads and click install

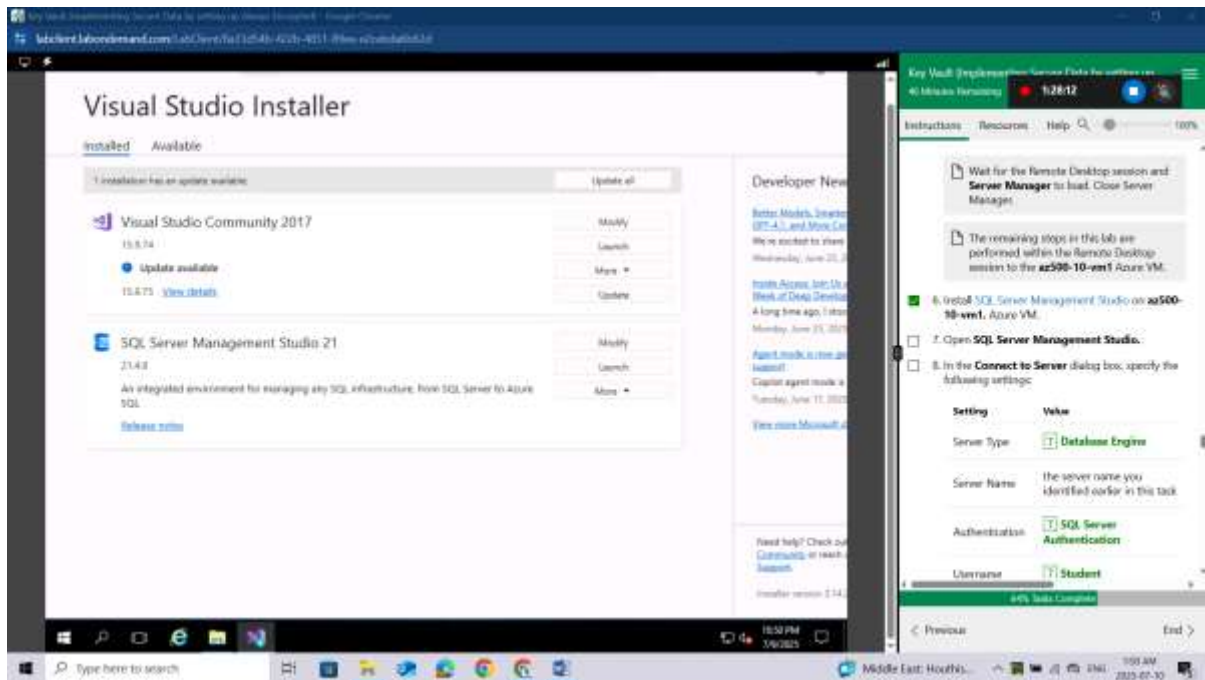


Click on Install and wait it to finish



Open **SQL Server Management Studio**.

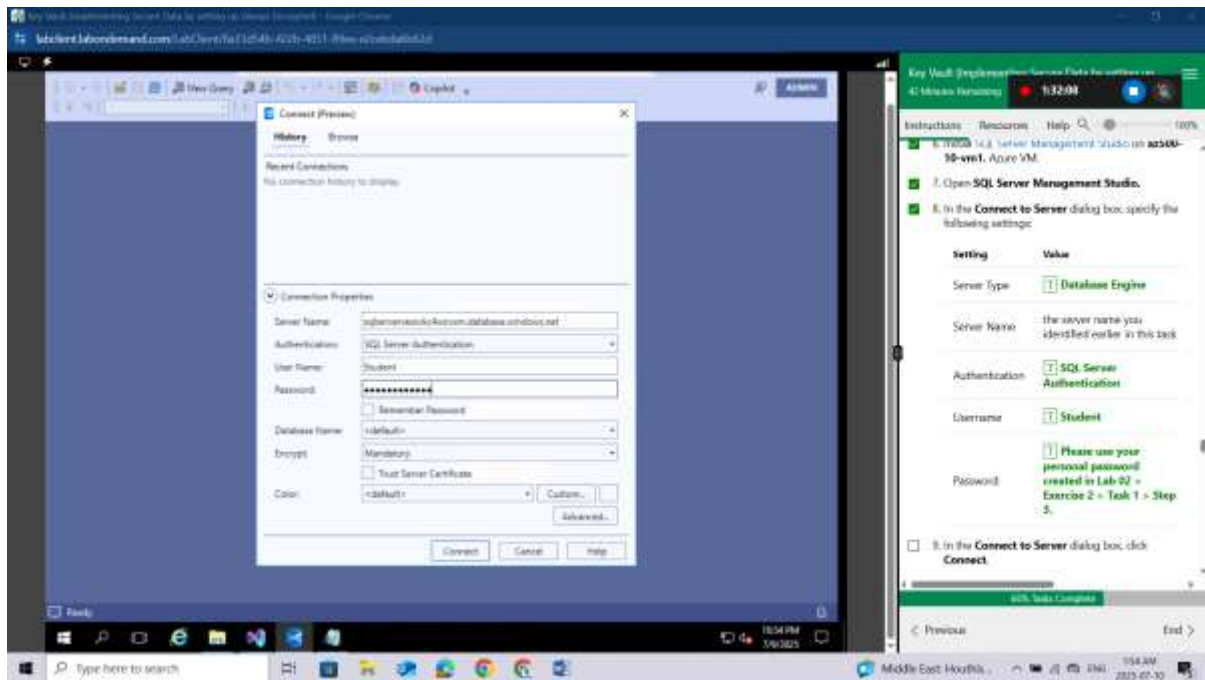
Click on Launch to open



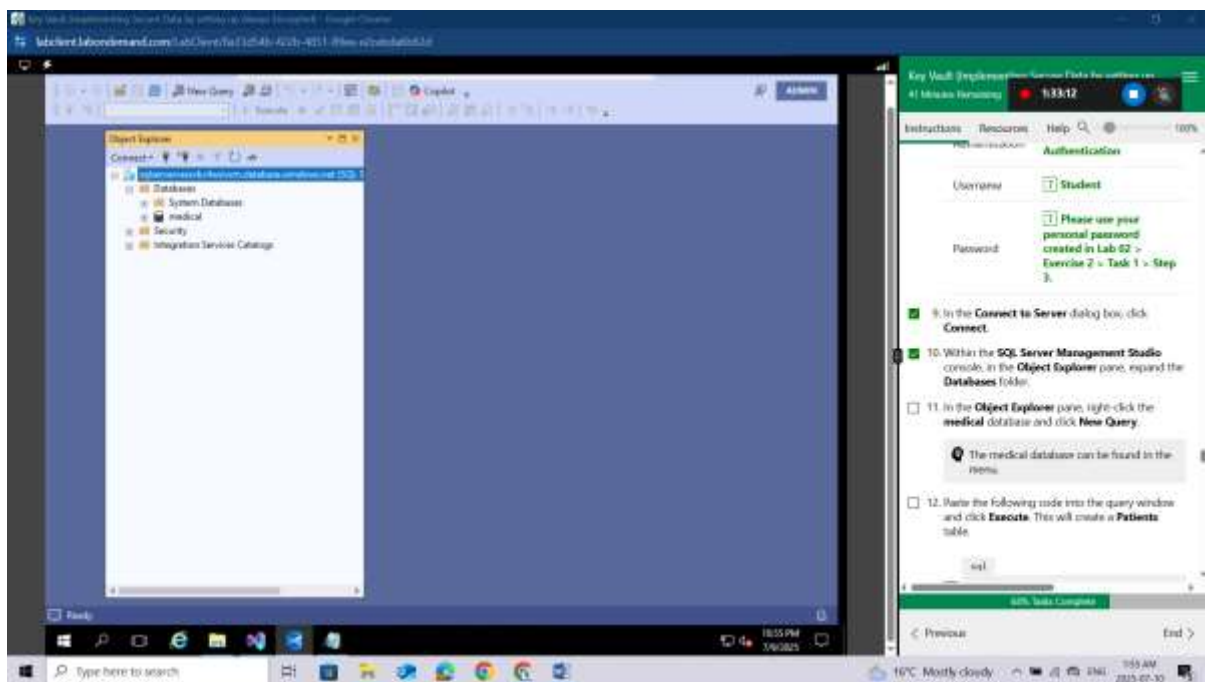
In the **Connect to Server** dialog box, specify the following settings:

Setting	Value
Server Type	Database Engine
Server Name	the server name you identified earlier in this task
Authentication	SQL Server Authentication
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.

In the **Connect to Server** dialog box, click **Connect**.

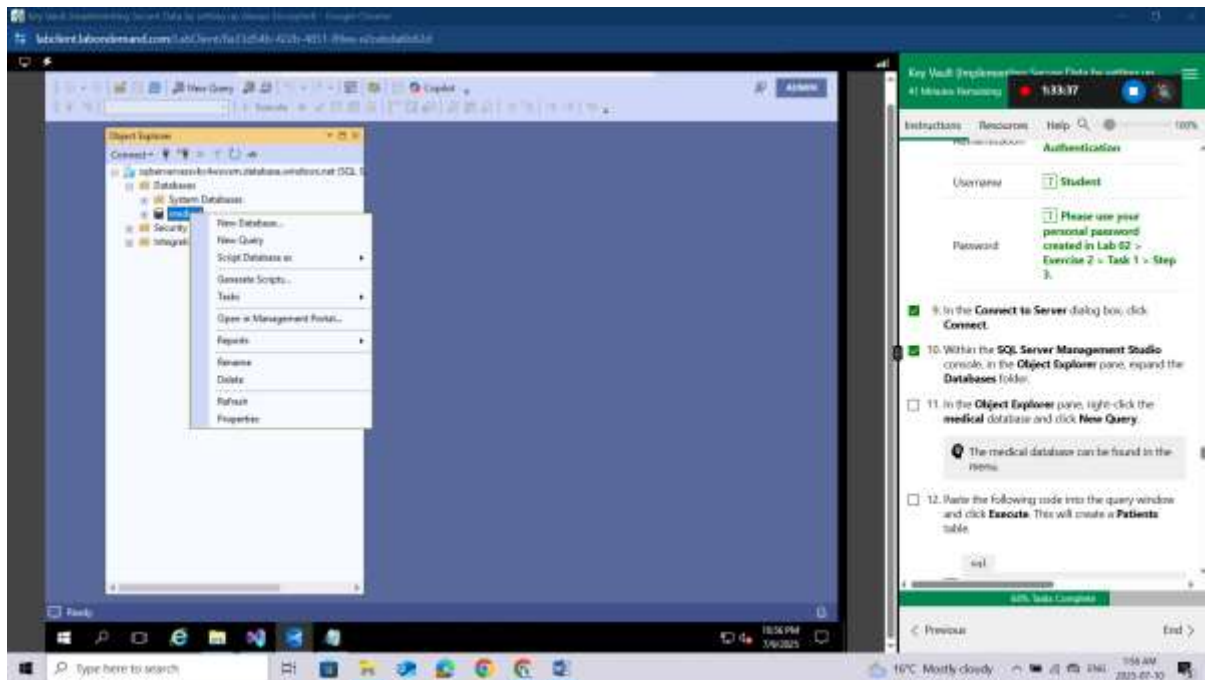


Within the **SQL Server Management Studio** console, in the **Object Explorer** pane, expand the **Databases** folder.



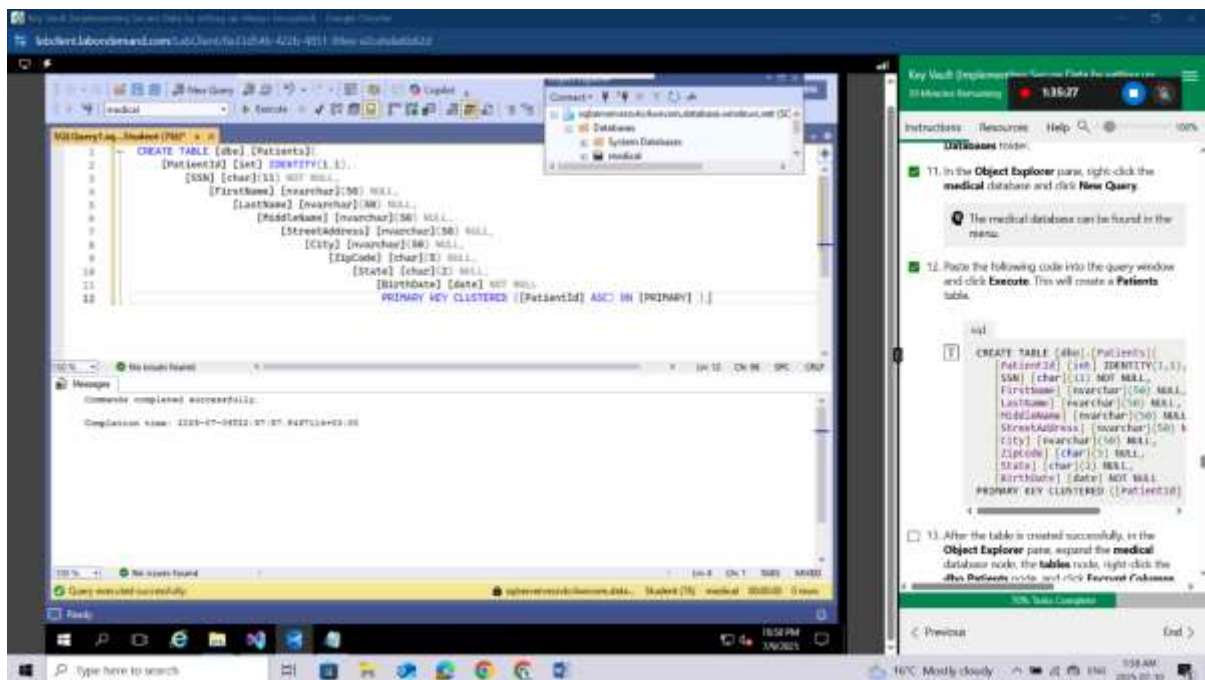
In the **Object Explorer** pane, right-click the **medical** database and click **New Query**.

The medical database can be found in the menu.

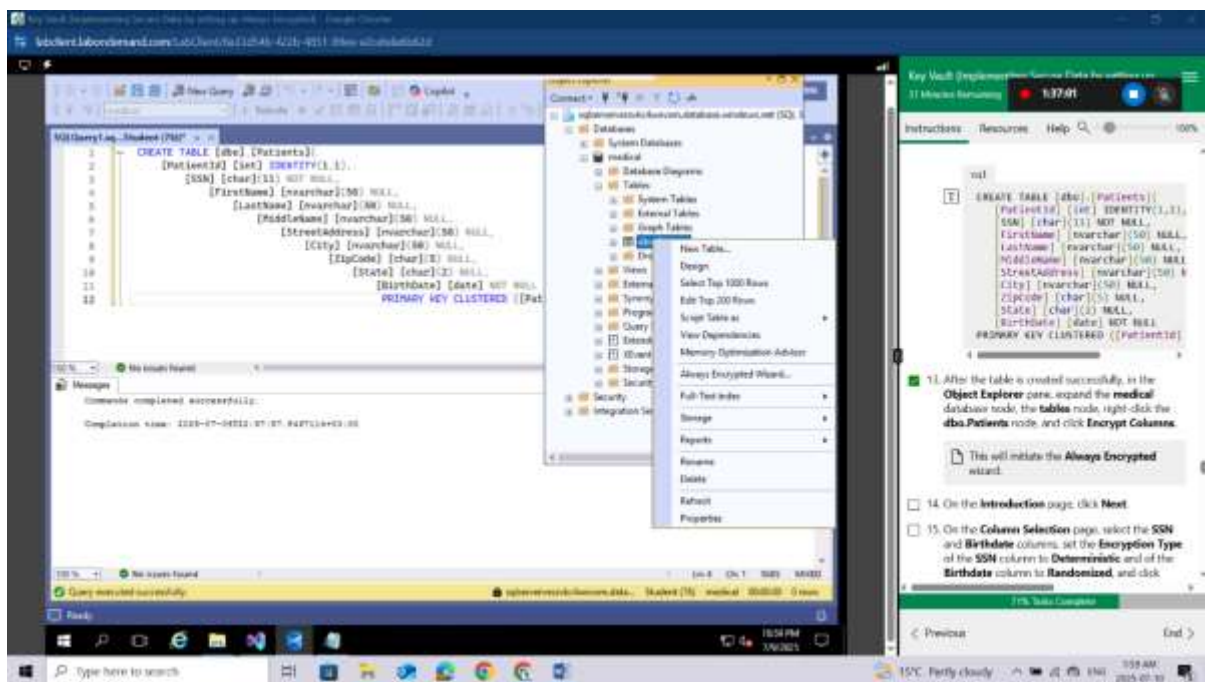


Paste the following code into the query window and click **Execute**. This will create a **Patients** table.

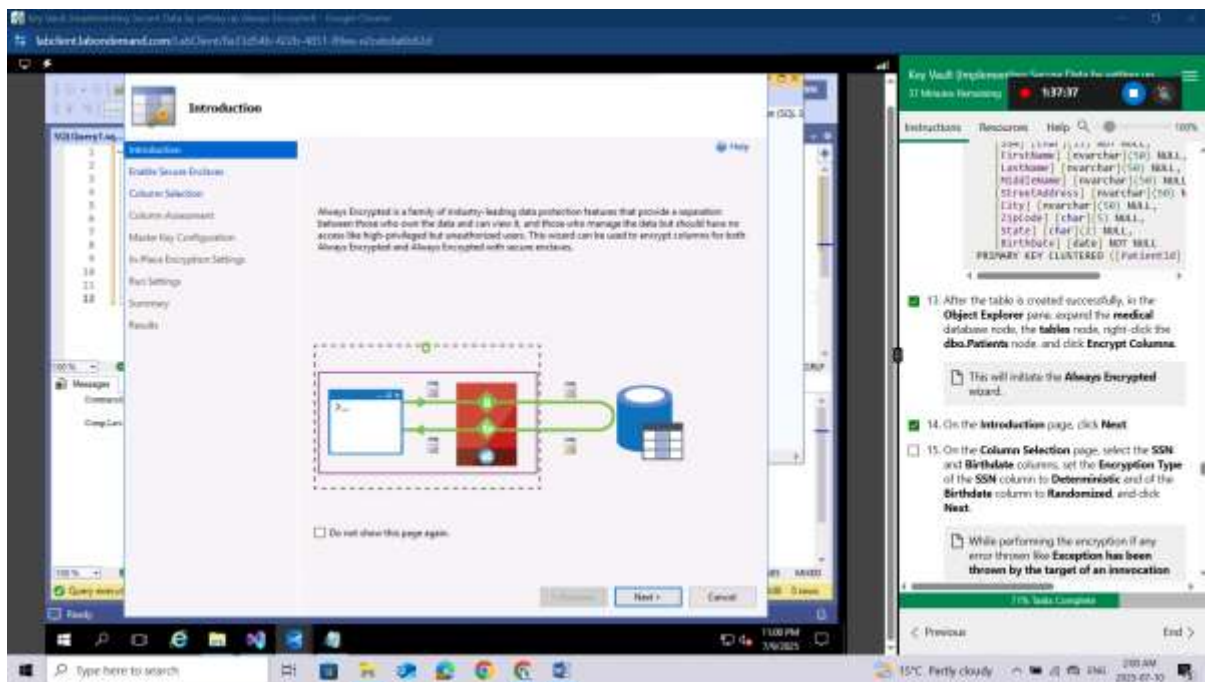
```
sql
CREATE TABLE [dbo].[Patients](
    [PatientId] [int] IDENTITY(1,1),
    [SSN] [char](11) NOT NULL,
    [FirstName] [nvarchar](50) NULL,
    [LastName] [nvarchar](50) NULL,
    [MiddleName] [nvarchar](50) NULL,
    [StreetAddress] [nvarchar](50) NULL,
    [City] [nvarchar](50) NULL,
    [ZipCode] [char](5) NULL,
    [State] [char](2) NULL,
    [BirthDate] [date] NOT NULL
    PRIMARY KEY CLUSTERED ([PatientId] ASC) ON [PRIMARY] );
```



After the table is created successfully, in the **Object Explorer** pane, expand the **medical** database node, the **tables** node, right-click the **dbo.Patients** node, and click **Encrypt Columns**. This will initiate the **Always Encrypted** wizard.

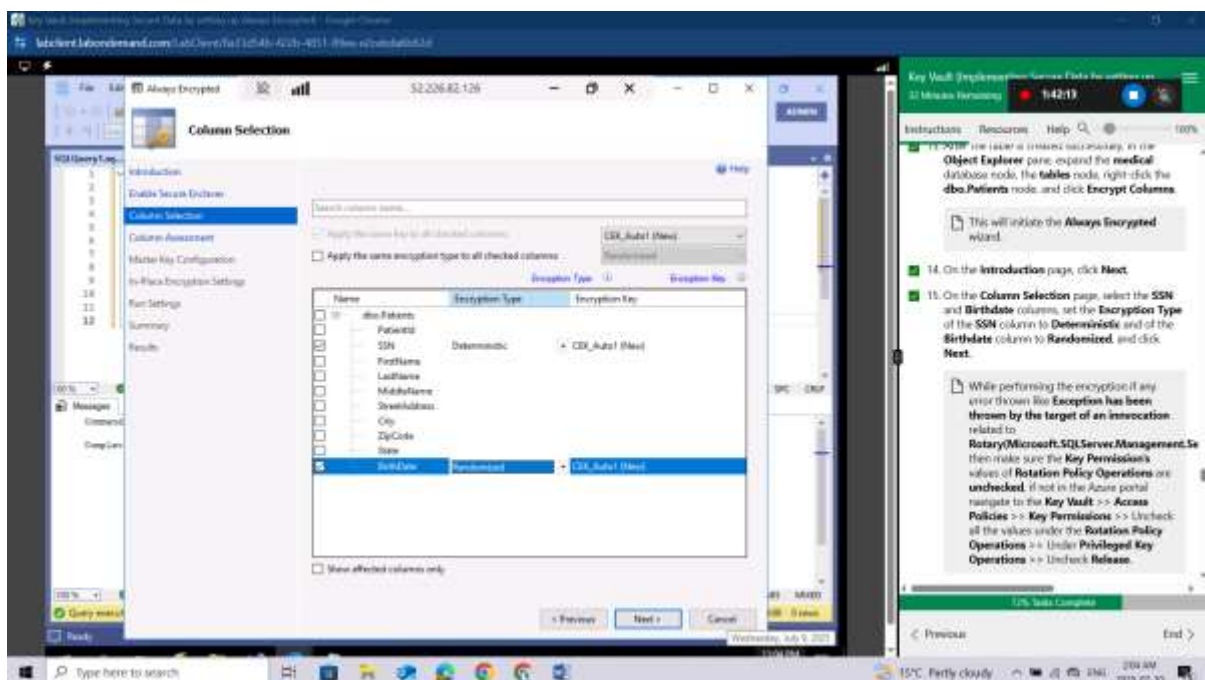


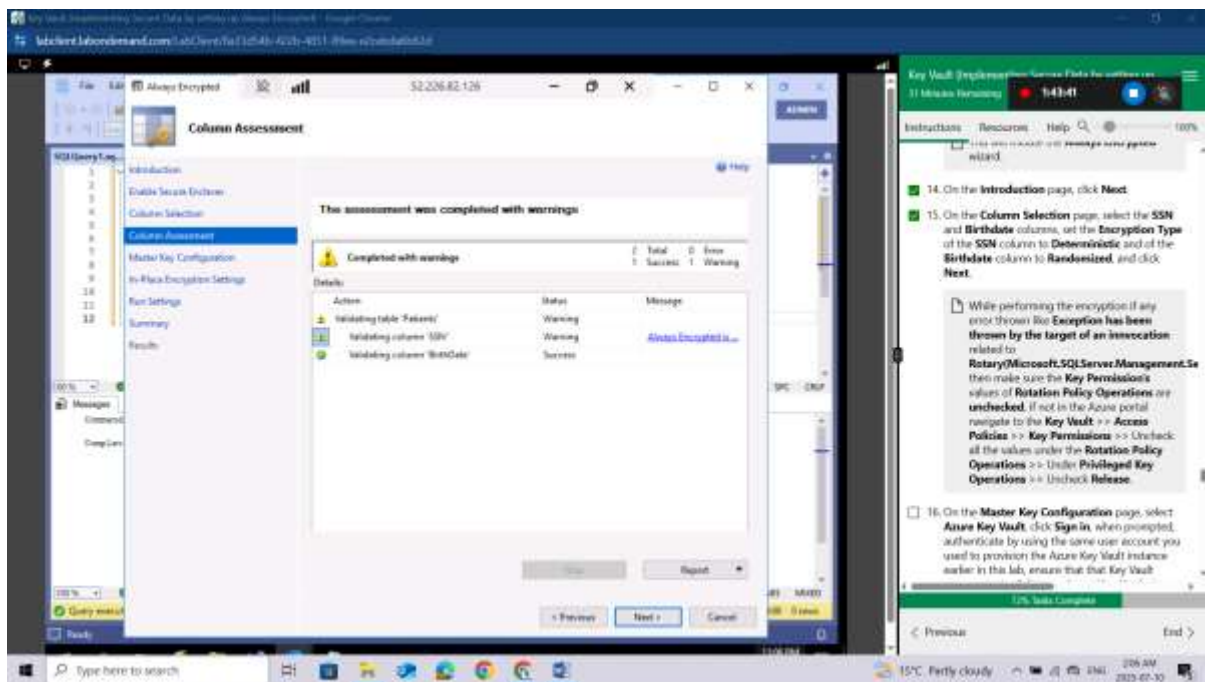
On the **Introduction** page, click **Next**.



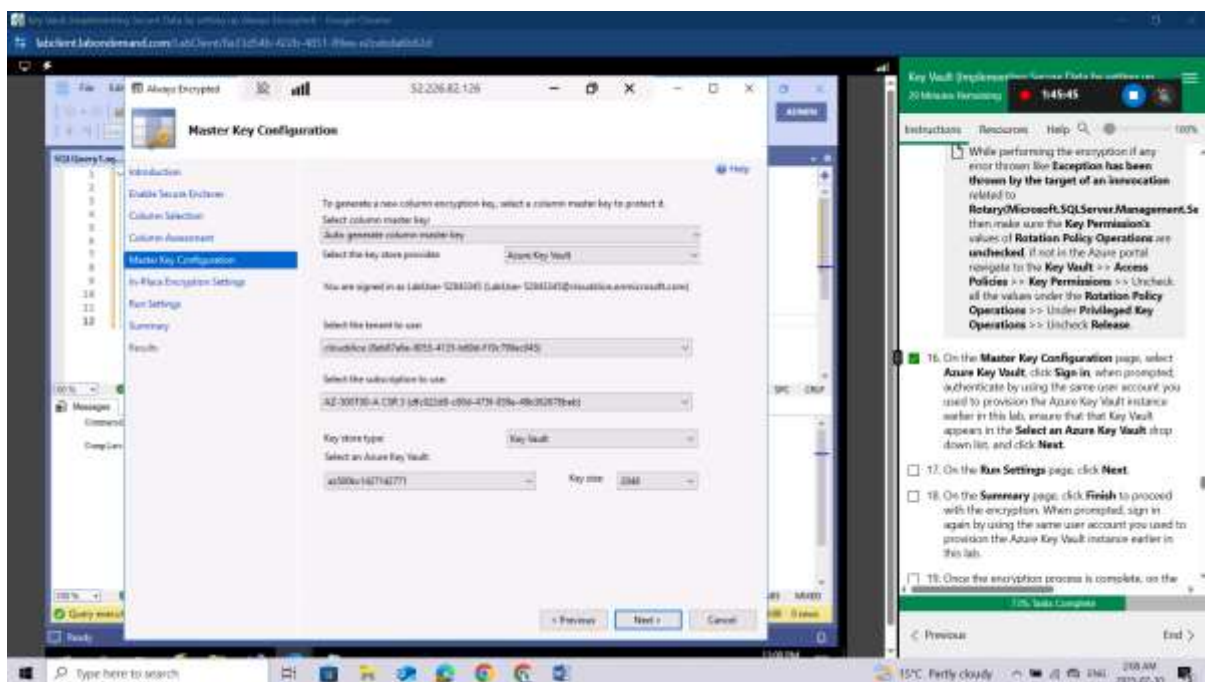
On the **Column Selection** page, select the **SSN** and **Birthdate** columns, set the **Encryption Type** of the **SSN** column to **Deterministic** and of the **Birthdate** column to **Randomized**, and click **Next**.

While performing the encryption if any error thrown like **Exception has been thrown by the target of an invocation** related to **Rotary(Microsoft.SqlServer.Management.ServiceManagement)** then make sure the **Key Permission's** values of **Rotation Policy Operations** are **unchecked**, if not in the Azure portal navigate to the **Key Vault >> Access Policies >> Key Permissions >> Uncheck all the values under the Rotation Policy Operations >> Under Privileged Key Operations >> Uncheck Release**.

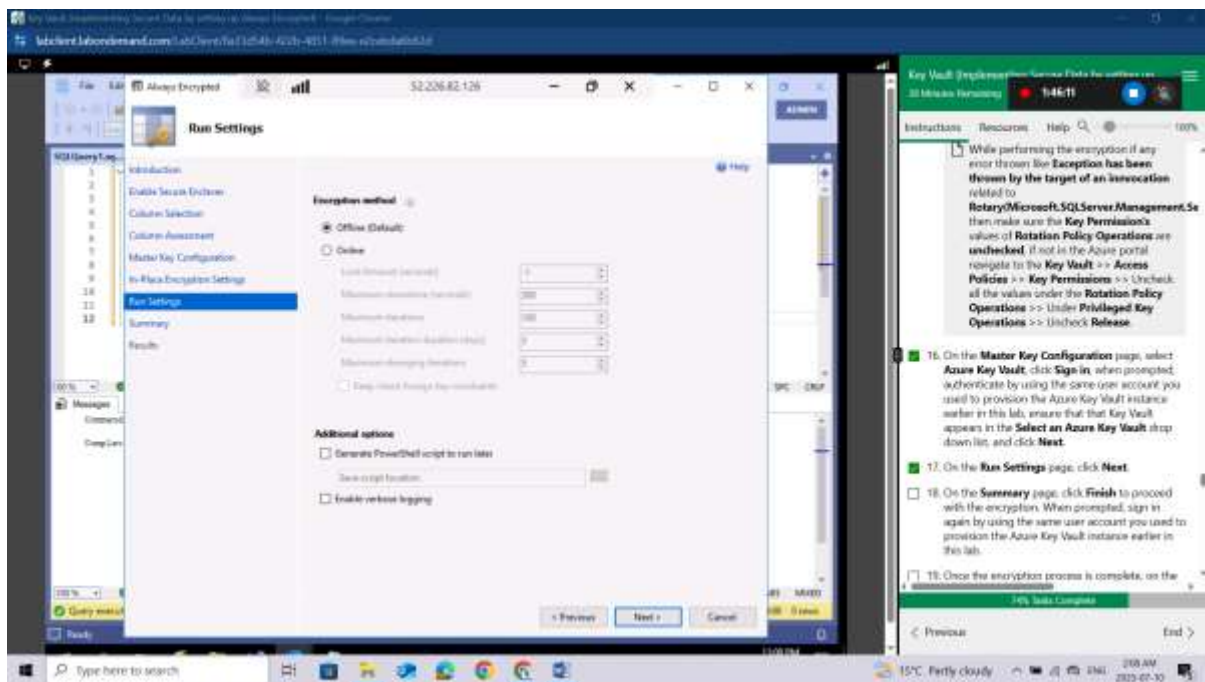




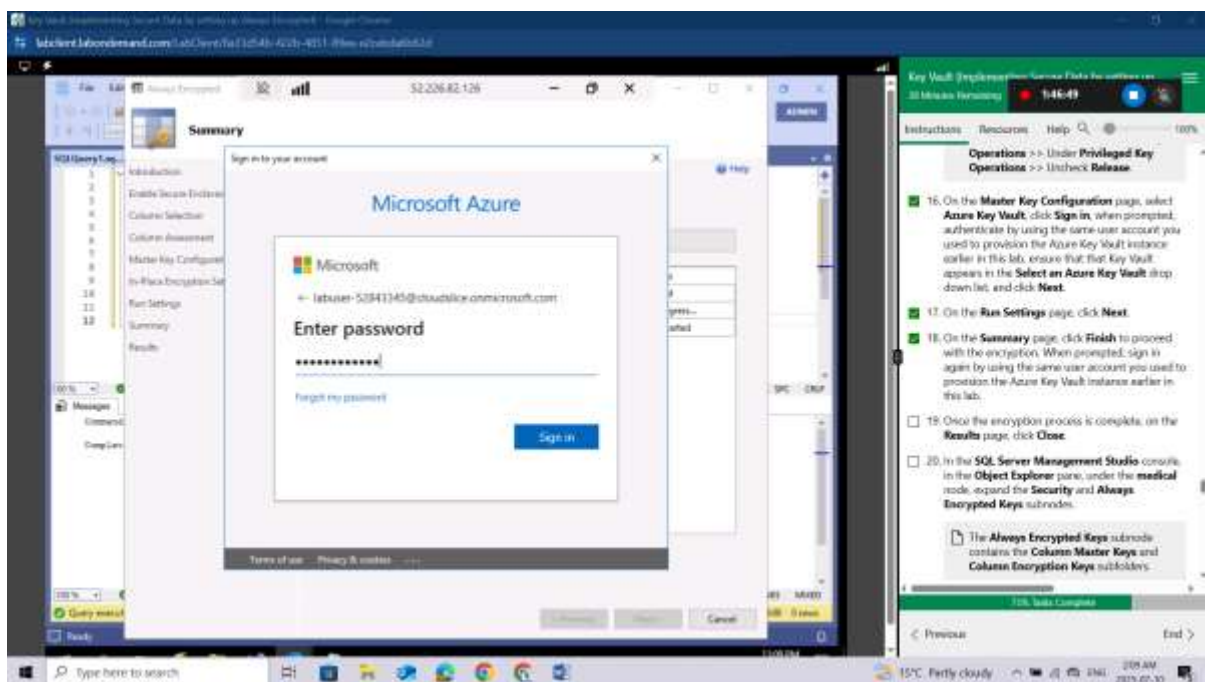
On the **Master Key Configuration** page, select **Azure Key Vault**, click **Sign in**, when prompted, authenticate by using the same user account you used to provision the Azure Key Vault instance earlier in this lab, ensure that that Key Vault appears in the **Select an Azure Key Vault** drop down list, and click **Next**.



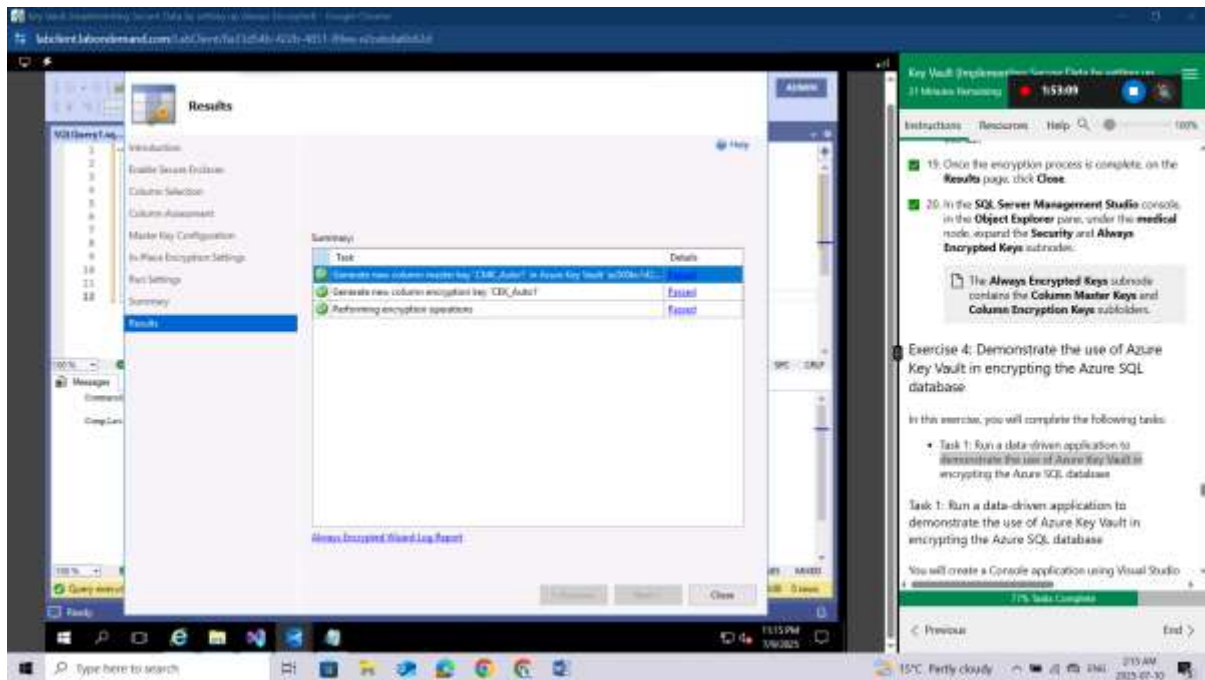
On the **Run Settings** page, click **Next**.



On the **Summary** page, click **Finish** to proceed with the encryption. When prompted, sign in again by using the same user account you used to provision the Azure Key Vault instance earlier in this lab.

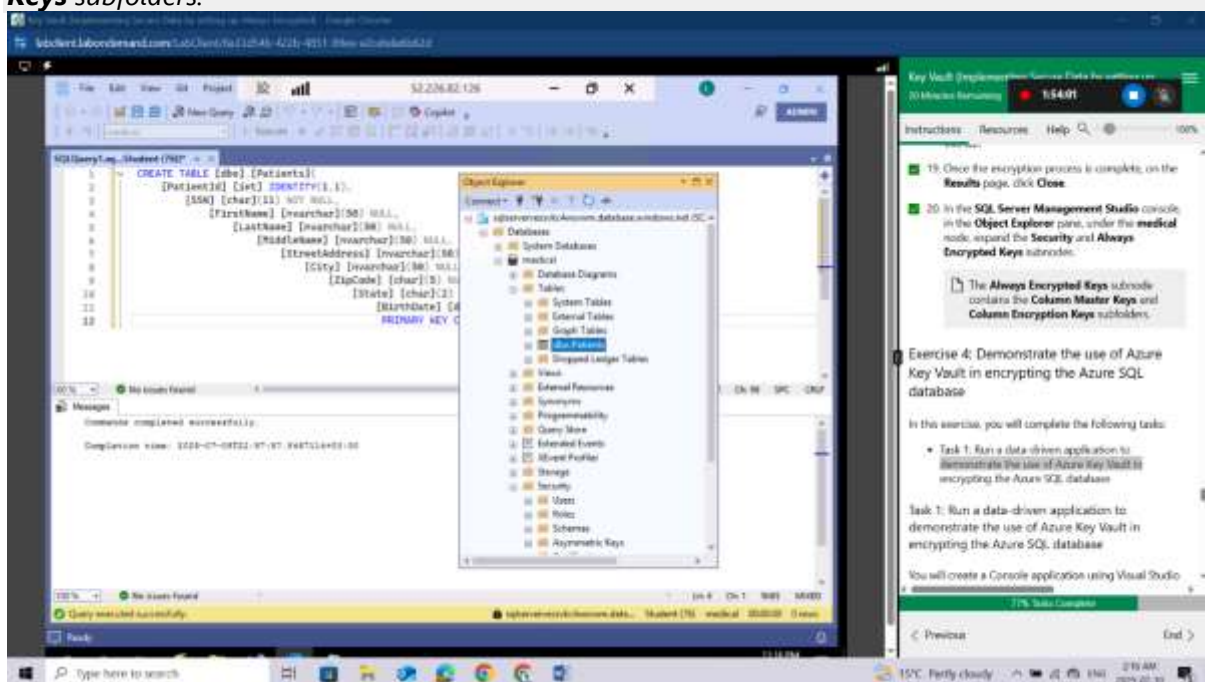


Once the encryption process is complete, on the **Results** page, click **Close**.



In the **SQL Server Management Studio** console, in the **Object Explorer** pane, under the **medical** node, expand the **Security** and **Always Encrypted Keys** subnodes.

The **Always Encrypted Keys** subnode contains the **Column Master Keys** and **Column Encryption Keys** subfolders.



Exercise 4: Demonstrate the use of Azure Key Vault in encrypting the Azure SQL database

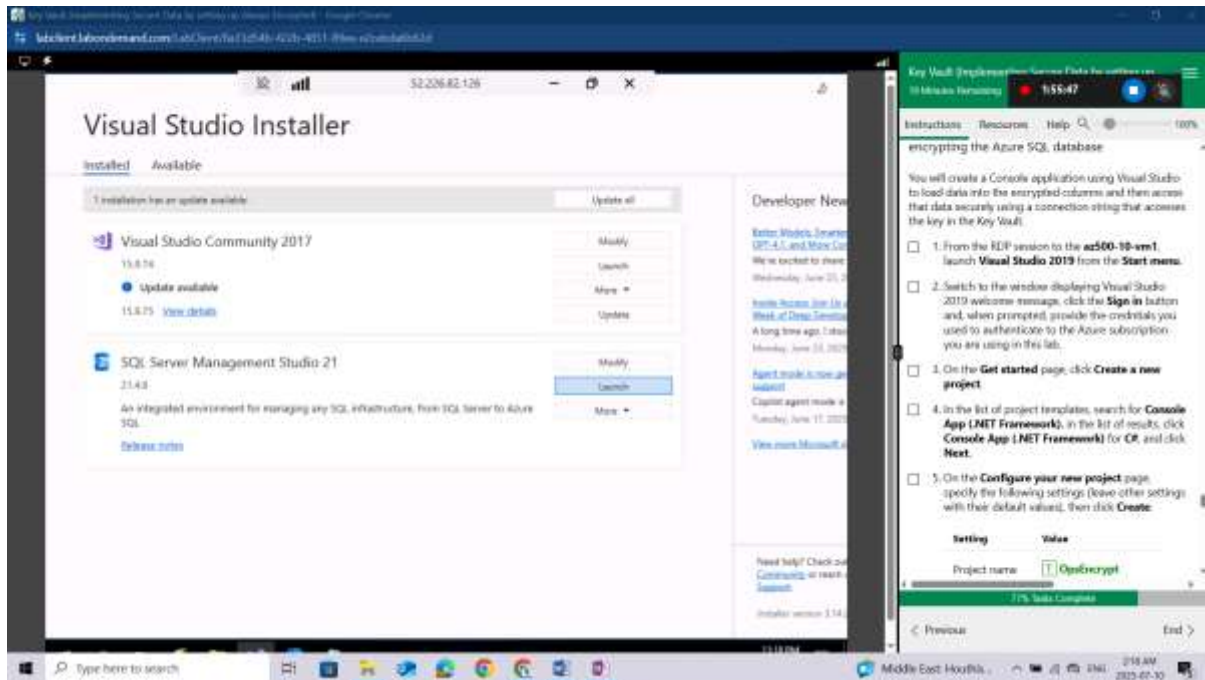
In this exercise, you will complete the following tasks:

- Task 1: Run a data-driven application to demonstrate the use of Azure Key Vault in encrypting the Azure SQL database

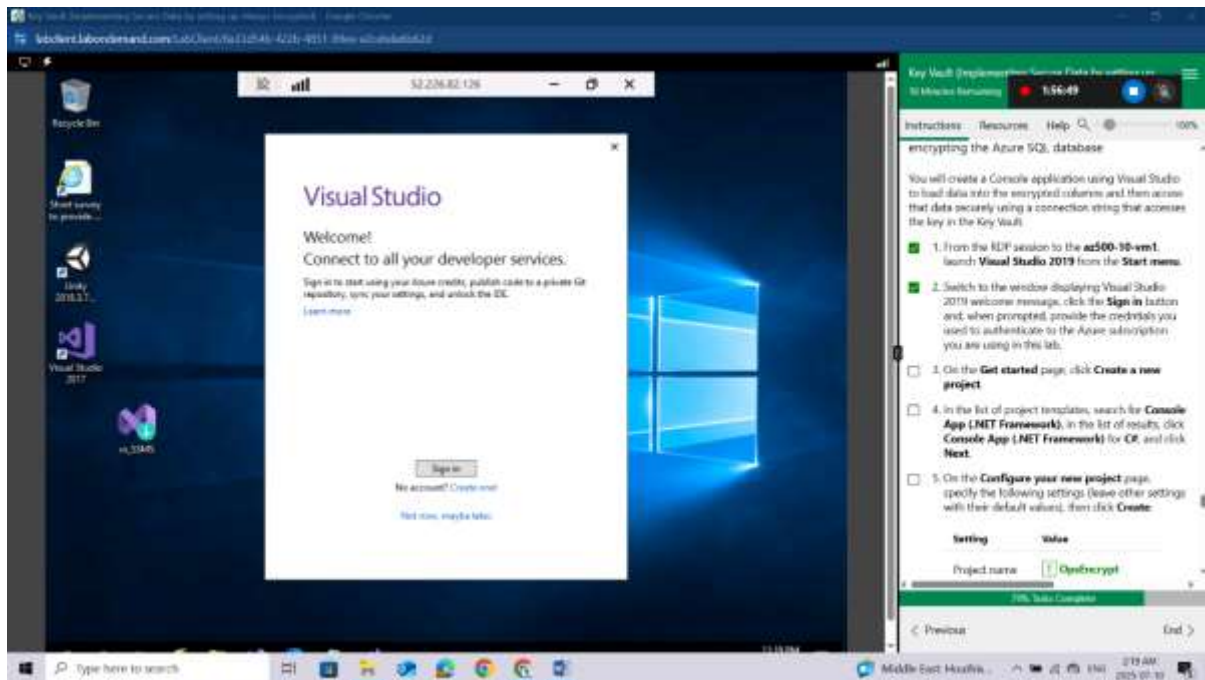
Task 1: Run a data-driven application to demonstrate the use of Azure Key Vault in encrypting the Azure SQL database

You will create a Console application using Visual Studio to load data into the encrypted columns and then access that data securely using a connection string that accesses the key in the Key Vault.

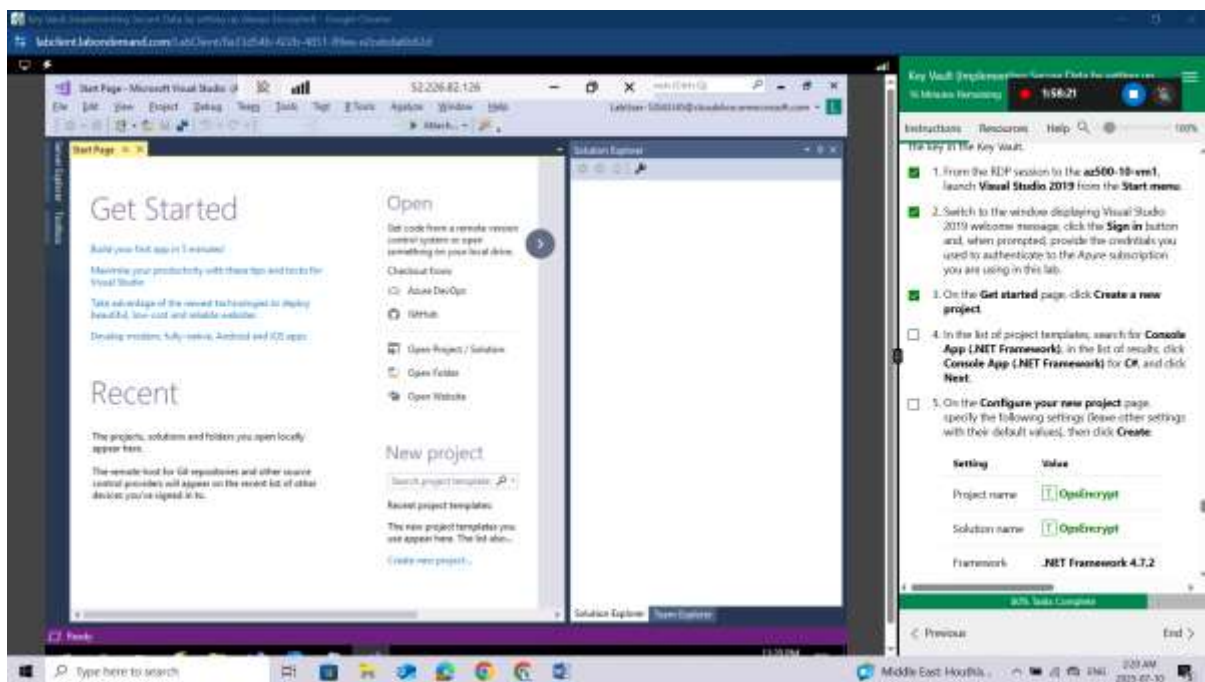
From the RDP session to the **az500-10-vm1**, launch **Visual Studio 2019** from the **Start menu**.



Switch to the window displaying Visual Studio 2019 welcome message, click the **Sign in** button and, when prompted, provide the credentials you used to authenticate to the Azure subscription you are using in this lab



On the **Get started** page, click **Create a new project**.

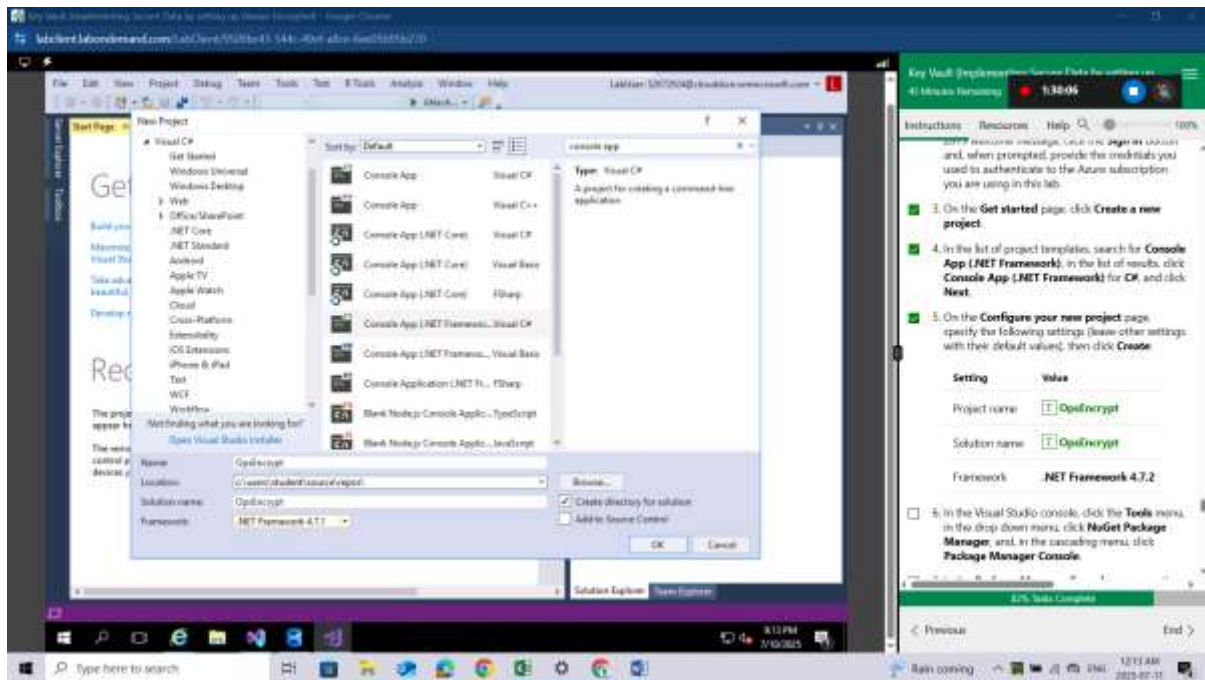


In the list of project templates, search for **Console App (.NET Framework)**, in the list of results, click **Console App (.NET Framework) for C#**,

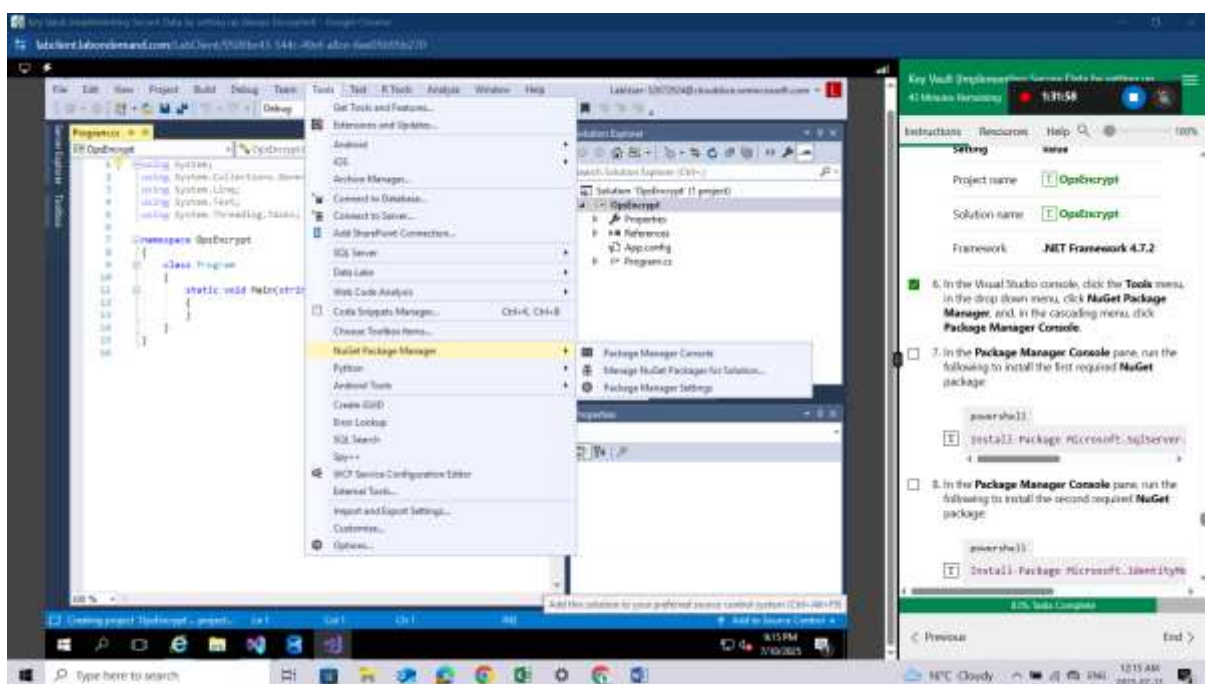
On the **Configure your new project** page, specify the following settings (leave other settings with their default values), then click **Create**:

Setting	Value
Project name	OpsEncrypt

Setting	Value
Solution name	OpsEncrypt
Framework	.NET Framework 4.7.2



In the Visual Studio console, click the **Tools** menu, in the drop down menu, click **NuGet Package Manager**, and, in the cascading menu, click **Package Manager Console**

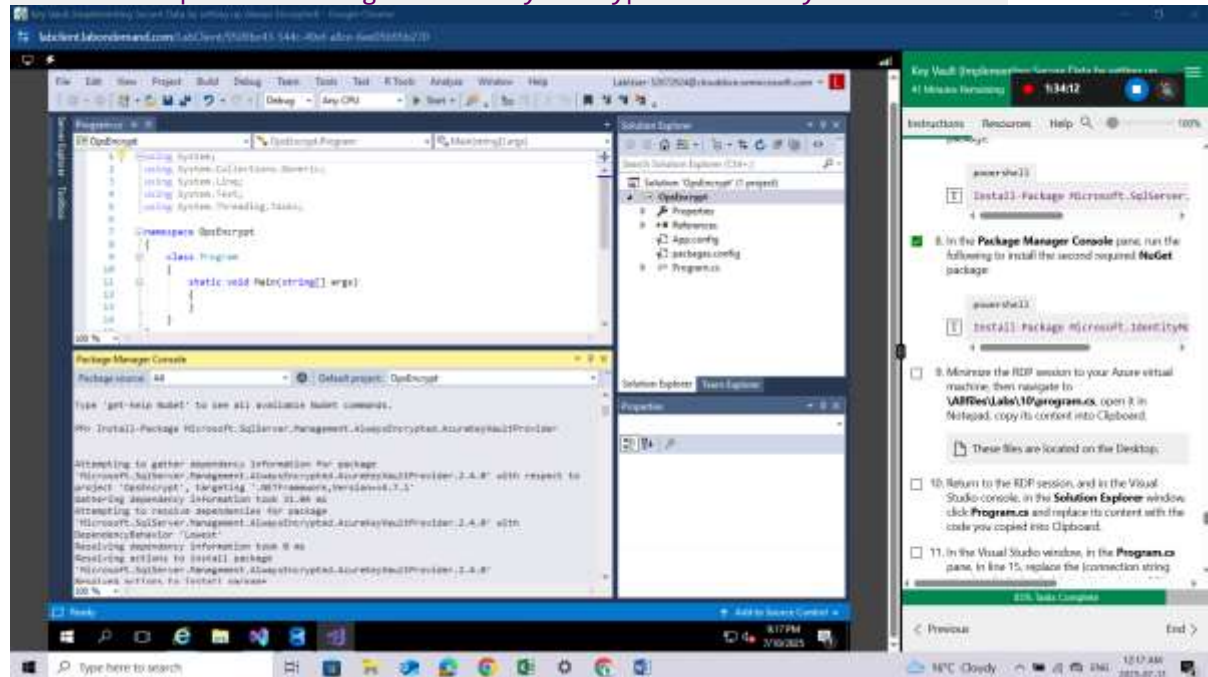


In the **Package Manager Console** pane, run the following to install the first required **NuGet** package:

powershell

Install-Package

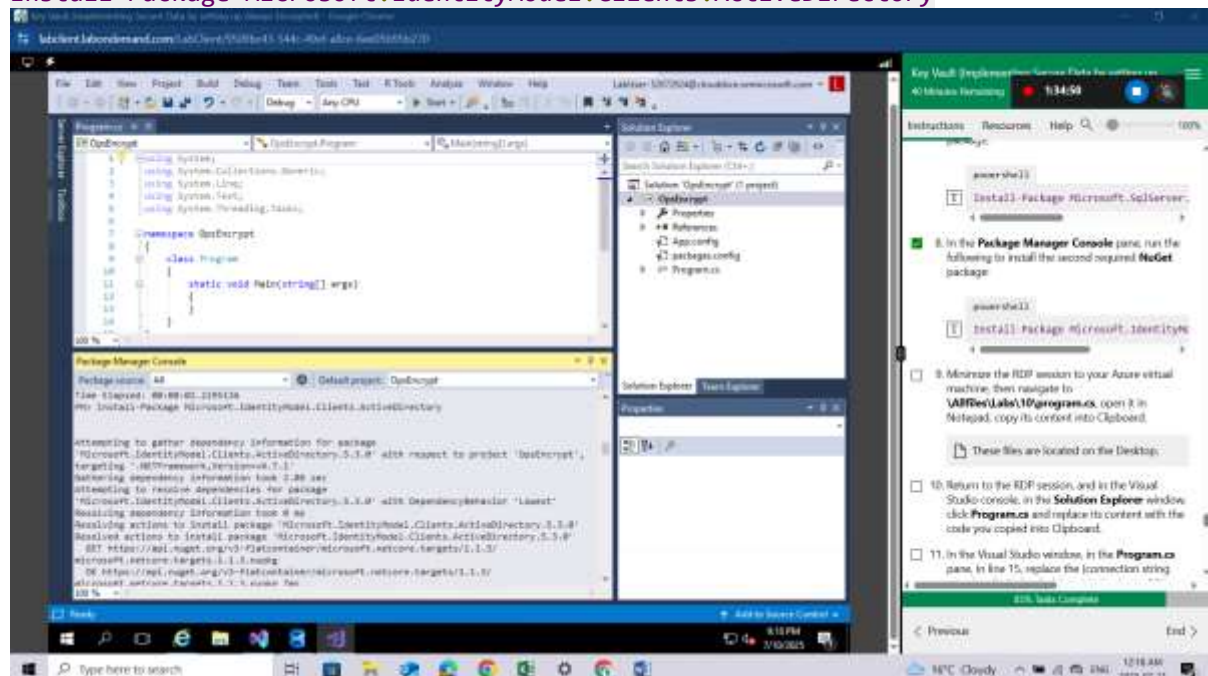
Microsoft.SqlServer.Management.AlwaysEncrypted.AzureKeyVaultProvider



In the **Package Manager Console** pane, run the following to install the second required NuGet package:

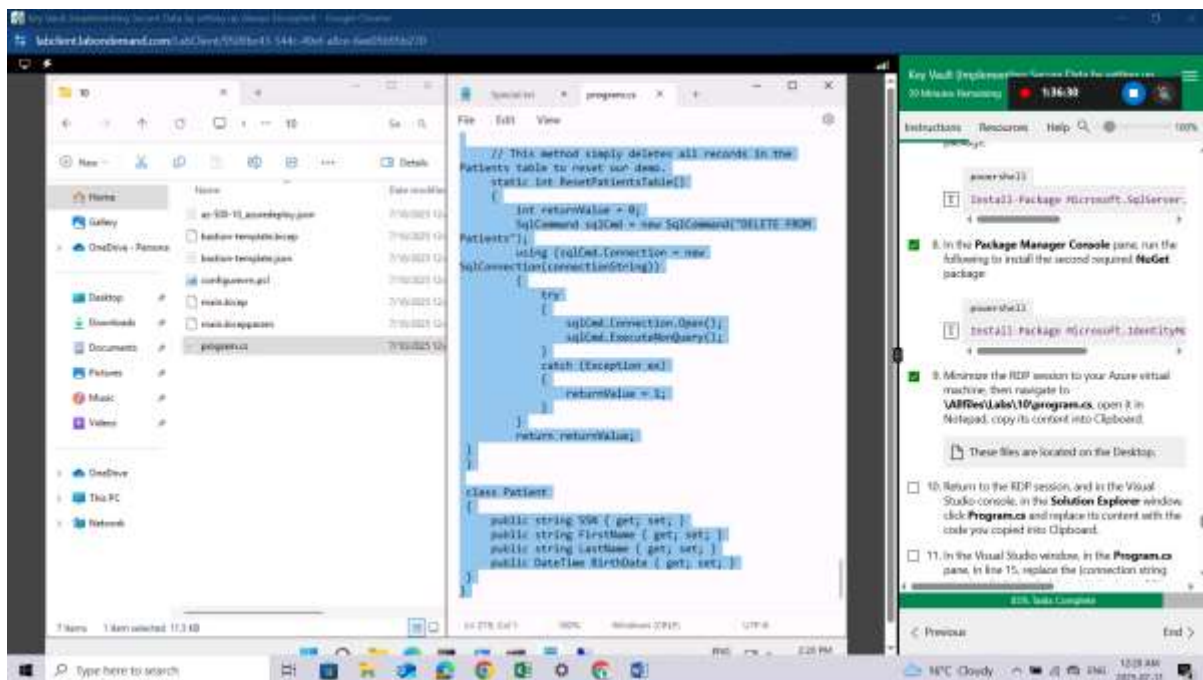
powershell

Install-Package Microsoft.IdentityModel.Clients.ActiveDirectory

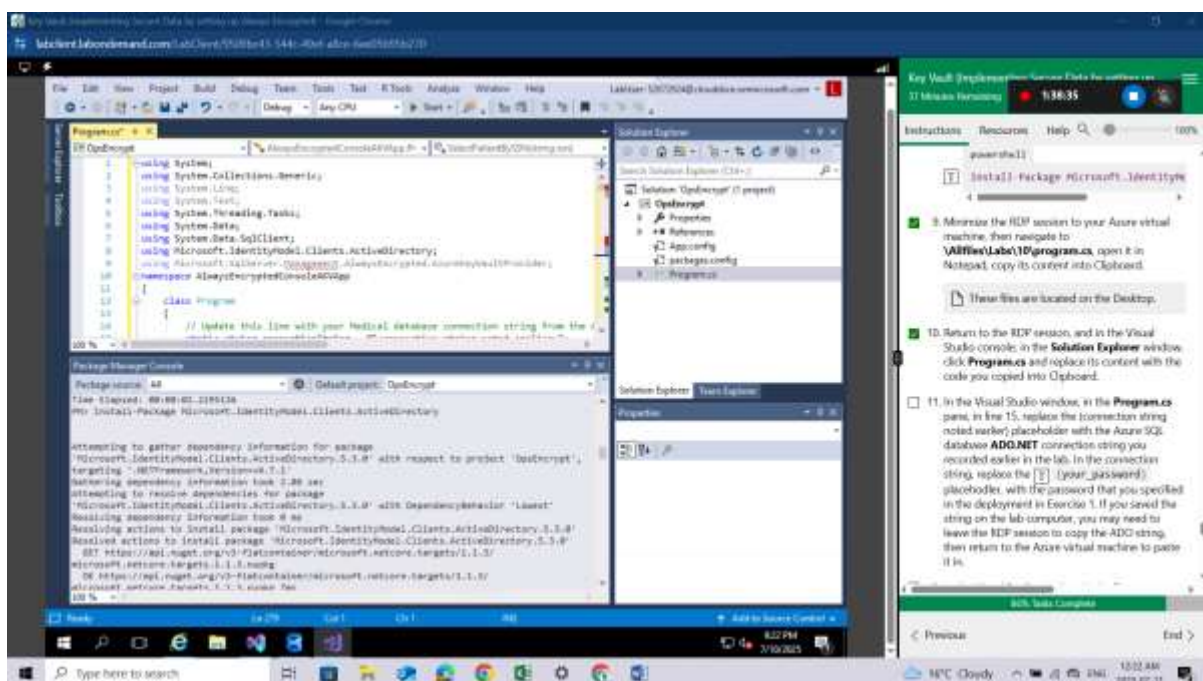


Minimize the RDP session to your Azure virtual machine, then navigate to **\\Allfiles\Labs\10\program.cs**, open it in Notepad, copy its content into Clipboard.

These files are located on the Desktop.



Return to the RDP session, and in the Visual Studio console, in the **Solution Explorer** window, click **Program.cs** and replace its content with the code you copied into Clipboard

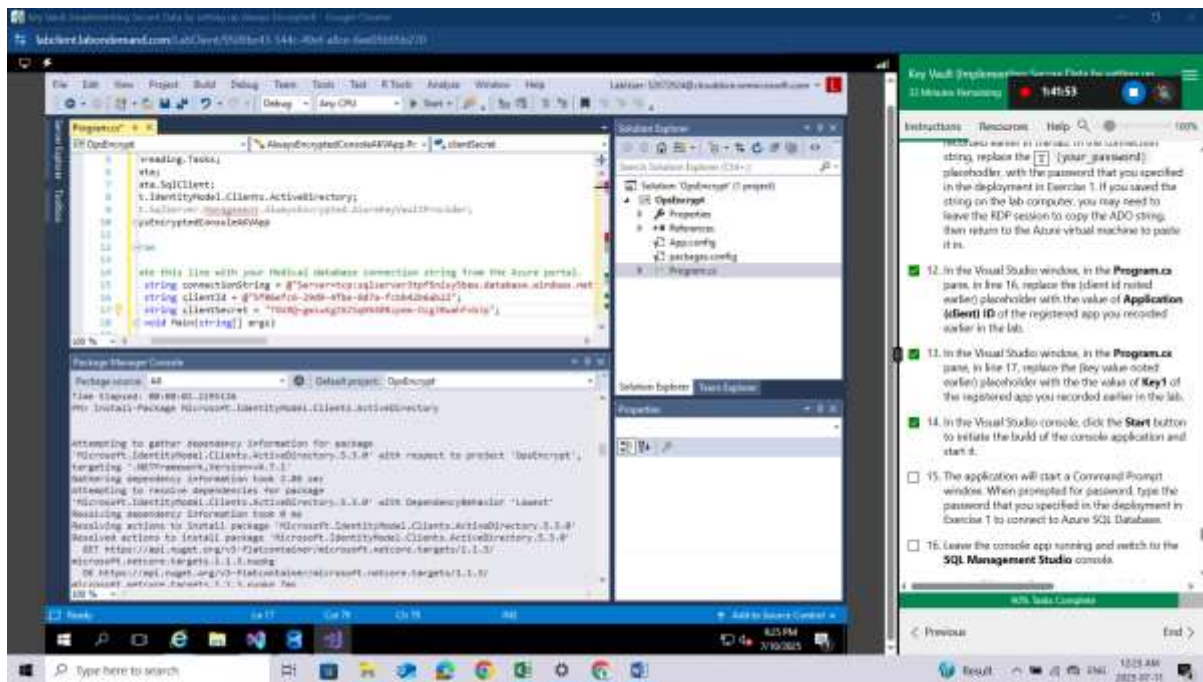


In the Visual Studio window, in the **Program.cs** pane, in line 15, replace the (connection string noted earlier) placeholder with the Azure SQL database **ADO.NET** connection string you recorded earlier in the lab. In the connection string, replace the `{your_password}` placeholder, with the password that you specified in the deployment in Exercise 1. If you saved the string on the lab computer, you may need to leave the RDP session to copy the ADO string, then return to the Azure virtual machine to paste it in.

In the Visual Studio window, in the **Program.cs** pane, in line 16, replace the (client id noted earlier) placeholder with the value of **Application (client) ID** of the registered app you recorded earlier in the lab.

In the Visual Studio window, in the **Program.cs** pane, in line 17, replace the (key value noted earlier) placeholder with the value of **Key1** of the registered app you recorded earlier in the lab

In the Visual Studio console, click the **Start** button to initiate the build of the console application and start it.



The application will start a Command Prompt window. When prompted for password, type the password that you specified in the deployment in Exercise 1 to connect to Azure SQL Database.

Leave the console app running and switch to the **SQL Management Studio** console

In the **Object Explorer** pane, right-click the **medical database** and, in the right-click menu, click **New Query**.

From the query window, run the following query to verify that the data that loaded into the database from the console app is encrypted.

```
sql
SELECT FirstName, LastName, SSN, BirthDate FROM Patients;
```

Switch back to the console application where you are prompted to enter a valid SSN. This will query the encrypted column for the data. At the Command Prompt, type the following and press the Enter key:

```
cmd
999-99-0003
```

Verify that the data returned by the query is not encrypted.

To terminate the console app, press the Enter key

Clean up resources

Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

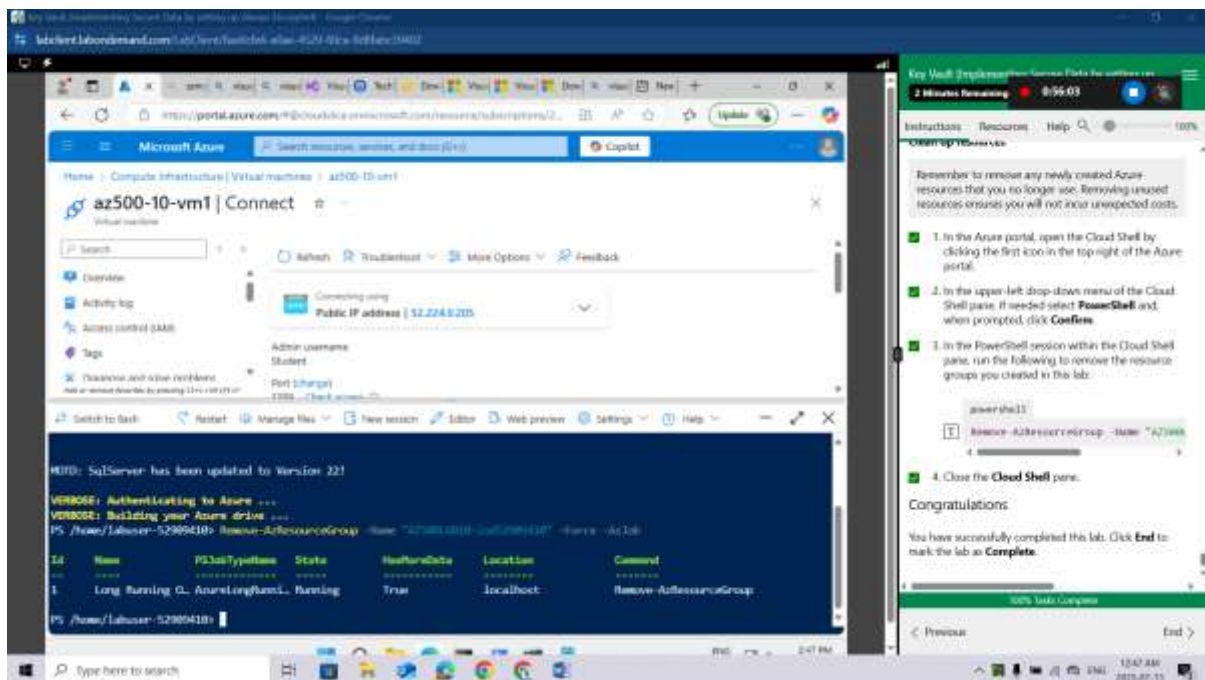
In the Azure portal, open the Cloud Shell by clicking the first icon in the top right of the Azure portal.

In the upper-left drop-down menu of the Cloud Shell pane, if needed select **PowerShell** and, when prompted, click **Confirm**

In the PowerShell session within the Cloud Shell pane, run the following to remove the resource groups you created in this lab:

```
powershell
Remove-AzResourceGroup -Name "AZ500LAB10-lod52843345" -Force -AsJob
```

Close the **Cloud Shell** pane.



Congratulations! You have successfully completed this lab. Click **End** to mark the lab as **Complete**.

Conclusion

In this lab, I successfully implemented secure data practices using Azure Key Vault and the Always Encrypted feature in Azure SQL Database. I began by deploying the base infrastructure, including an Azure virtual machine and a SQL database, using an ARM template. I then configured the Azure Key Vault by creating both a key and a secret essential for encryption operations. After that, I set up the Azure SQL database along with a sample application to demonstrate how sensitive data can be encrypted and accessed securely. Finally, I integrated the Key Vault with the database to enable Always Encrypted, ensuring that sensitive information remains protected even during data processing and is accessible only to authorized users. This lab provided a practical understanding of

how Azure Key Vault enhances data protection and helps meet compliance requirements through encryption.